

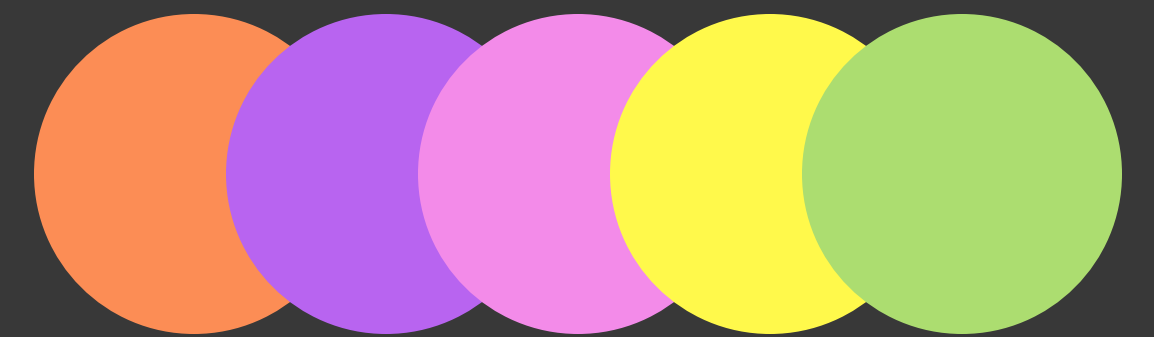
Jeff Tao



9/21/2026

CIS 5450: Big Data Analytics

Data Visualization





speculative.tech

Jeff Tao

CIS PhD Student (4th year)

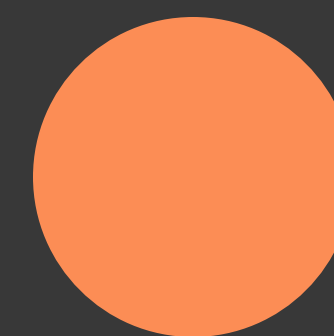
Current Research: Data-oriented tools for debugging code, ML-based query optimization

Previously: MS Thesis @ Columbia
Thesis topic: *An Execution Engine for Physical Visualization Design*

Pre-previously: 5 years senior software engineer @ MongoDB, Microsoft

- 
1. Why Visualize?
 2. How to Visualize?
 3. Case Studies
 4. Viz Research (briefly)

Why Visualize?



Your Brain is Good at Pattern Recognition

```
[10] :
```

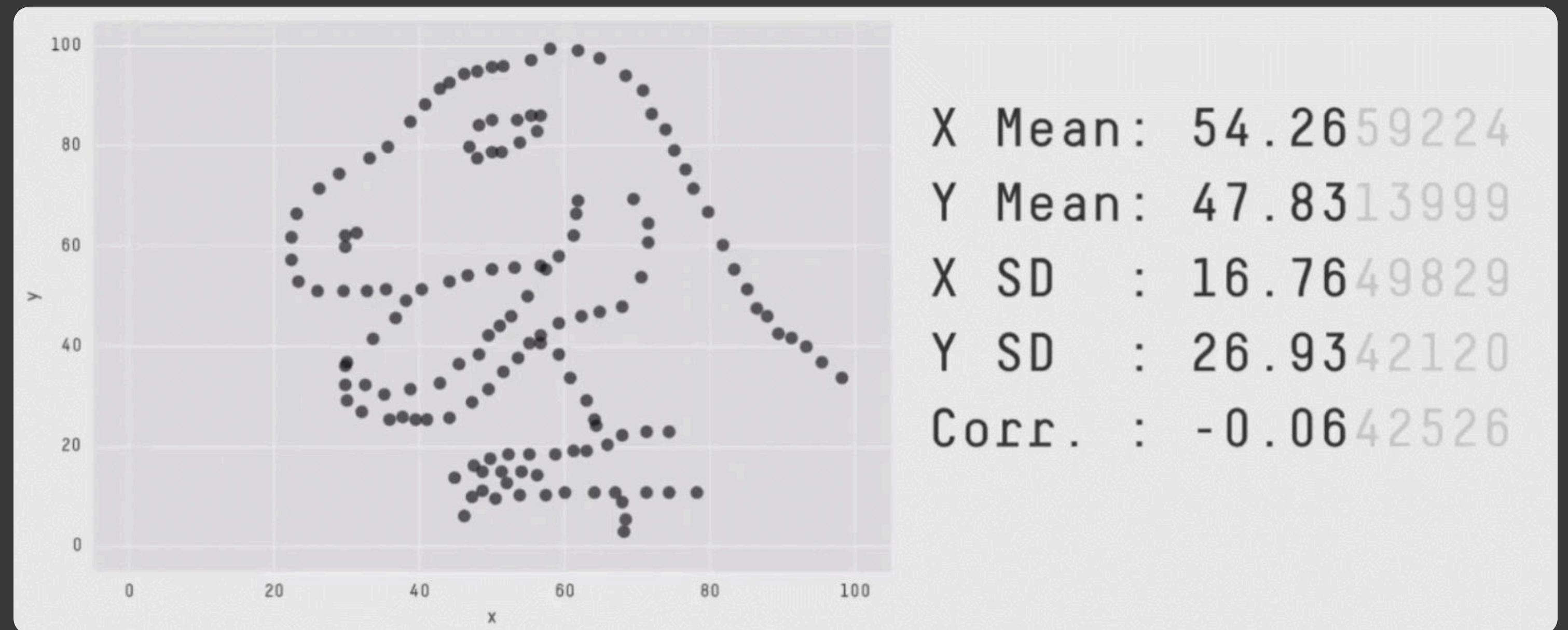
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0	55.3846	97.1795
1	51.5385	96.0256
2	46.1538	94.4872
3	42.8205	91.4103
4	40.7692	88.3333
...	...	...
137	39.4872	25.3846
138	91.2821	41.5385
139	50.0000	95.7692
140	47.9487	95.0000
141	44.1026	92.6923

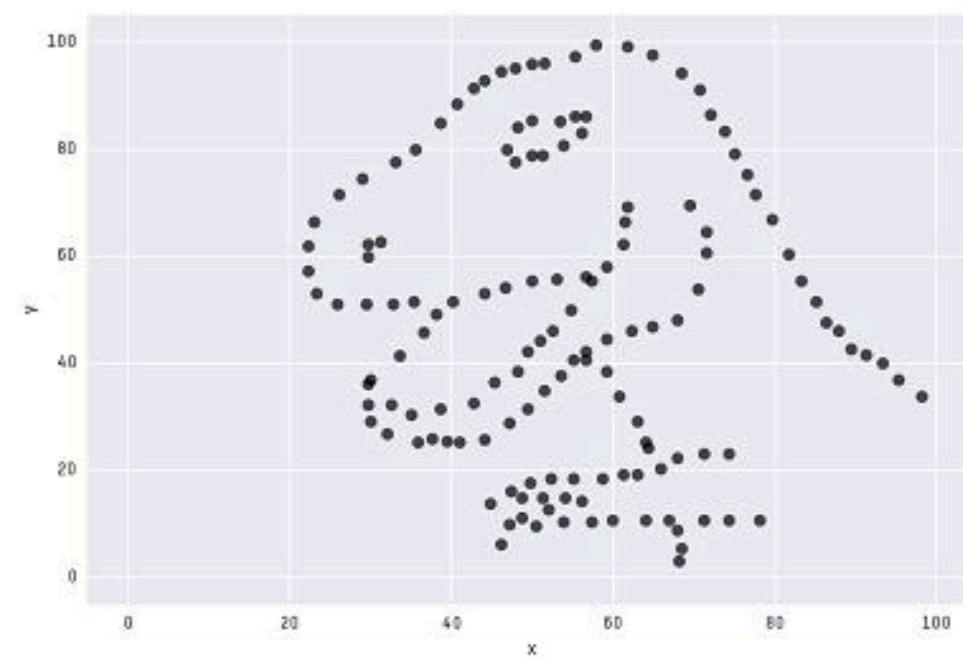
142 rows × 2 columns

Your Brain is Good at Pattern Recognition

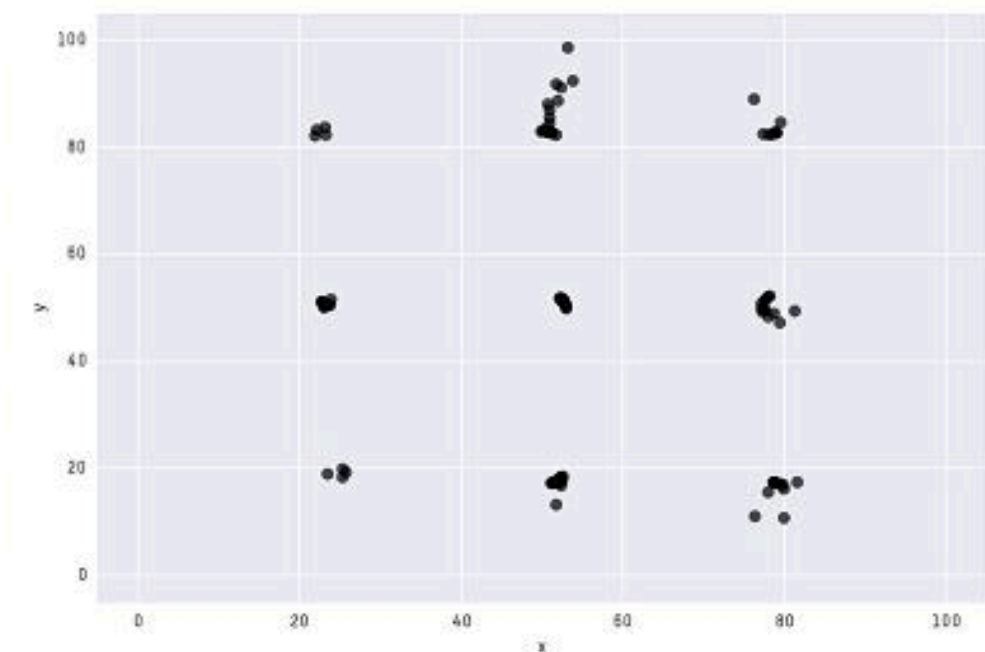
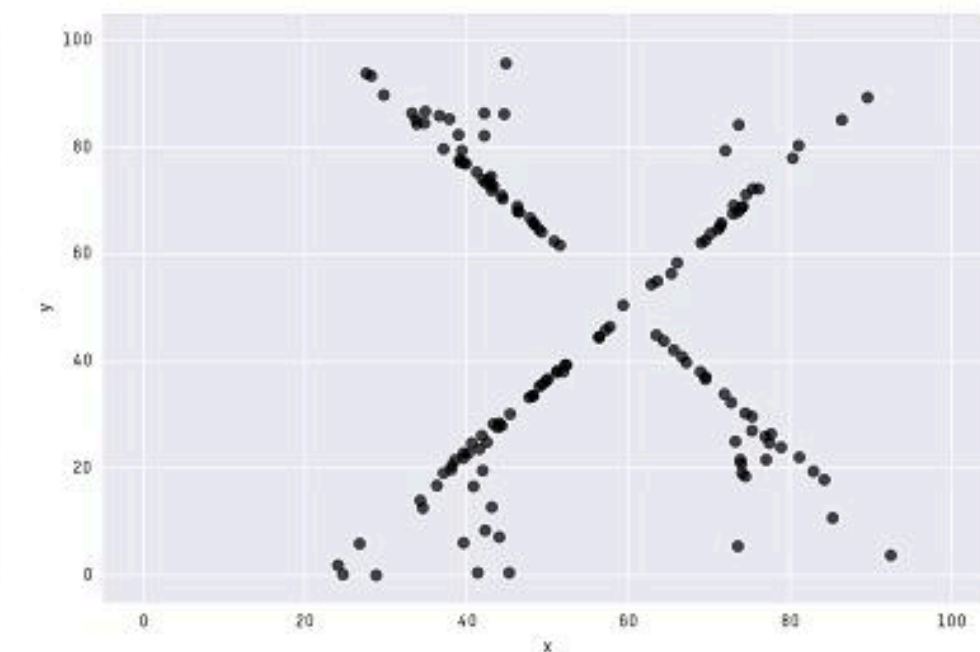
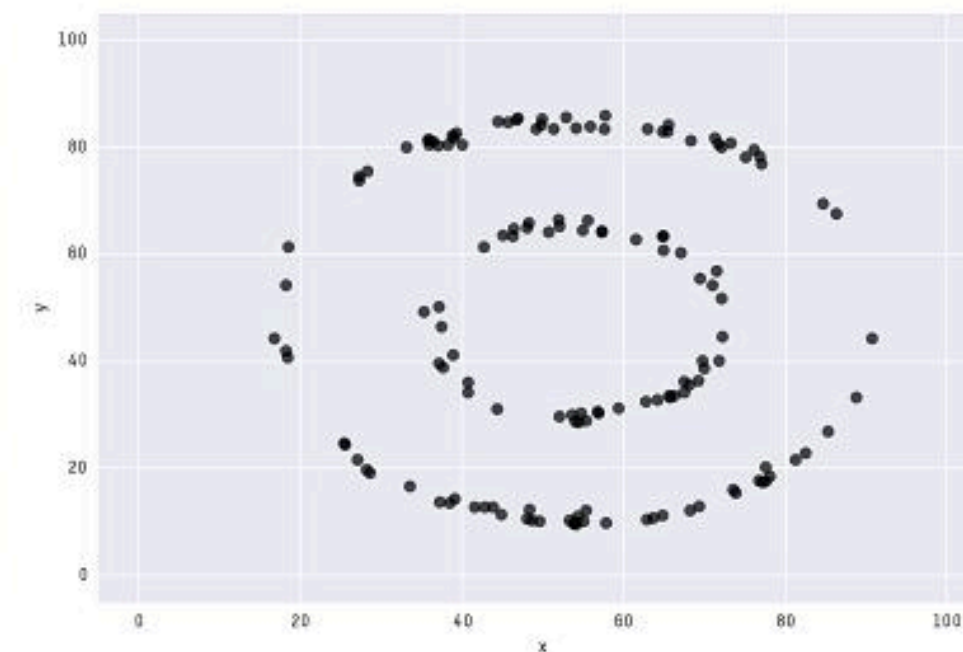
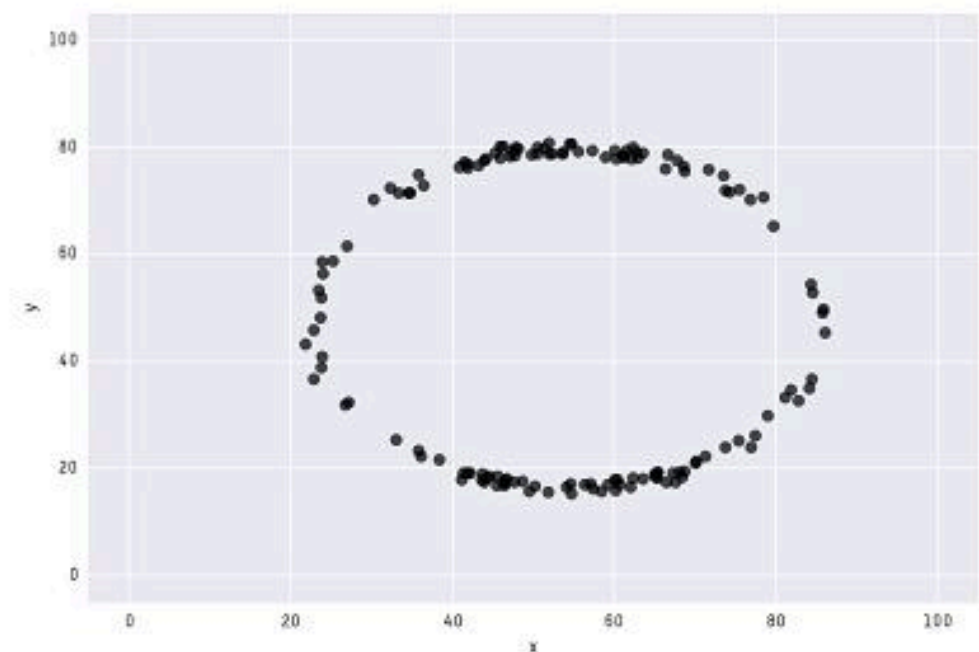
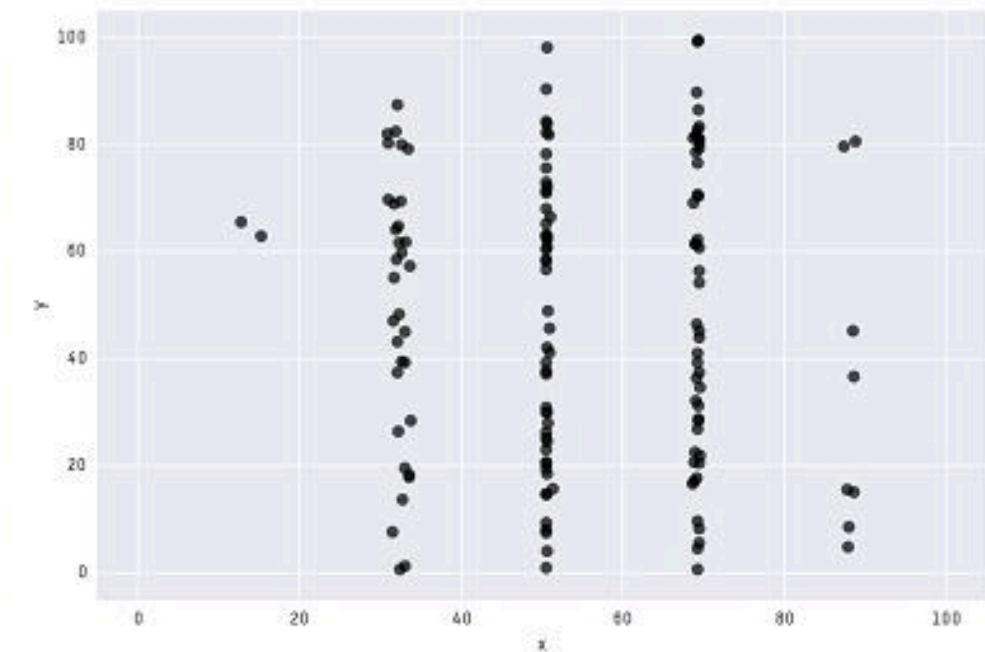
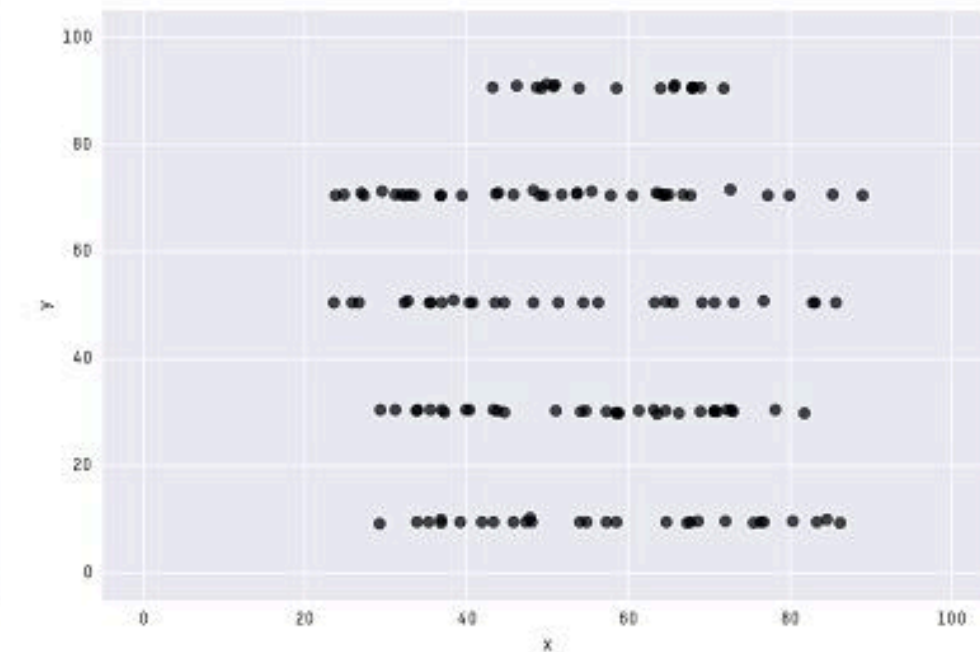
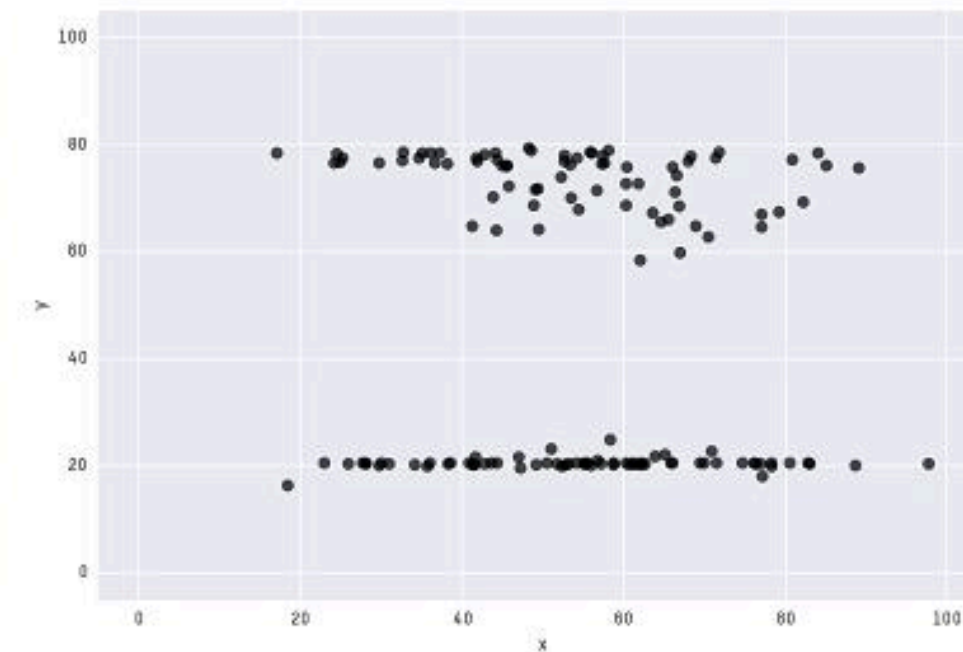
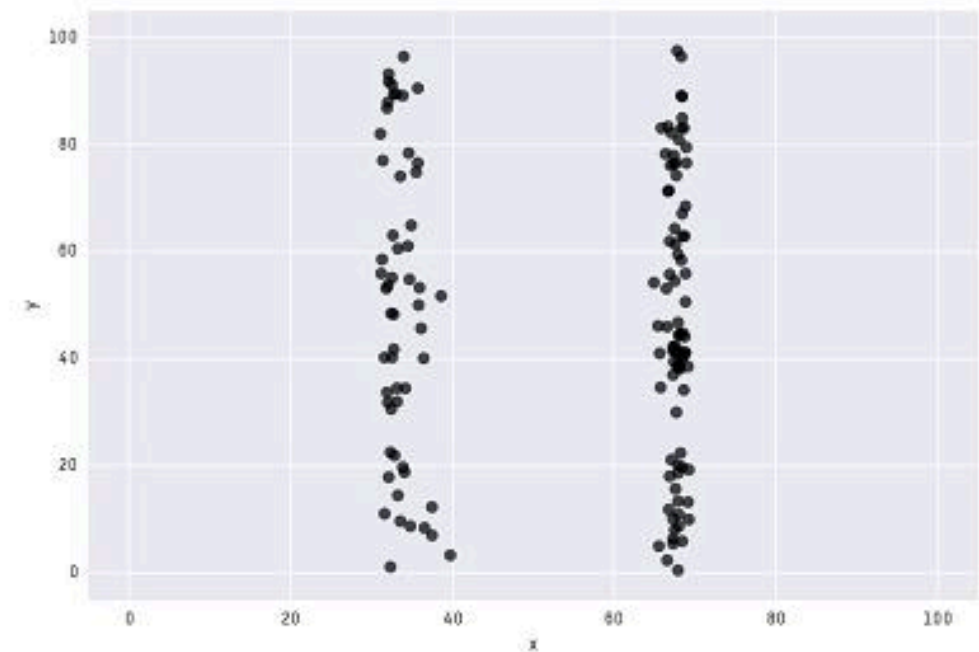
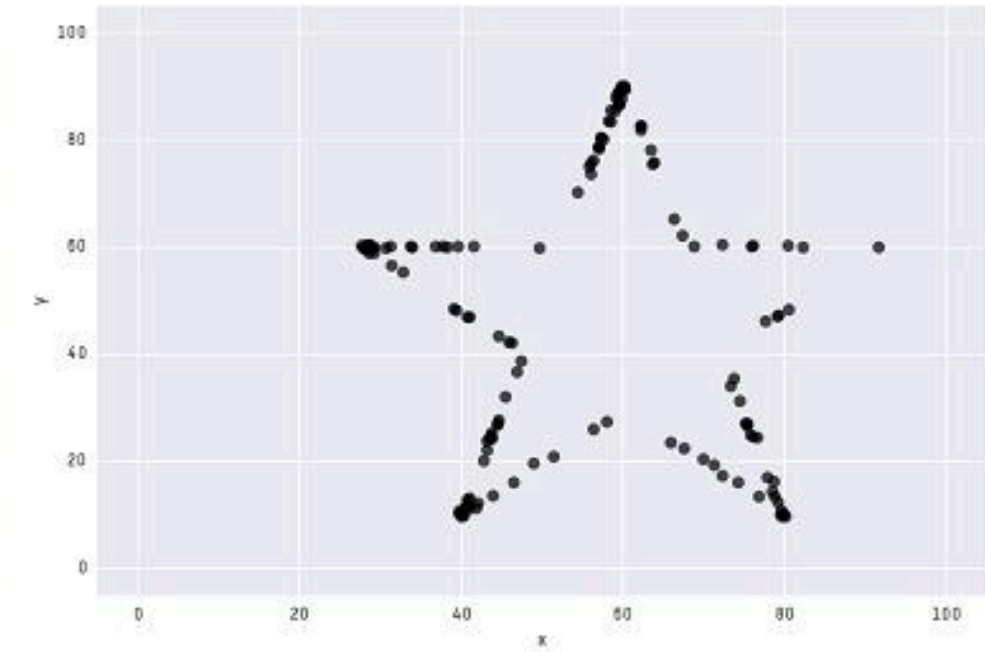
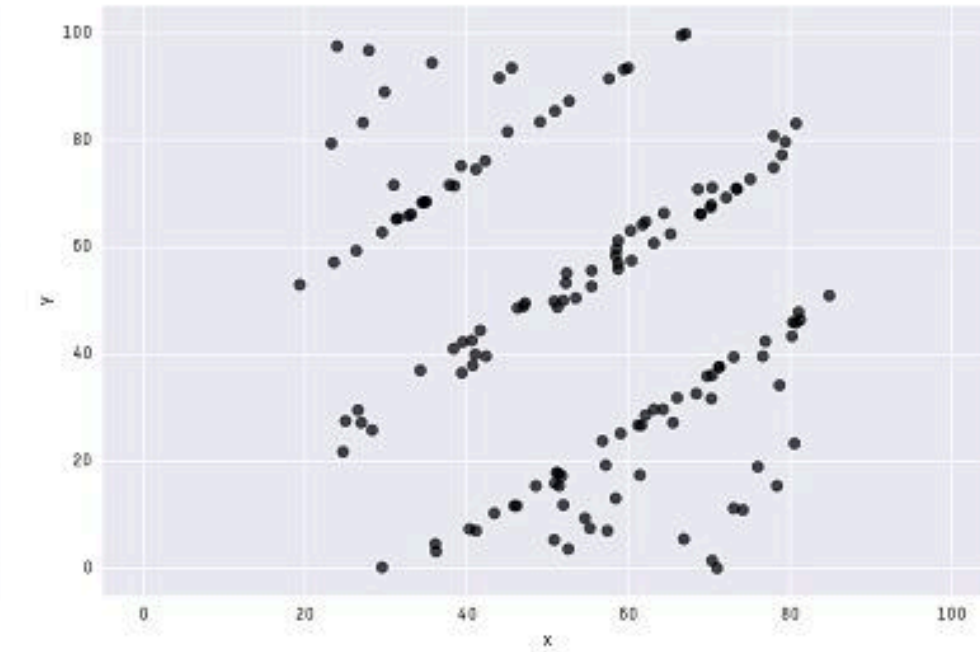
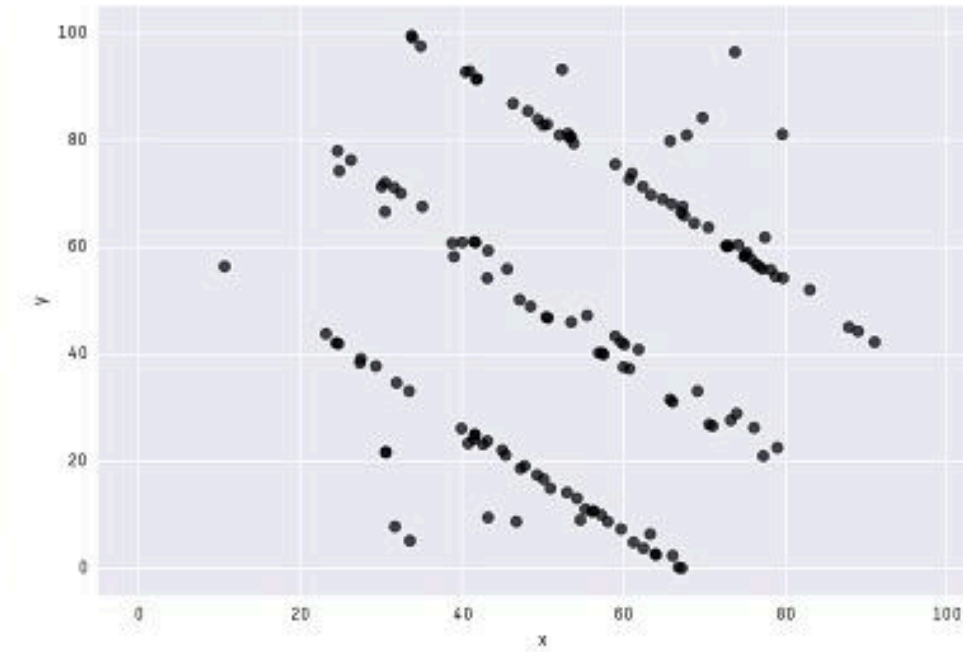
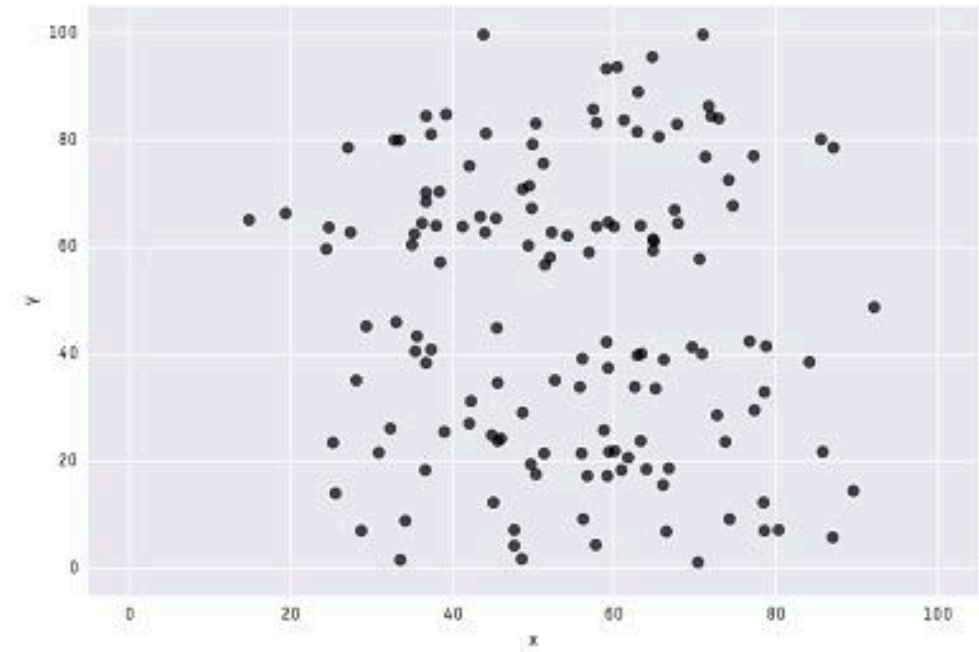
```
X Mean: 54.2659224
Y Mean: 47.8313999
X SD   : 16.7649829
Y SD   : 26.9342120
Corr.  : -0.0642526
```

Your Brain is Good at Pattern Recognition





X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06



Use Cases



To Explore

Your brain is really good at noticing visual patterns!

To Debug

“What changed about the data after I ran this code?”

To Present

Help *other people* understand what you found in the data.

Visualizations Are
Everywhere!

↑ Today February 2025						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Jan 26	27 10:15 AM Type Systems 2 PM Feedback section	28 1:45 PM Writing and Speak 4 PM Code search copilot 5:15 PM Beginning Canton 5:30 PM [Tentative] CIS 54	29 10:15 AM Type Systems 1 PM Ryan / Jeff 1-1 3 PM Jessica / Jeff talk about	30 11 AM Office Hours 1:45 PM Writing and Speak 3:30 PM Bayes opt for que 5:15 PM Beginning Canton	31 11 AM [C] PennHCI Group 11 AM Andrew group all-h 2 PM Jeff / Andrew - 1-on- 3:30 PM FIG: Formula Inter	Feb 1
2	3 10:15 AM Type Systems	4 1:45 PM Writing and Speak 4 PM Code search copilot 5:15 PM Beginning Canton 5:30 PM [Tentative] CIS 54	5 10:15 AM Type Systems 1 PM Ryan / Jeff 1-1	6 11 AM Office Hours 1:45 PM Writing and Speak 3:30 PM Bayes opt for que 5:15 PM Beginning Canton	7 11 AM [C] PennHCI Group 11 AM Andrew group all-h 2 PM Jeff / Andrew - 1-on- 3:30 PM Hold for FIG: Form	8
9	10 10:15 AM Type Systems	11 1:45 PM Writing and Speak 4 PM Code search copilot 5:15 PM Beginning Canton 5:30 PM [Tentative] CIS 54	12 10:15 AM Type Systems 1 PM Ryan / Jeff 1-1	13 11 AM Office Hours 1:45 PM Writing and Speak 3:30 PM Bayes opt for que 5:15 PM Beginning Canton	14 11 AM [C] PennHCI Group 11 AM Andrew group all-h 2 PM Jeff / Andrew - 1-on- 3:30 PM Hold for FIG: Form	15
16	17 PLATEAU 10:15 AM Type Systems	18 1:45 PM Writing and Speak 4 PM Code search copilot 5:15 PM Beginning Canton 5:30 PM [Tentative] CIS 54	19 10:15 AM Type Systems 1 PM Ryan / Jeff 1-1	20 11 AM Office Hours 1:45 PM Writing and Speak 3:30 PM Bayes opt for que 5:15 PM Beginning Canton	21 11 AM [C] PennHCI Group 11 AM Andrew group all-h 2 PM Jeff / Andrew - 1-on- 3:30 PM Hold for FIG: Form	22
23	24 10:15 AM Type Systems	25 1:45 PM Writing and Speak 4 PM Code search copilot 5:15 PM Beginning Canton 5:30 PM [Tentative] CIS 54	26 10:15 AM Type Systems 1 PM Ryan / Jeff 1-1	27 11 AM Office Hours 1:45 PM Writing and Speak 3:30 PM Bayes opt for que 5:15 PM Beginning Canton	28 11 AM [C] PennHCI Group 11 AM Andrew group all-h 2 PM Jeff / Andrew - 1-on- 3:30 PM Hold for FIG: Form	Mar 1

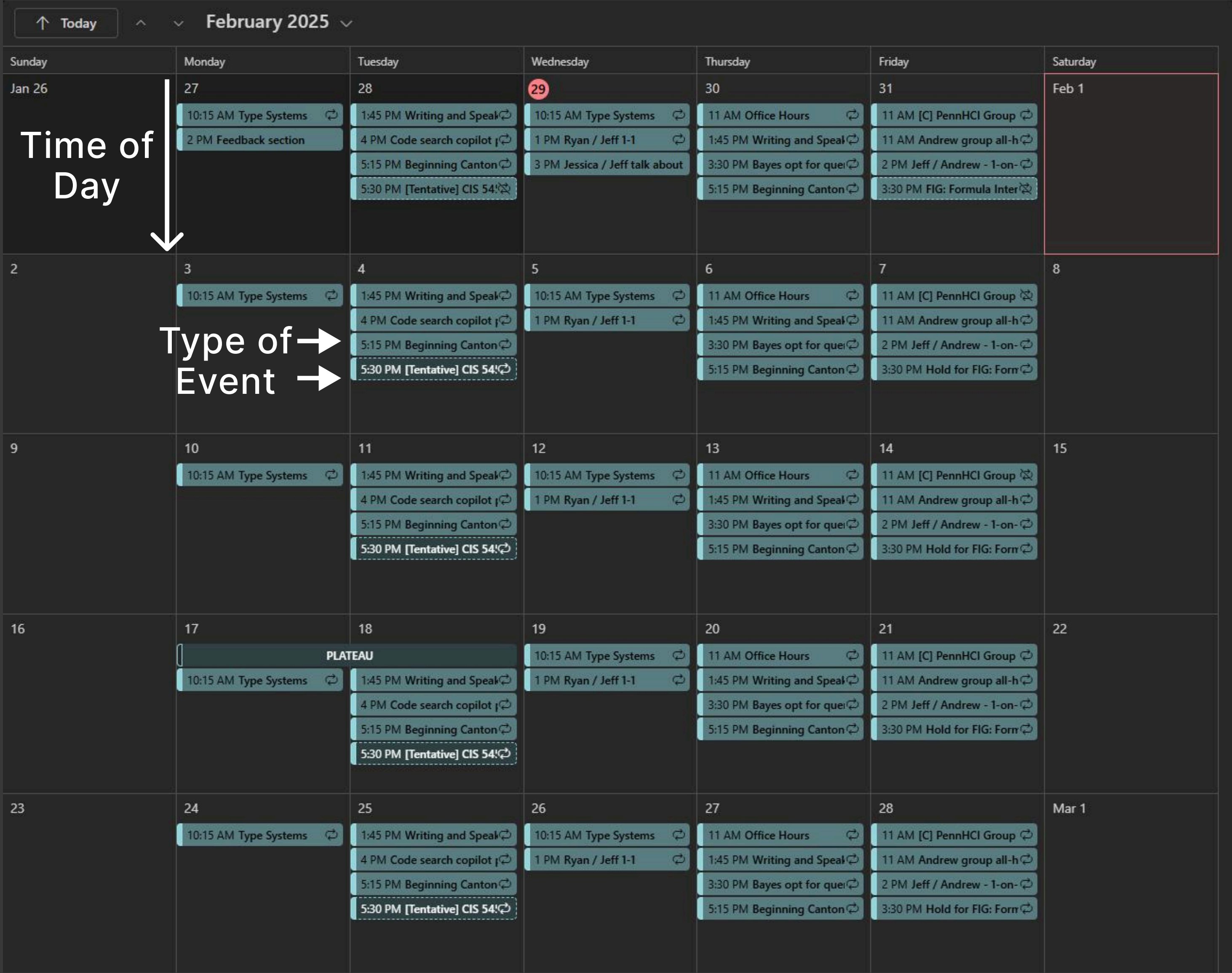
Visualizations Are Everywhere!

Week of Month

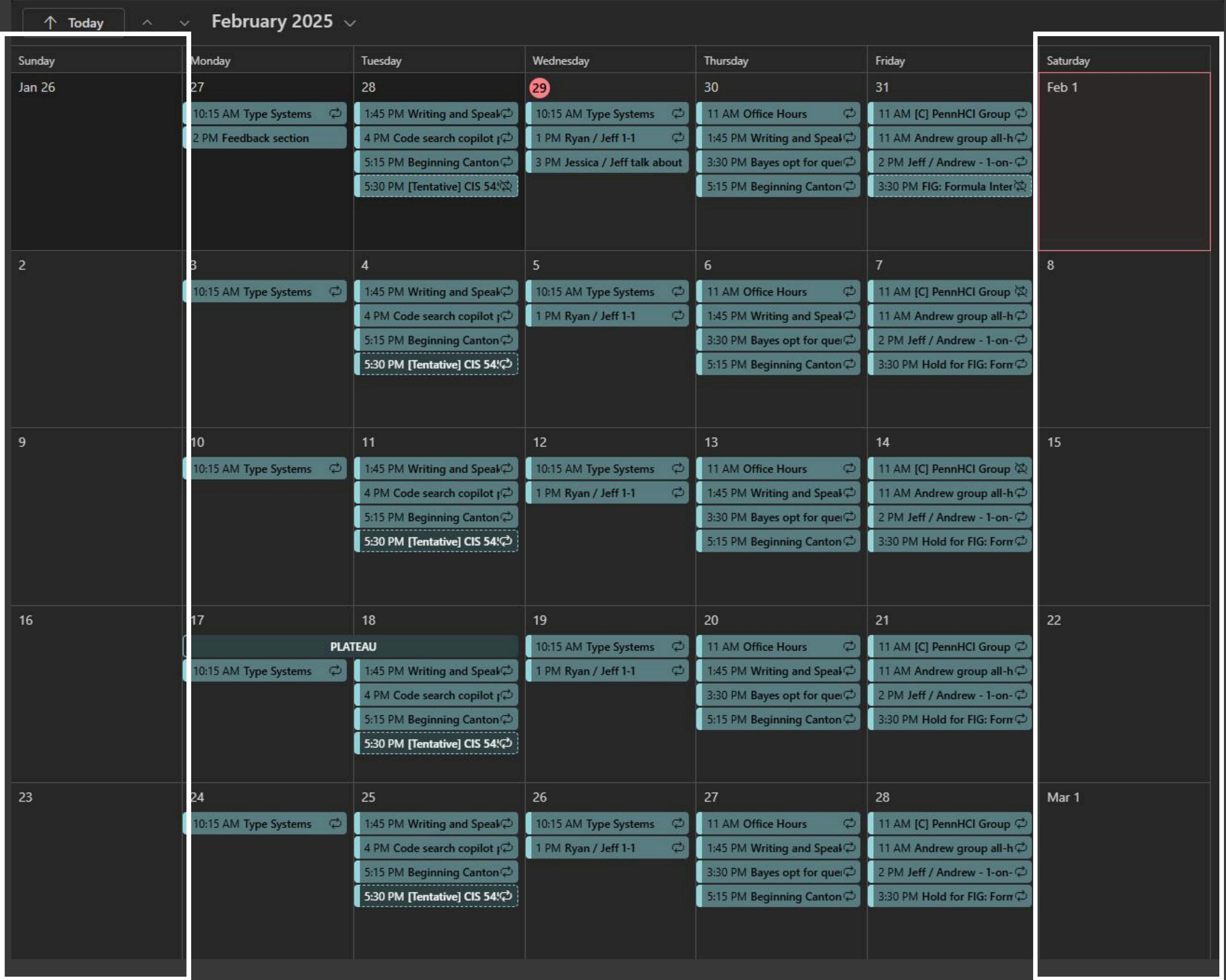
Time of Day

Type of Event

Day of Week



Visualizations Are Everywhere!



Visualizations Are Everywhere!

Today

Wed, Jan 29

51° / 29°

10:15 AM

1h 30m

Type Systems

Towne 327

1:00 PM

1h

Ryan / Jeff 1-1

rcmarcus@seas.upenn.edu

Towne 219C or Online

3:00 PM

30m

Jessica / Jeff talk about proof assistants

jwshi@seas.upenn.edu

Tomorrow

Thu, Jan 30

42° / 33°

11:00 AM

2h

Office Hours

1:45 PM

1h 30m

Writing and Speaking with Style

3:30 PM

1h

Bayes opt for query optimization

Levine 460

5:15 PM

1h 30m

Beginning Cantonese II

Fri, Jan 31

48° / 37°

11:00 AM

1h

[C] PennHCI Group Meeting

<https://upenn.zoom.us/j/91976100977?p...>

11:00 AM

15m

Andrew group all-hands

Levine 303

2:00 PM

30m

Jeff / Andrew - 1-on-1

head@seas.upenn.edu

Levine 303

3:30 PM

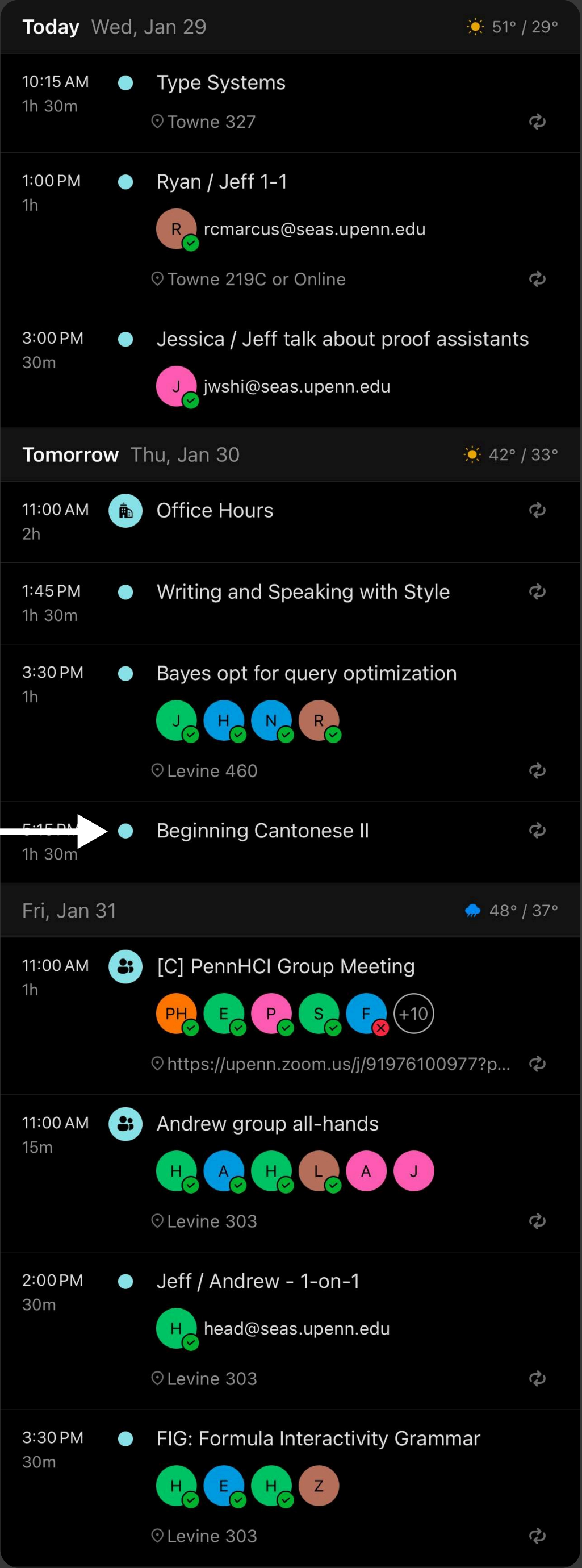
30m

FIG: Formula Interactivity Grammar

Levine 303

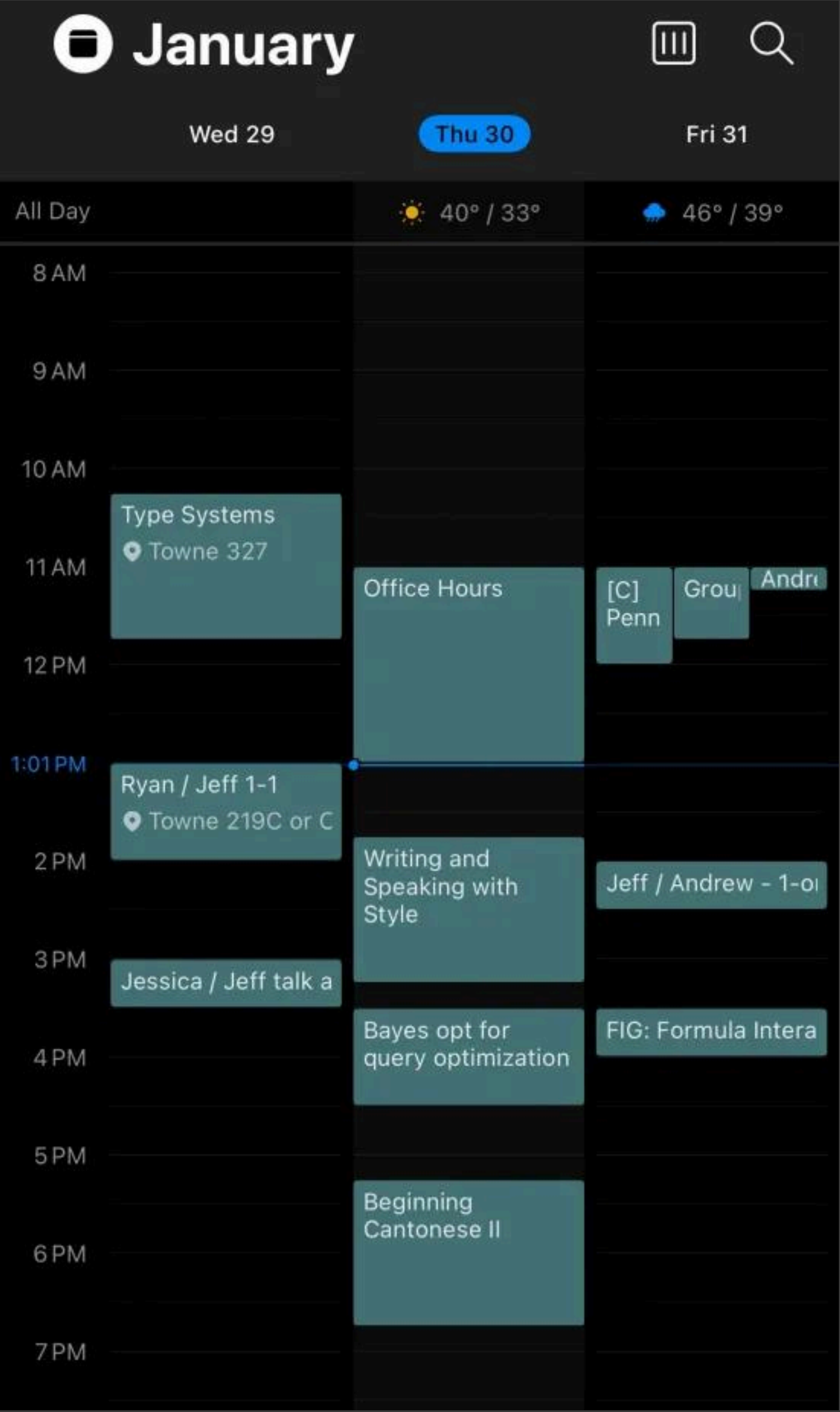
Visualizations Are Everywhere!

Calendar category →

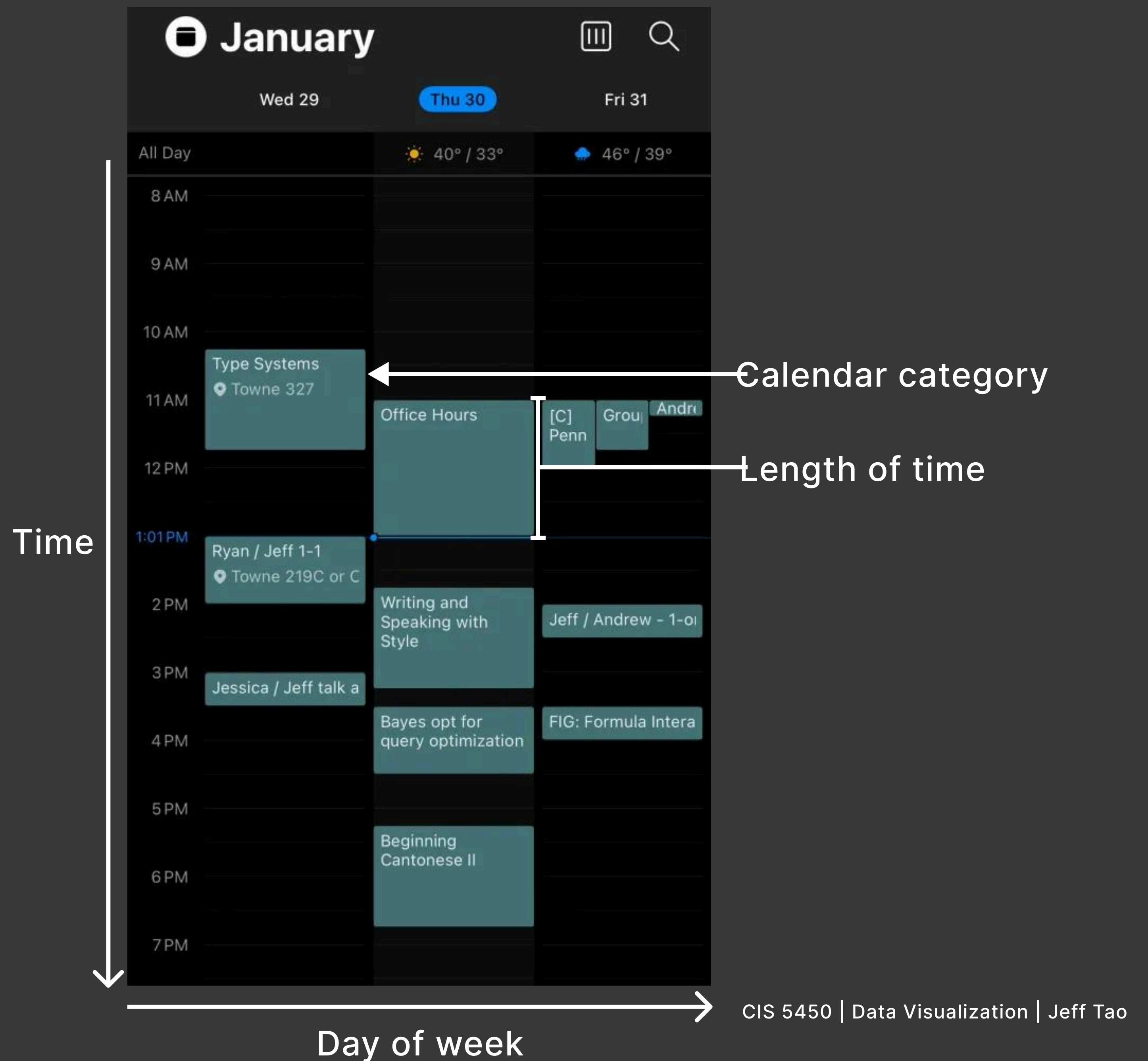


Time

Visualizations Are Everywhere!



Visualizations Are Everywhere!



Visualization leverages
your *perception* and
pattern recognition to
help explain data.



How to Visualize?



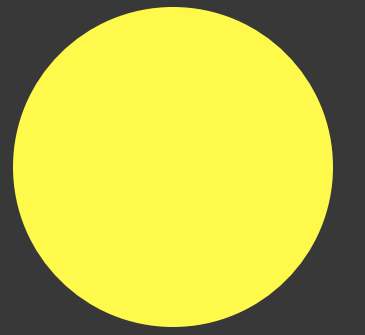
What Type of Data?



Your data has a *type*.

Each visualization pattern only works with certain types of data.

Nominal/Categorical



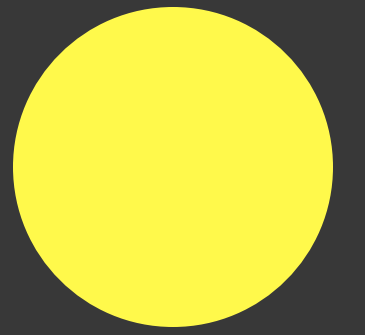
Can only compare equal/non-equal

- States, Countries
- Type of Student (undergrad, masters, phd)
- Basically anything that isn't written with numbers

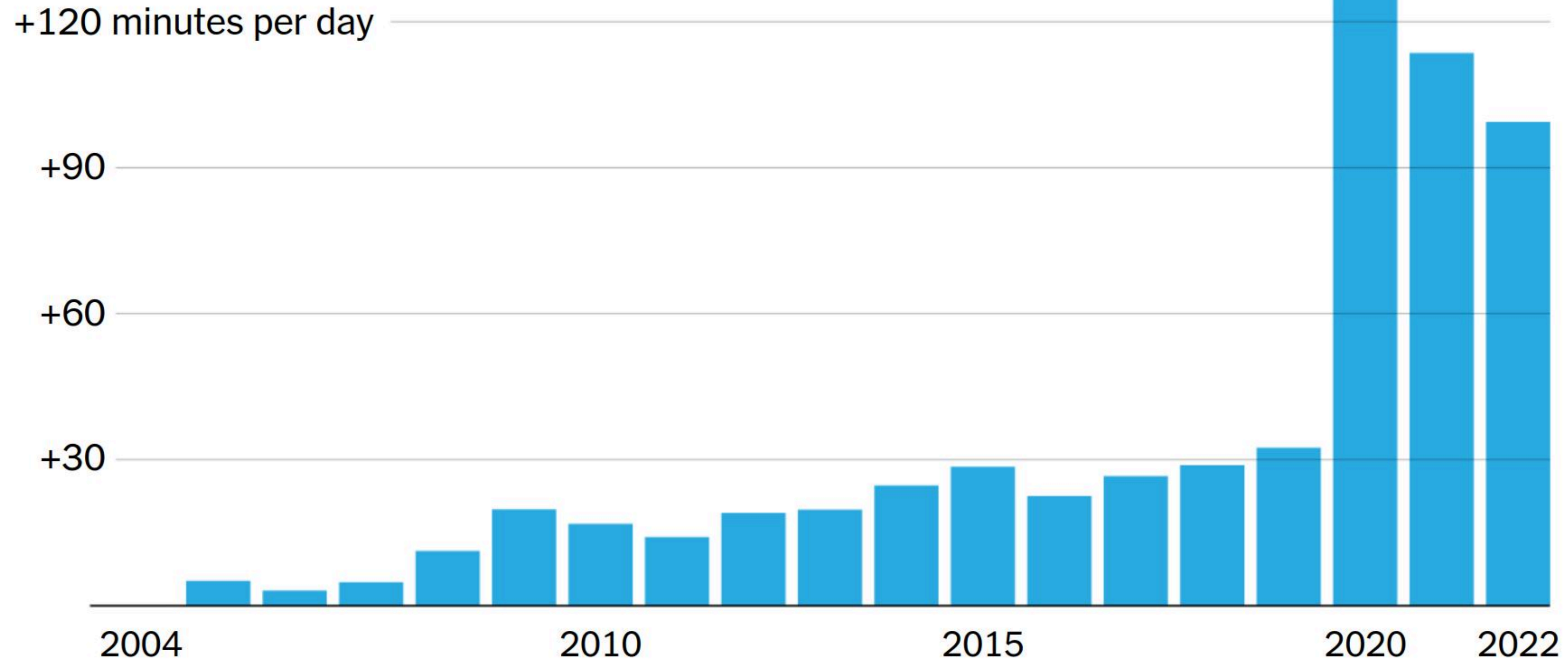
Quantitative

Inherently numeric, you can do math to it

- \$ of Revenue
- # of customers
- Intervals
- Ratios
- Temporal (time)



Change since 2003 in average time spent at home



Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

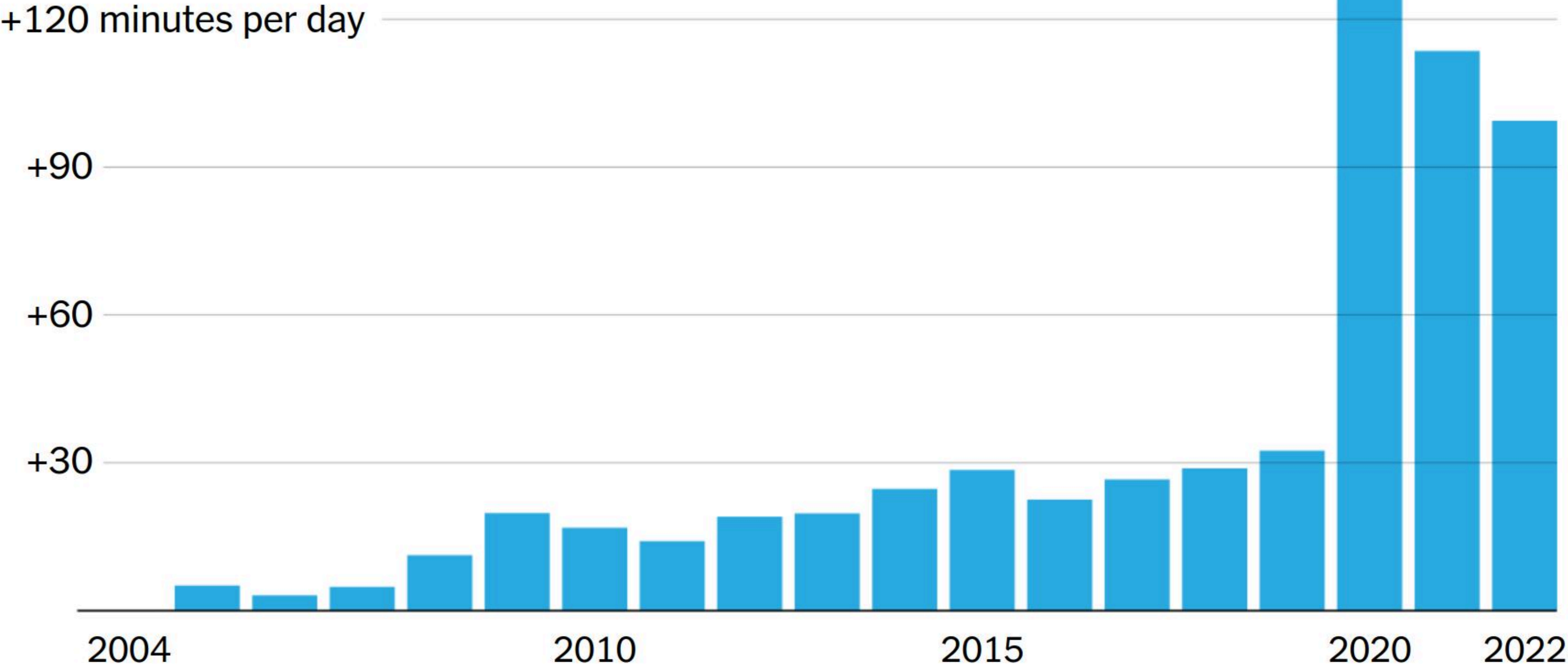
Anatomy of a Chart

Source: NYT

Title



Change since 2003 in average time spent at home



Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**

+120 minutes per day

+90

Gridlines →

+60

+30

2004

2010

2015

2020

2022

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**

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2004

2010

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2020

2022

← Tick marks

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

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Tick marks →

→ Caption

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**

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+90

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+60

+30

← Bars

2004

2010

2015

2020

2022

← Tick marks

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

← Caption

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**

+120 minutes per day

+90

Gridlines →

+60

+30

Bars
Marks?

2004

2010

2015

2020

2022

Tick marks →

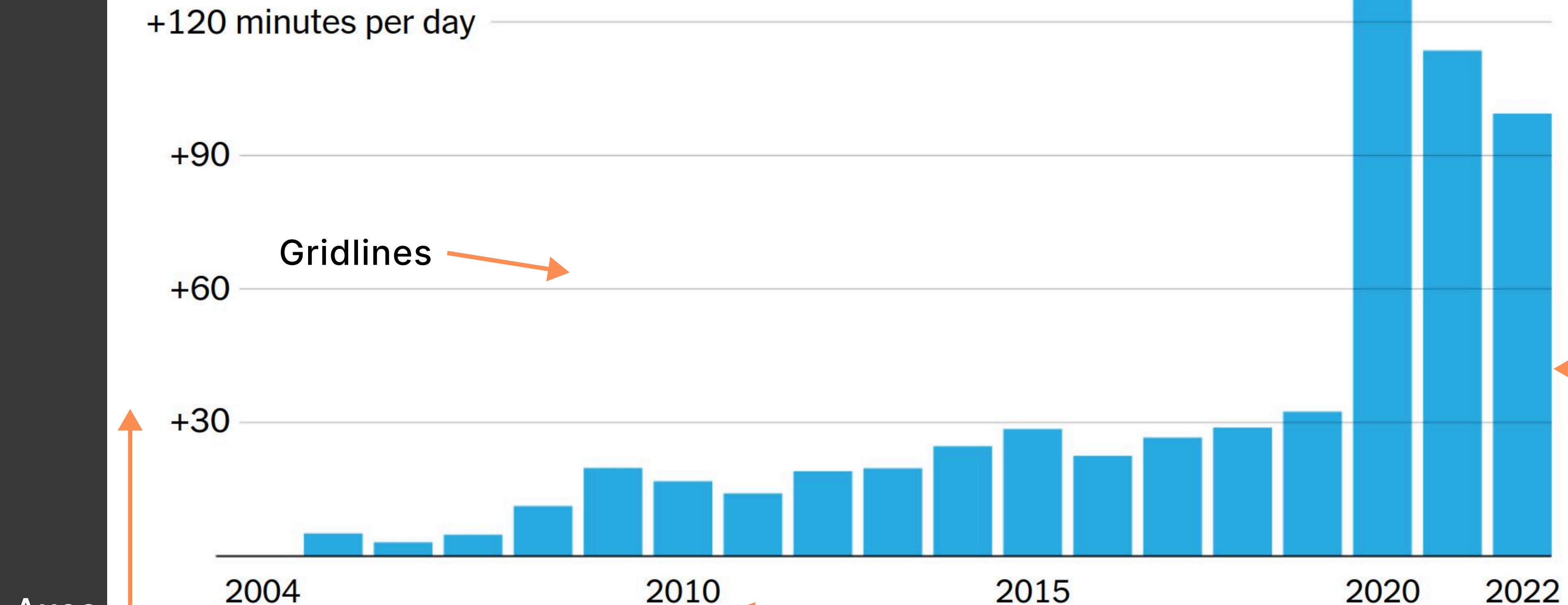
Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

→ Caption

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**



Axes →

→ Tick marks

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

→ Caption

Anatomy of a Chart

Source: NYT

Title → **Change since 2003 in average time spent at home**

+120 minutes per day

Gridlines →

+60

+30

Bars
Marks?

Axes
Encodings?

2004

2010

2015

2020

2022

Tick marks

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

Caption

Anatomy of a Chart

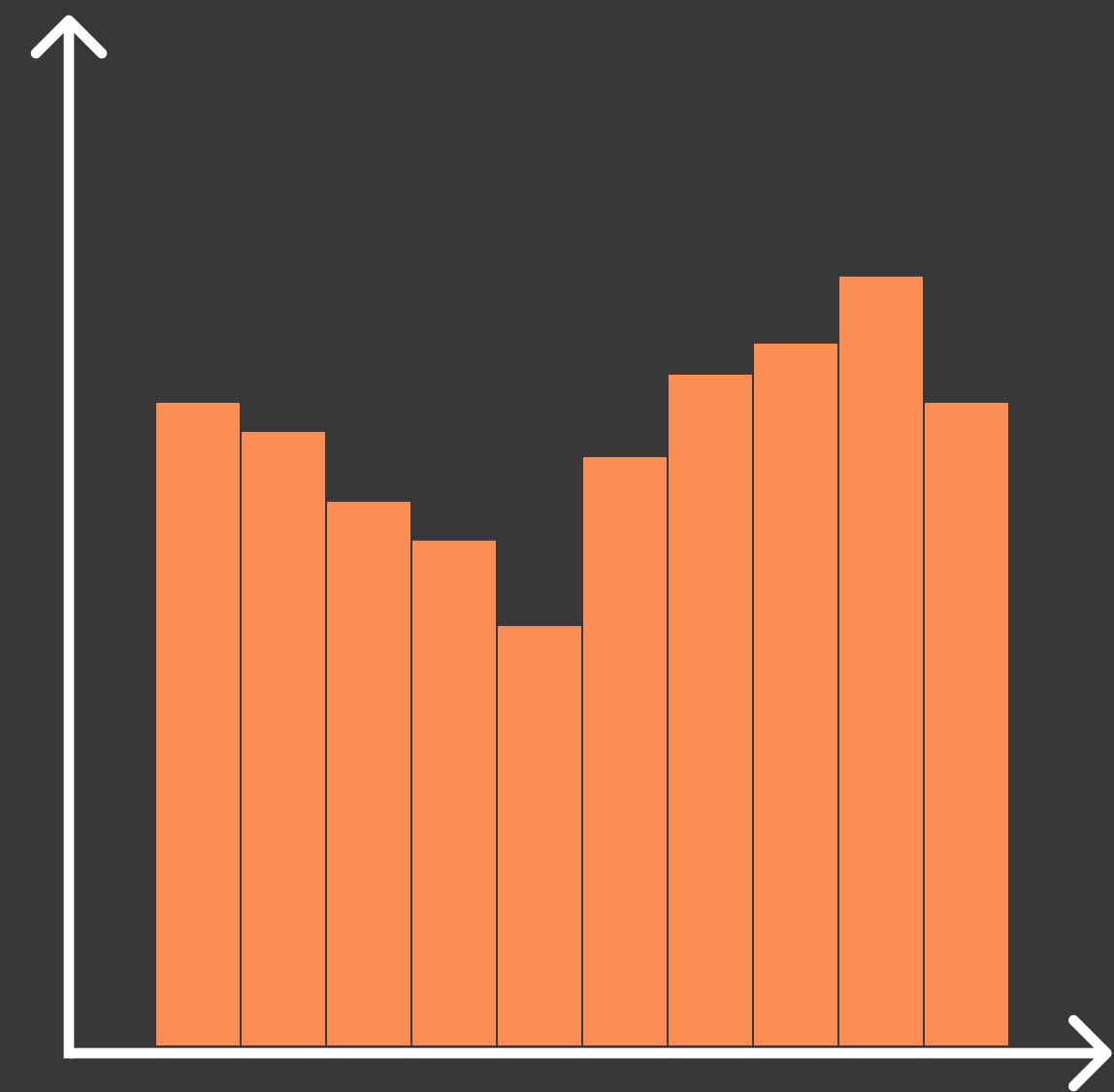
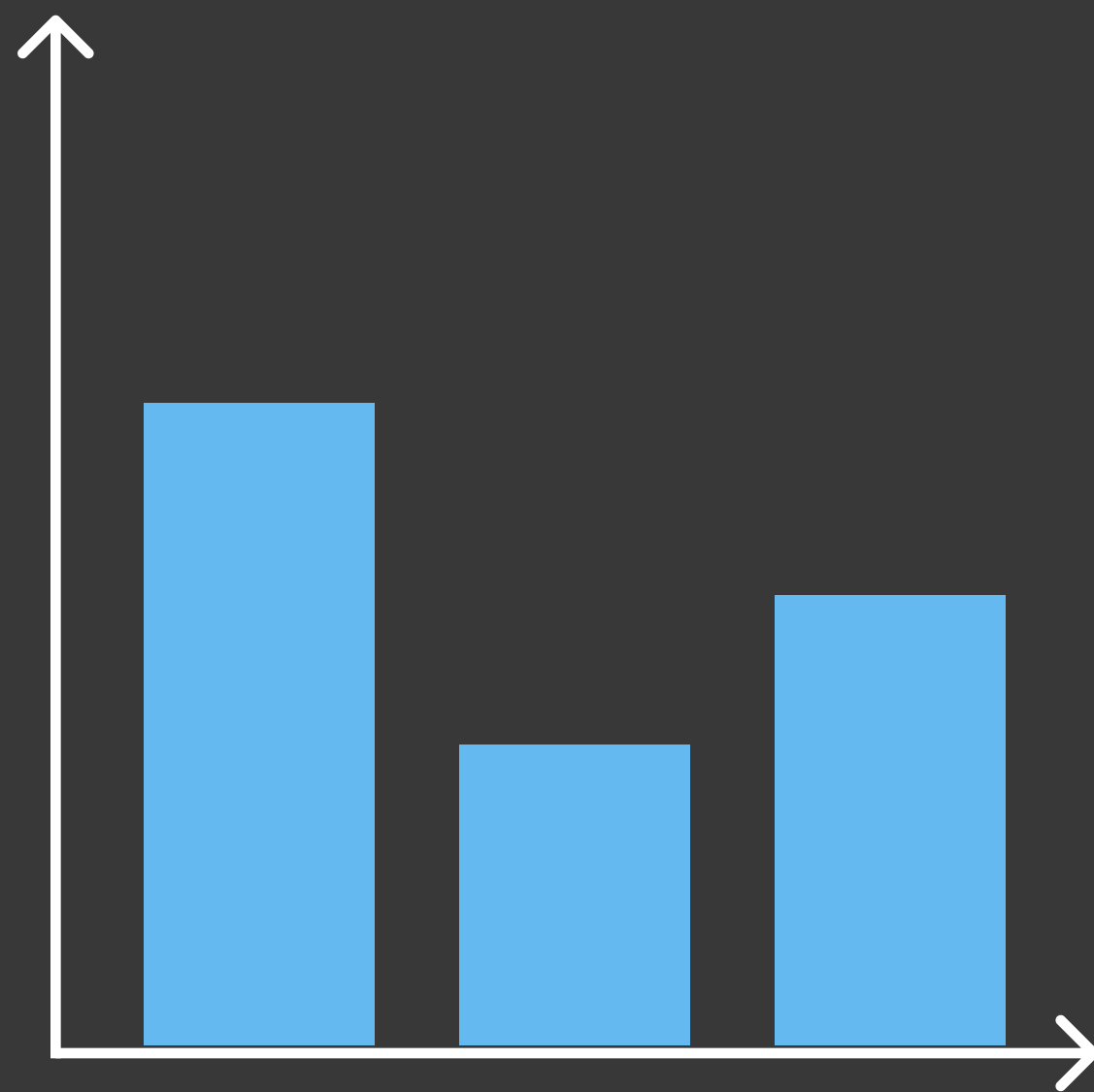
Source: NYT

Marks

Graphical shapes that represents some part of the data



Bars: Compare Position, Length

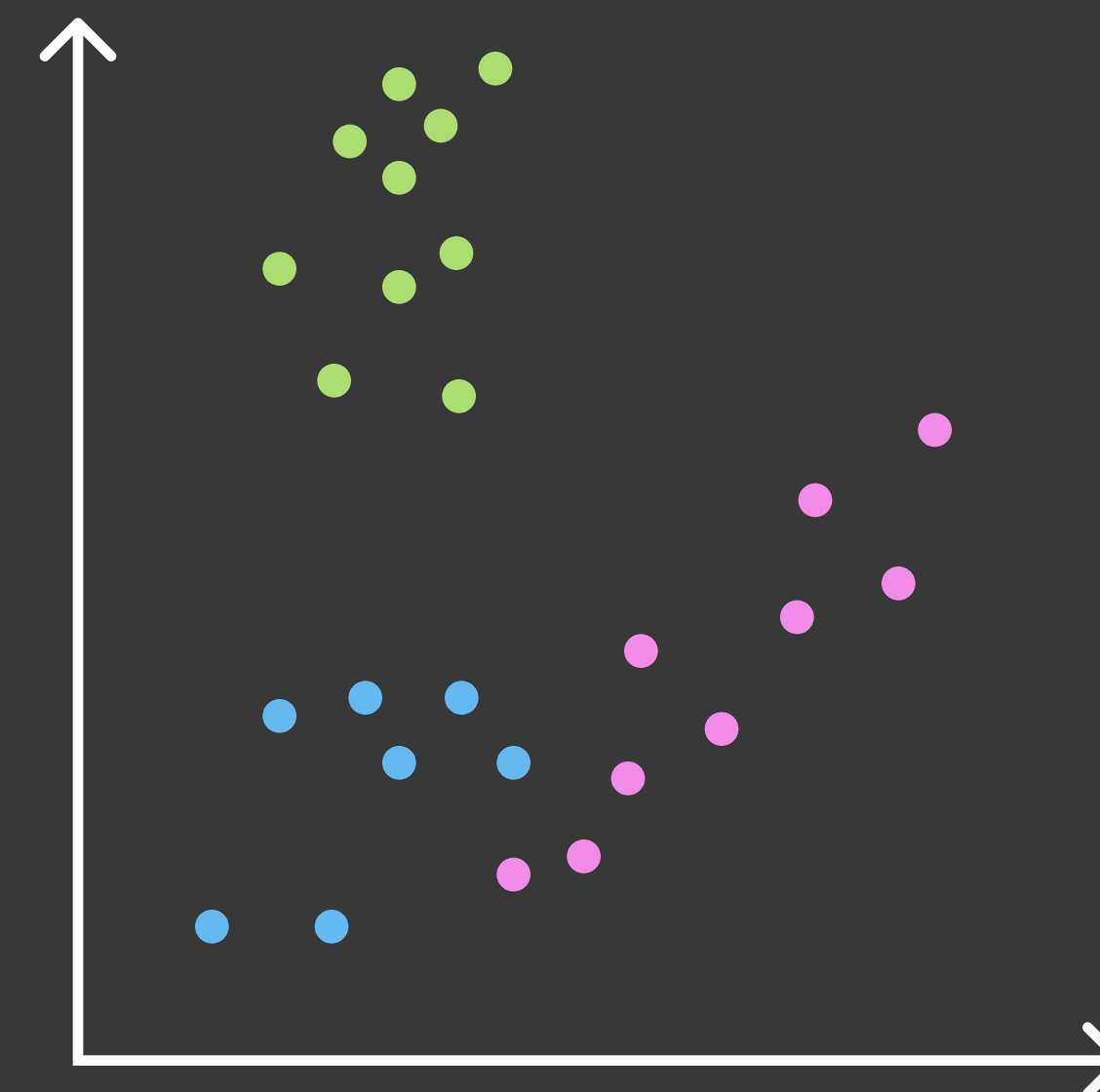
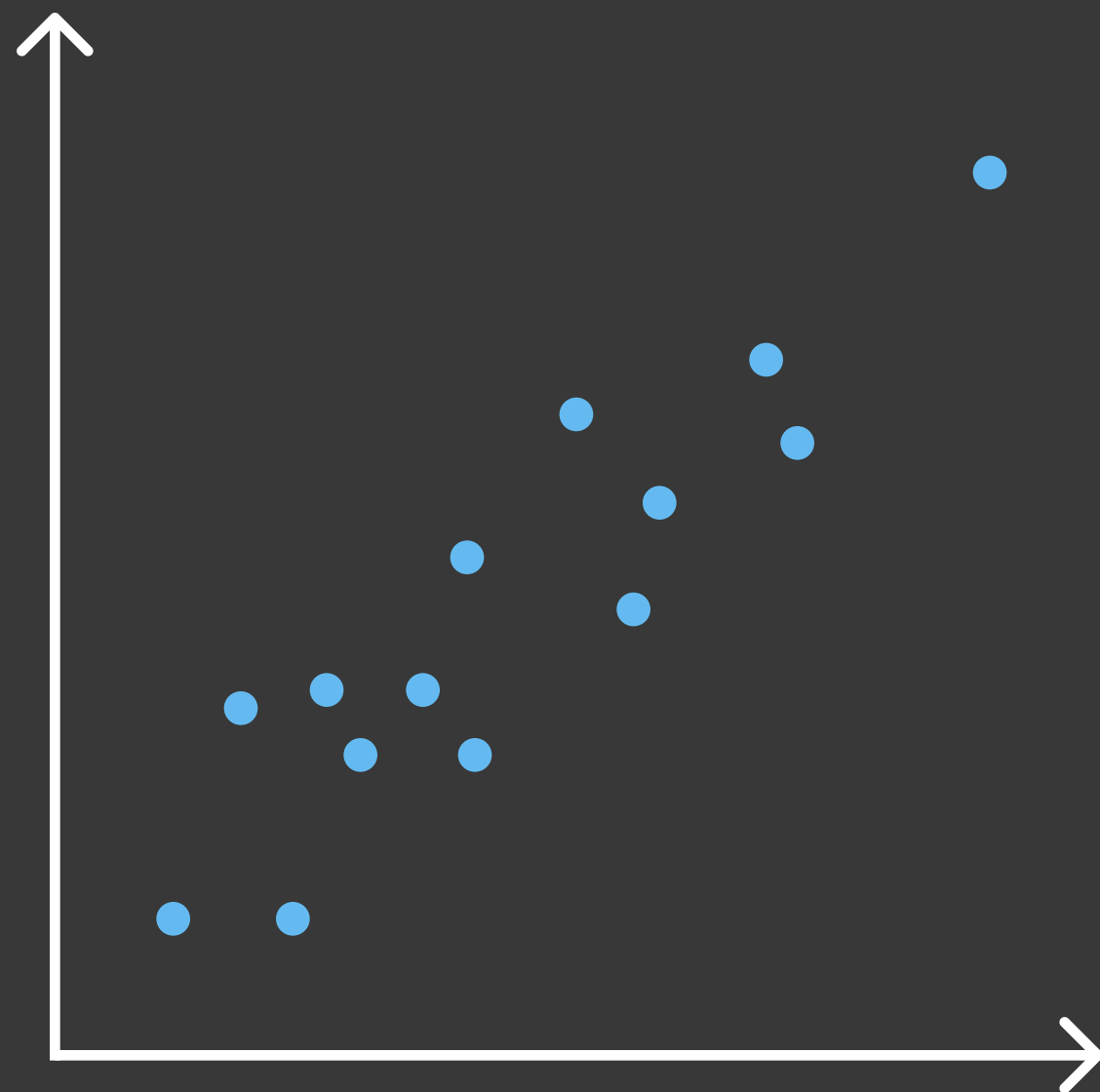


Marks

Graphical shapes that represents some part of the data



Points: Compare Position

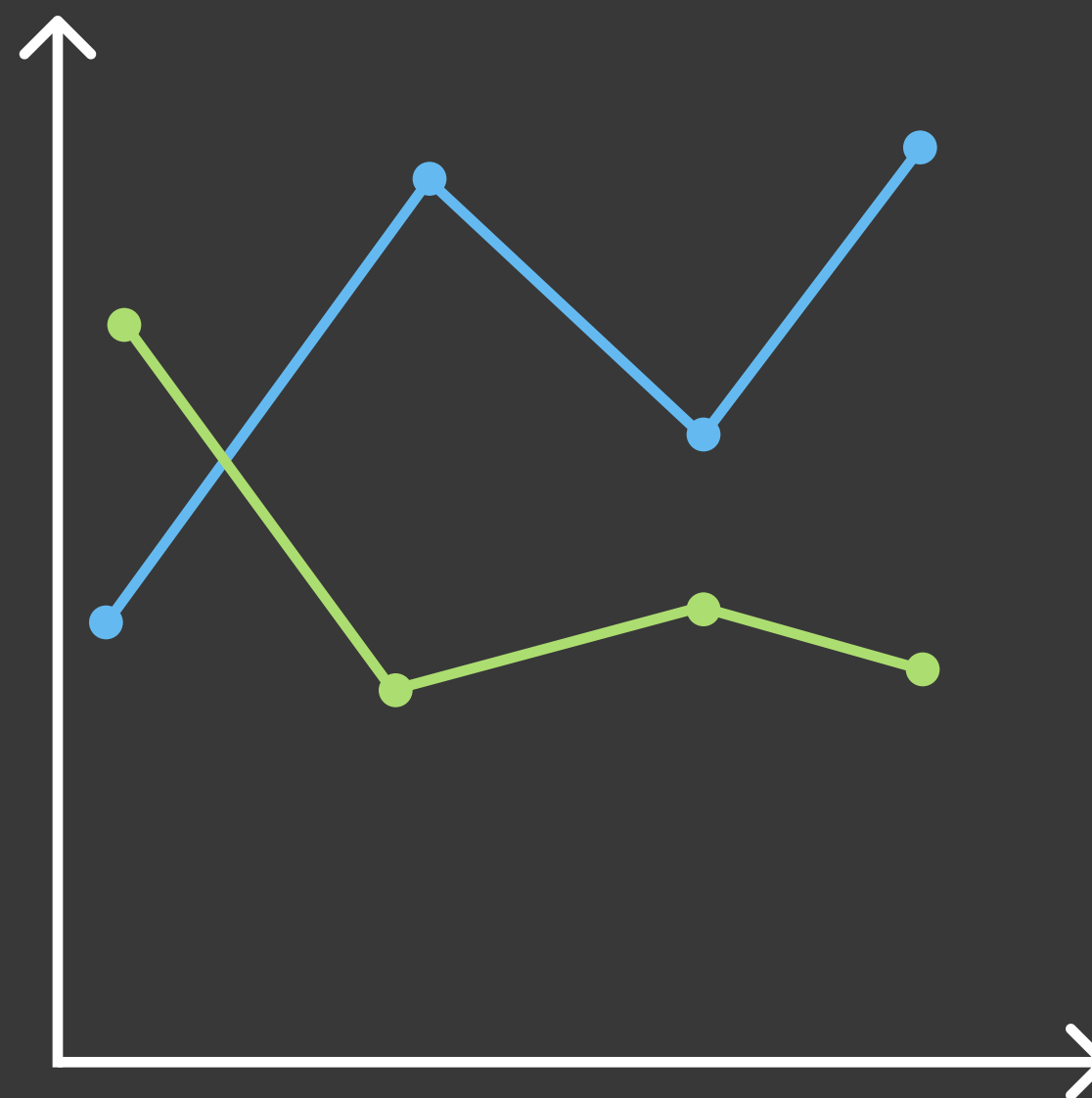


Marks

Graphical shapes that represents some part of the data



Lines: Compare Position, Length

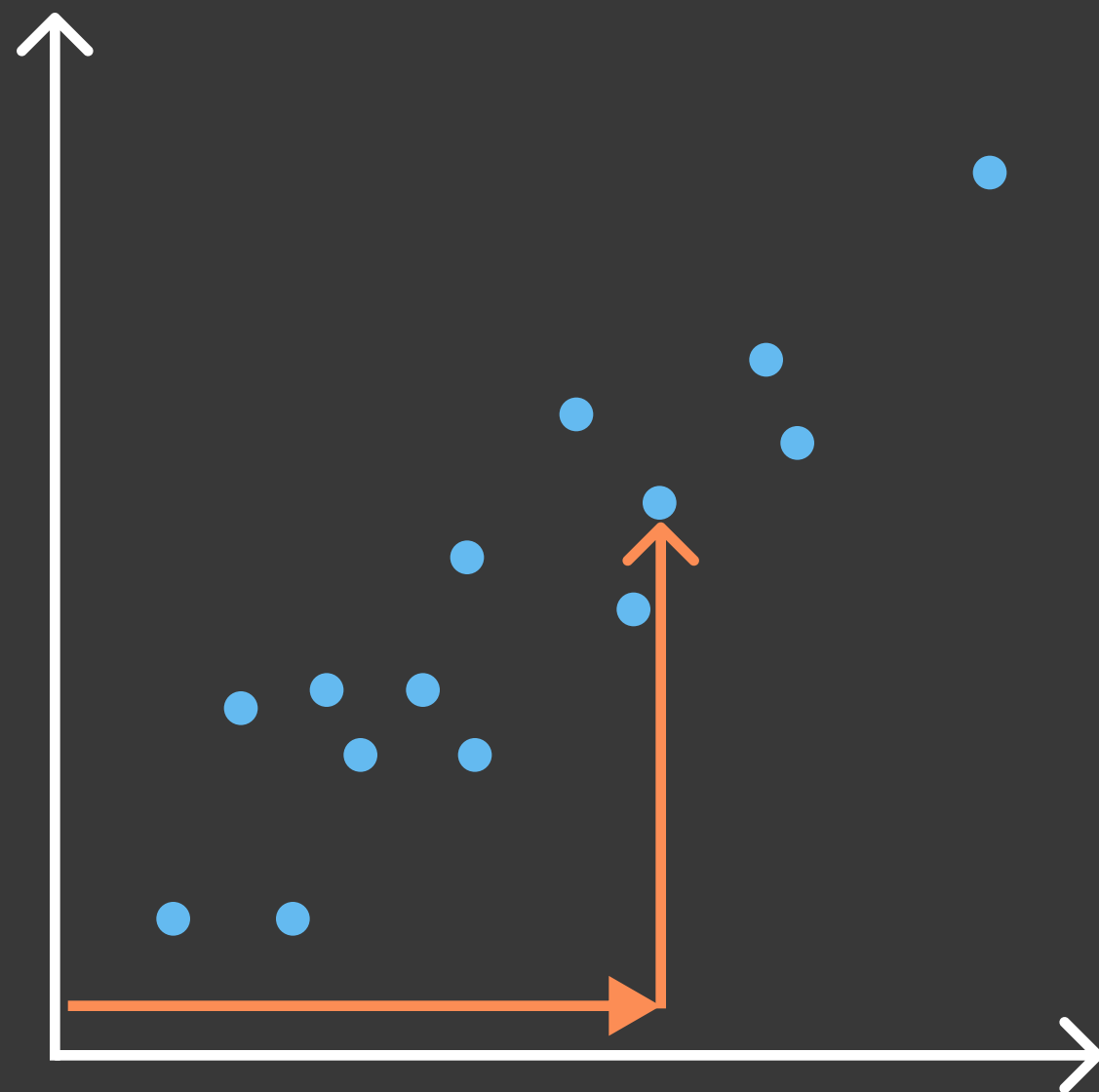


Encodings

How data maps to perceptual “channels”



Position: Quantitative, Categorical

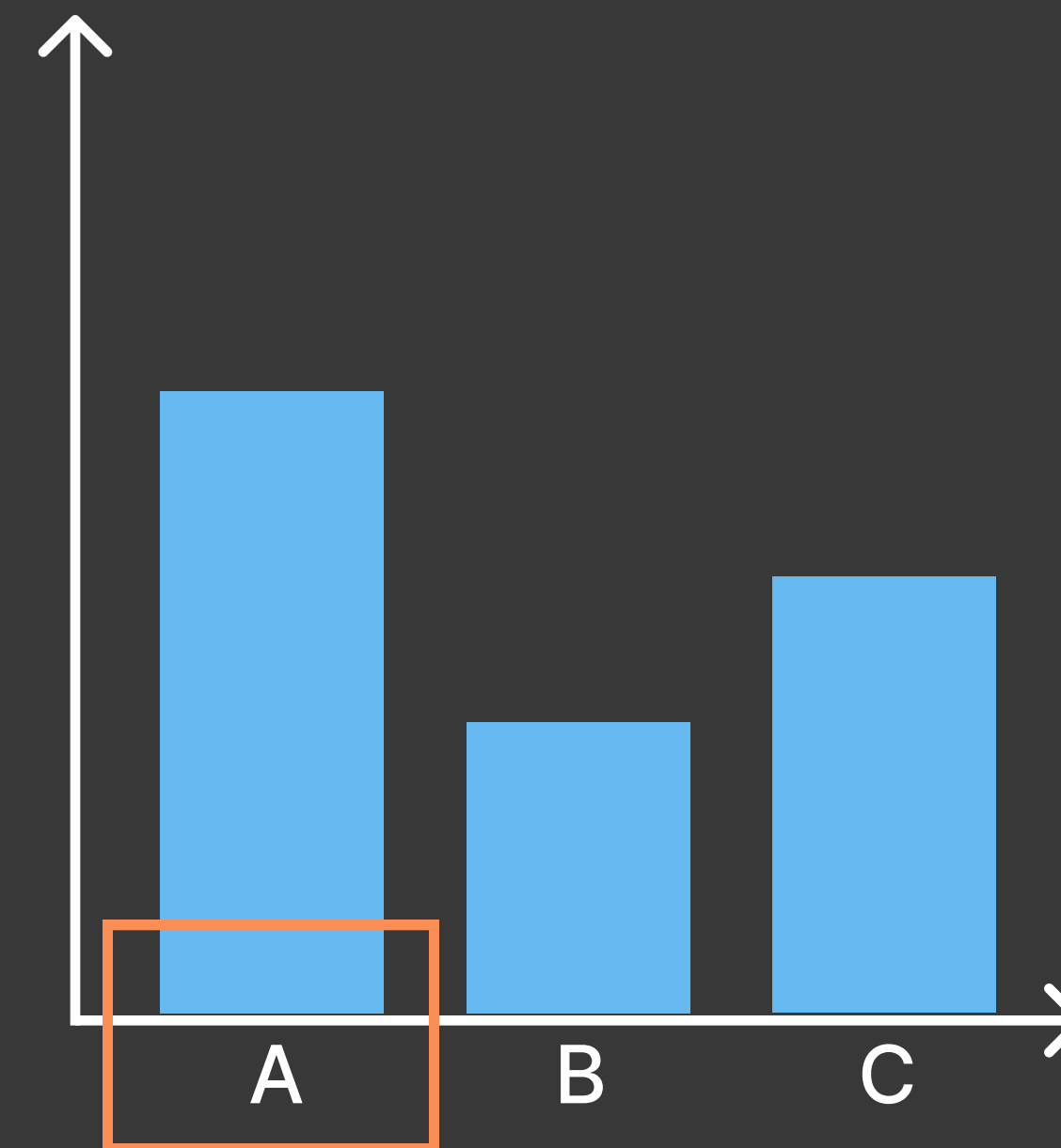
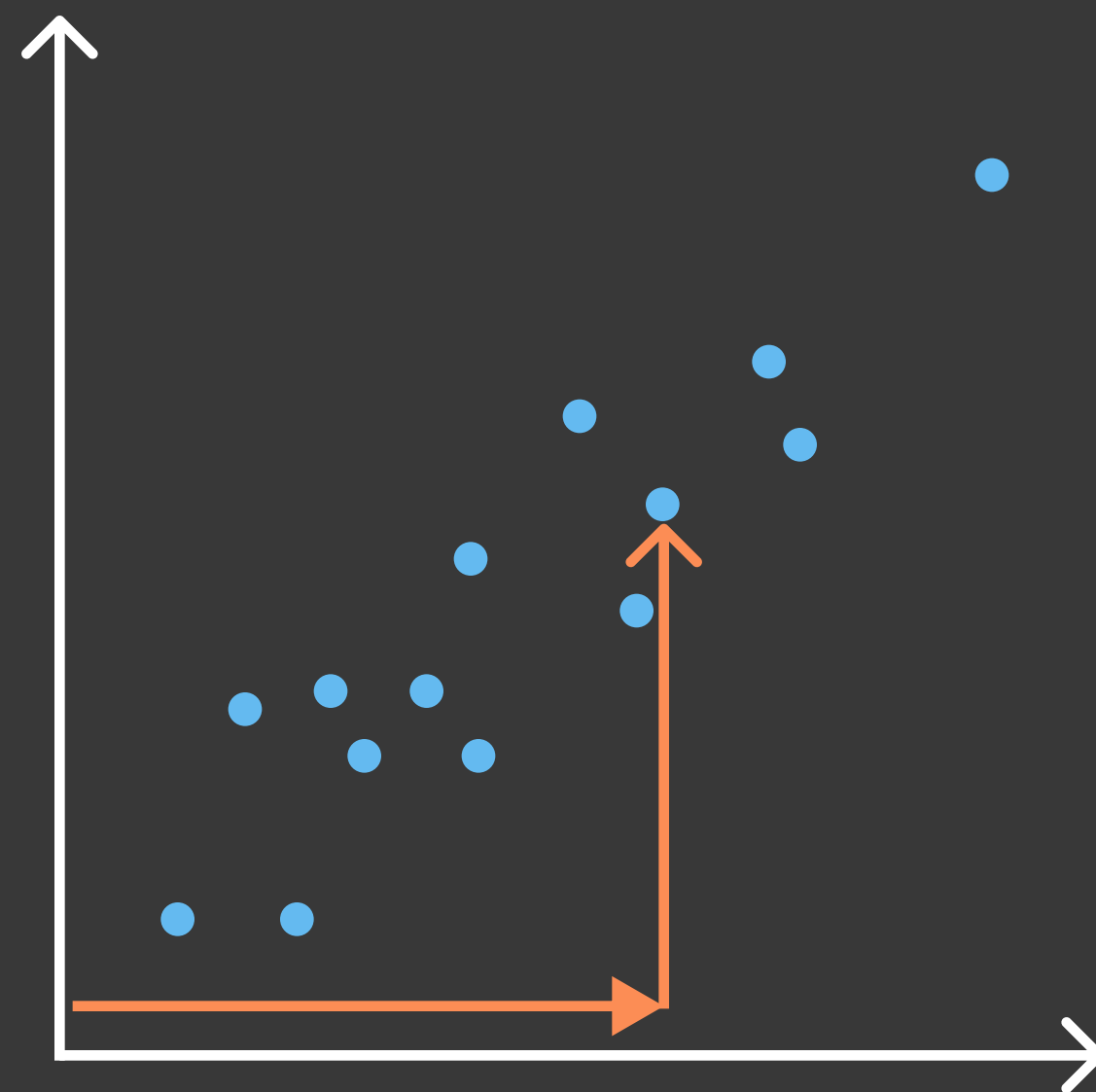


Encodings

How data maps to perceptual “channels”



Position: Quantitative, Categorical

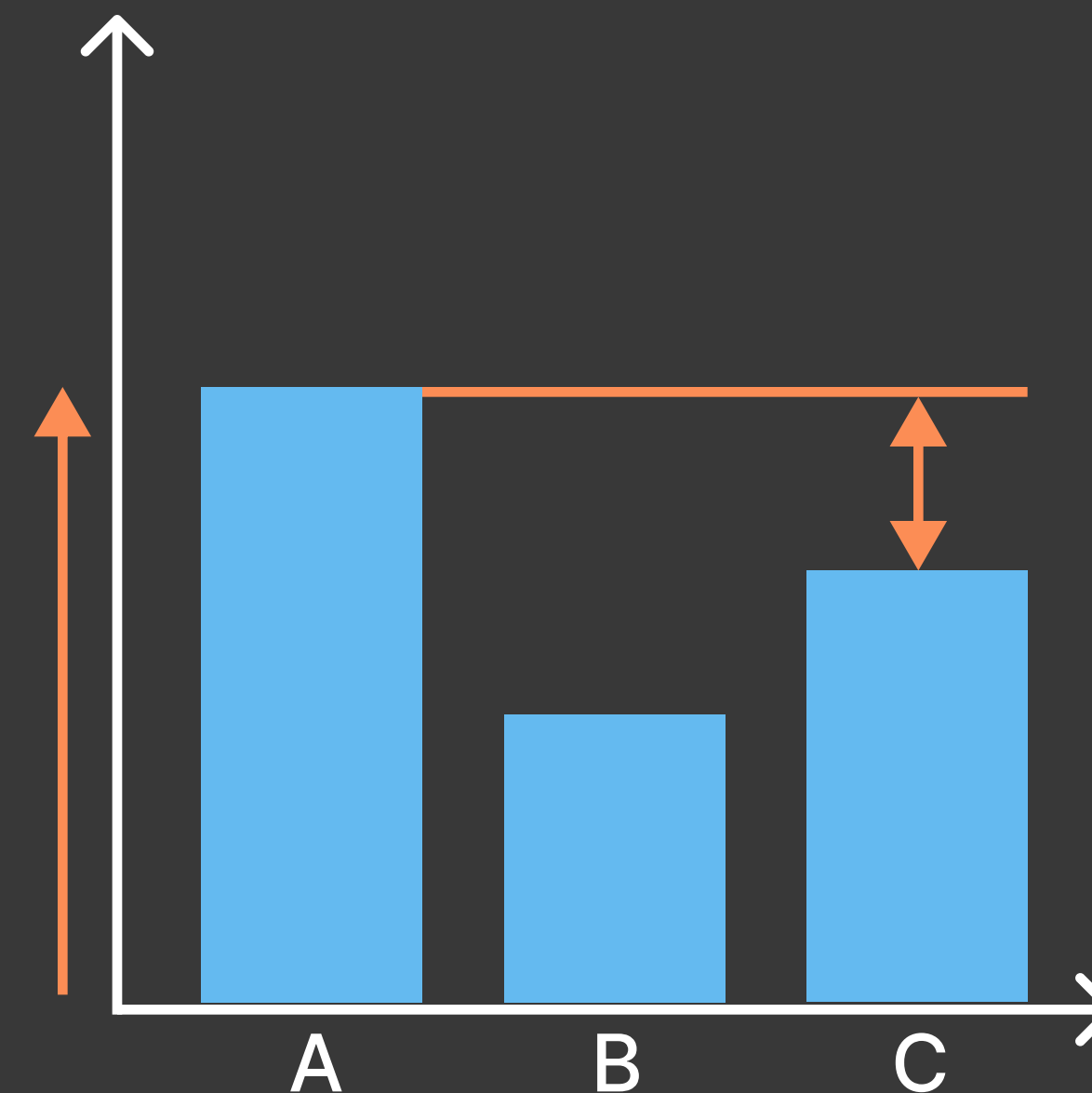


Encodings

How data maps to perceptual “channels”



Length: Quantitative

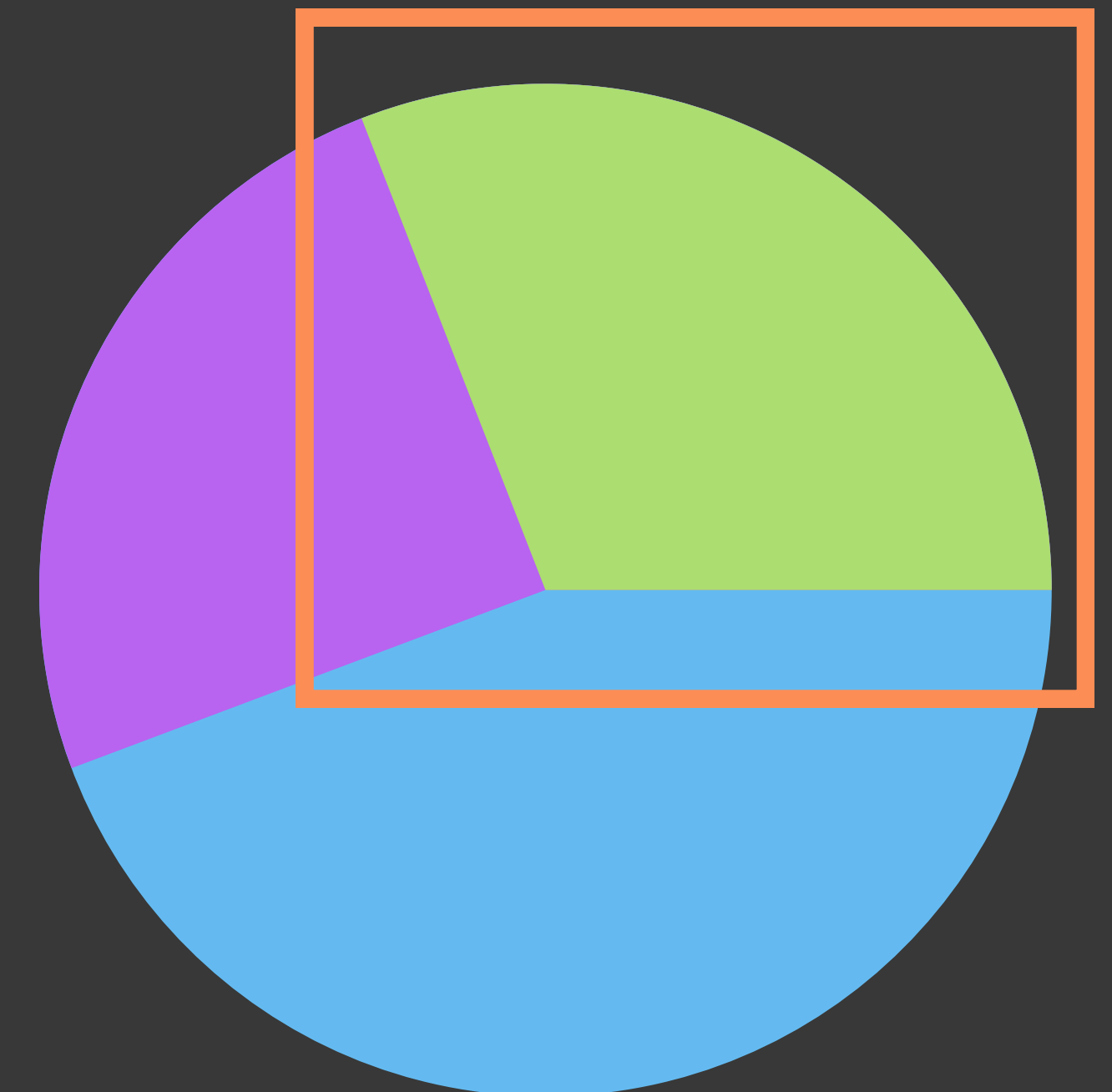
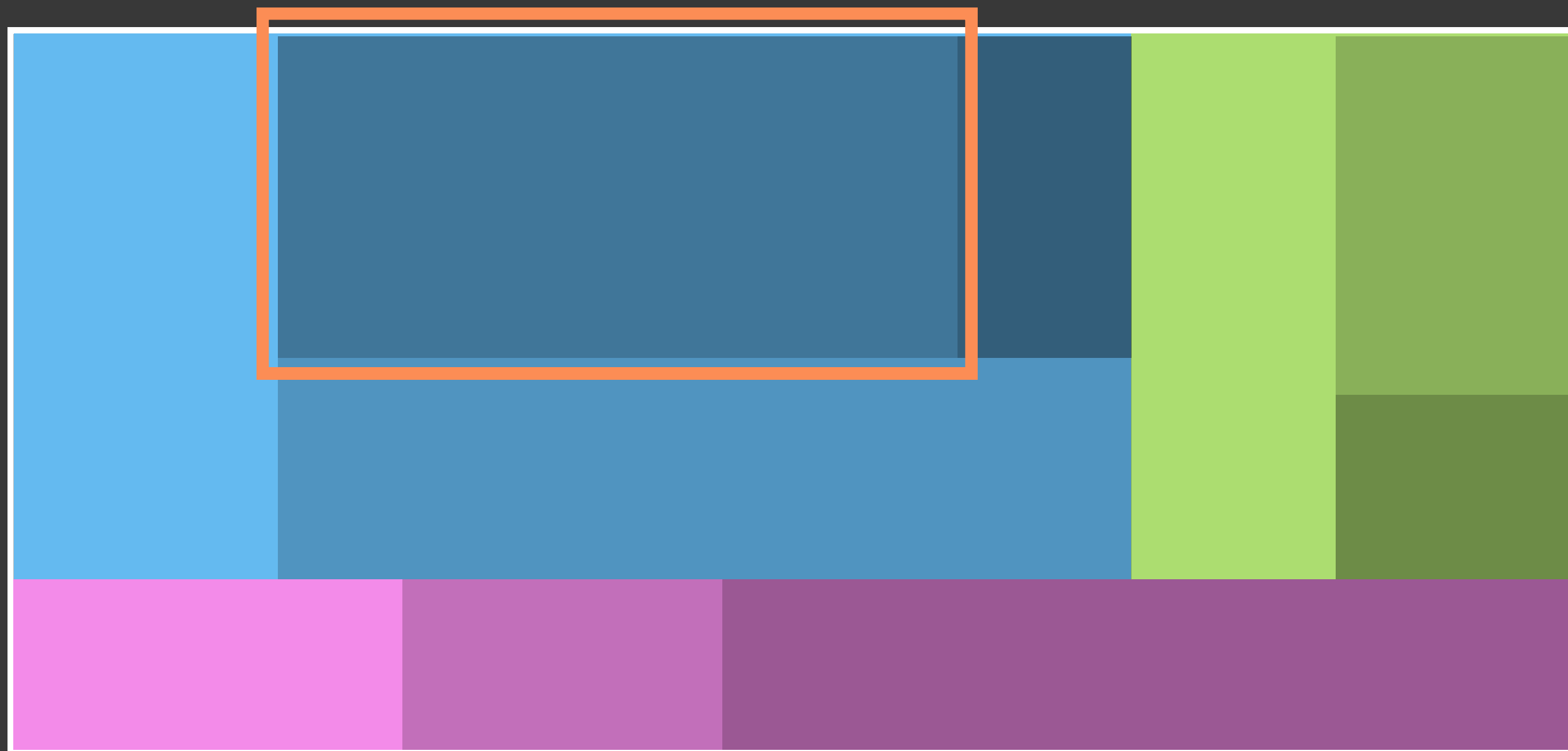


Encodings

How data maps to perceptual “channels”



Area: Quantitative

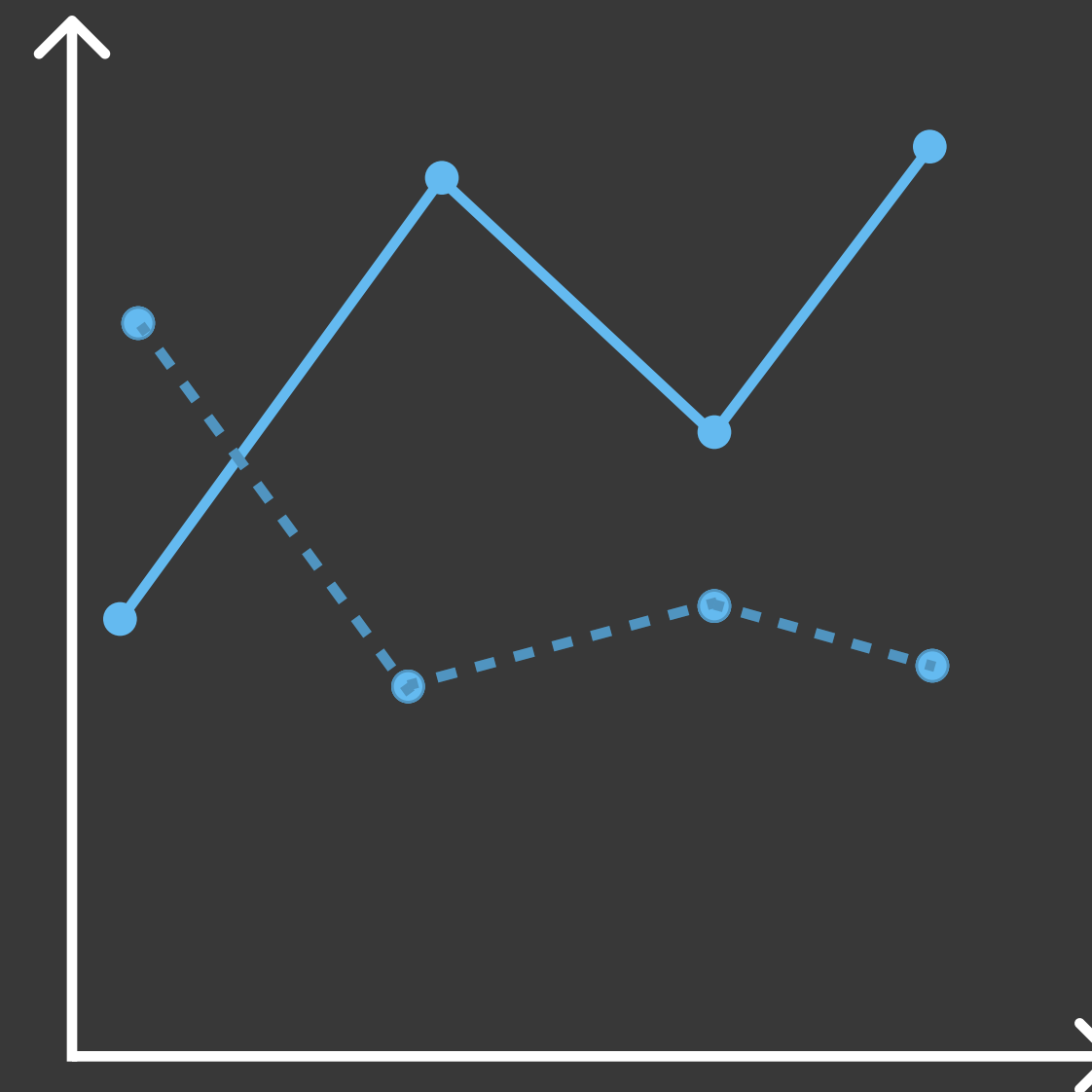
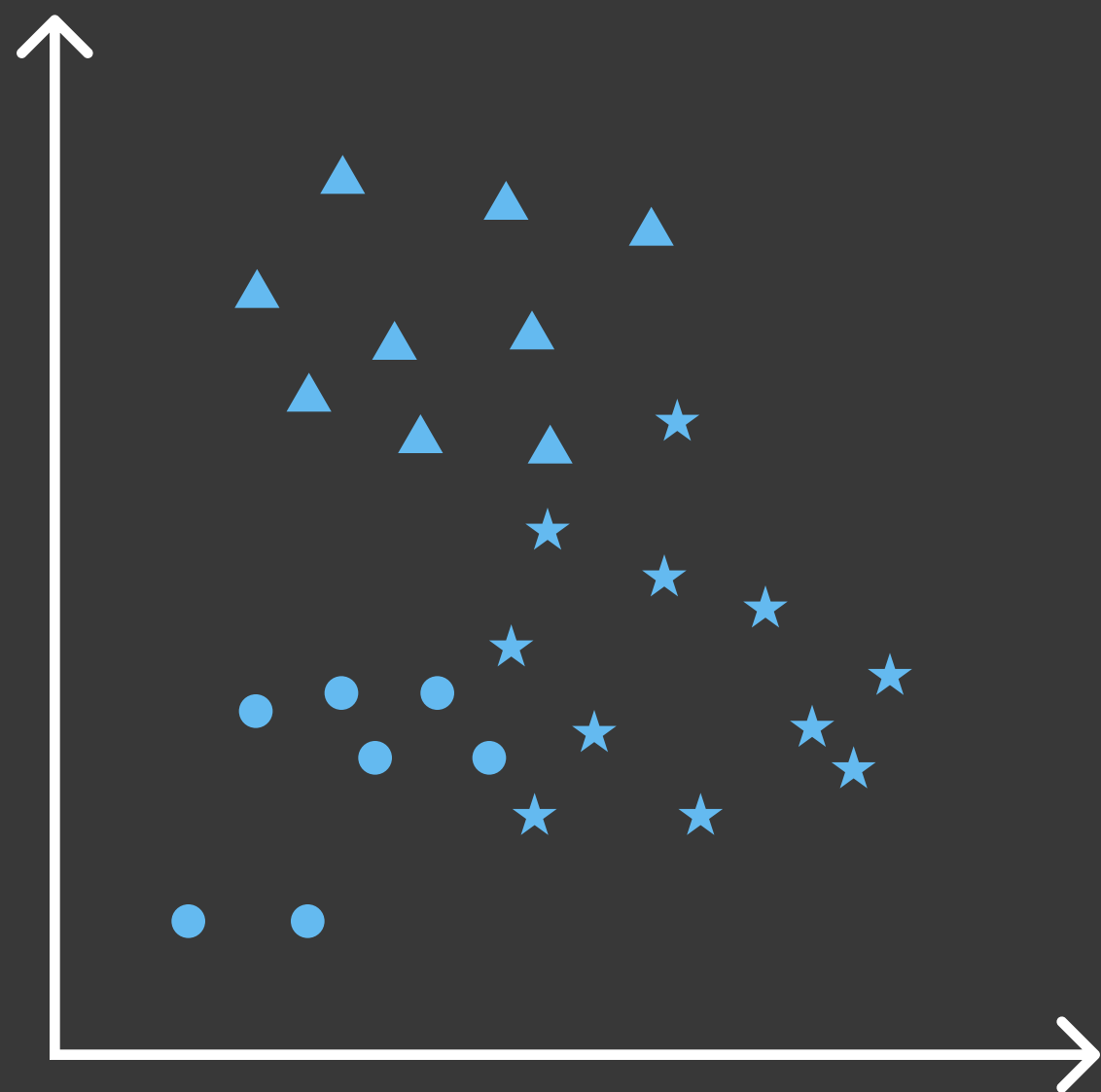


Encodings

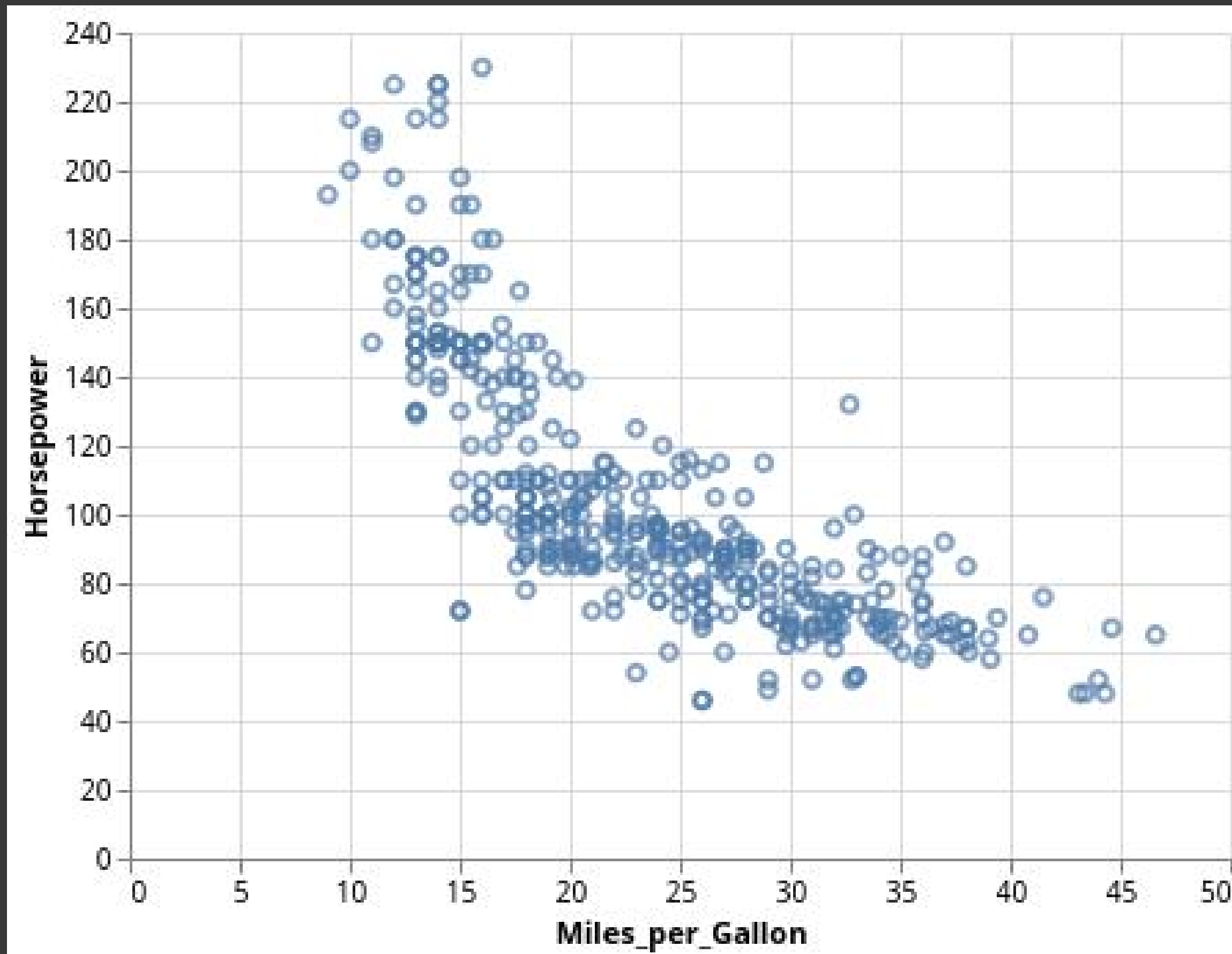
How data maps to perceptual “channels”



Shape, Style: Categorical

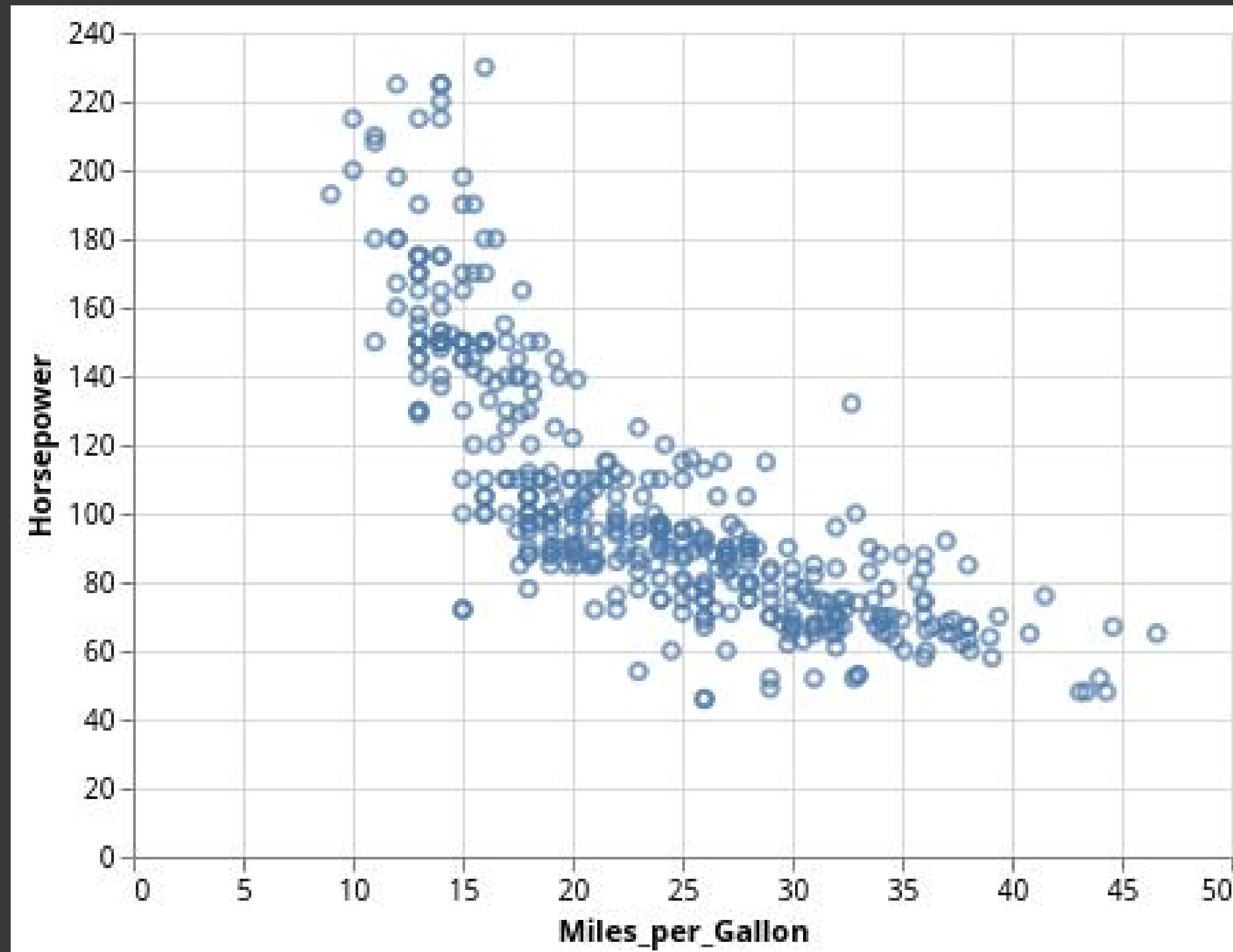
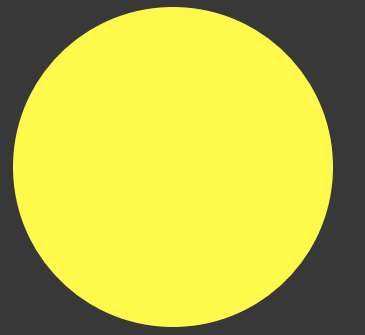


altair Code



```
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  .encode(
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    y='Horsepower'
  )
```

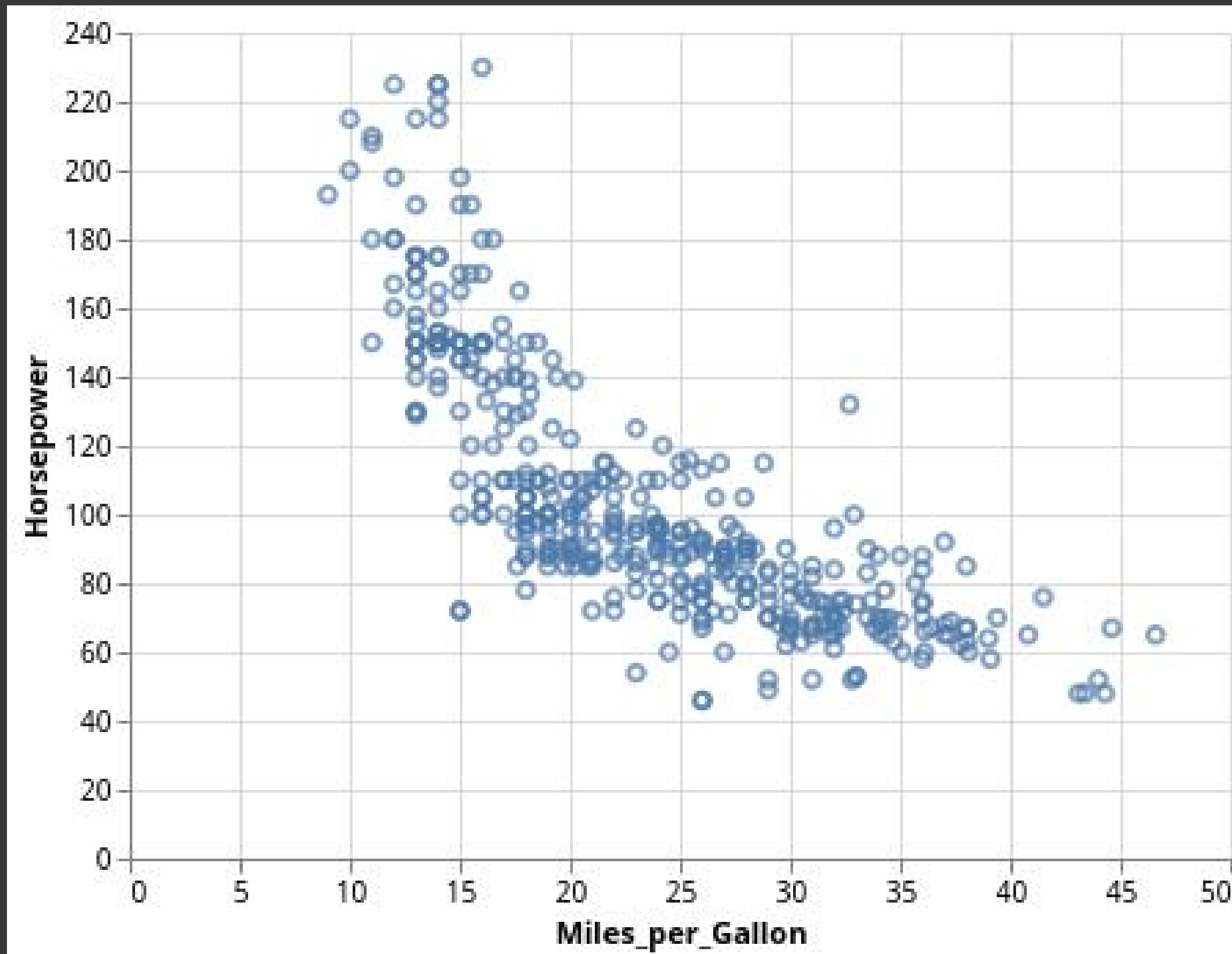
altair Code



dataframe

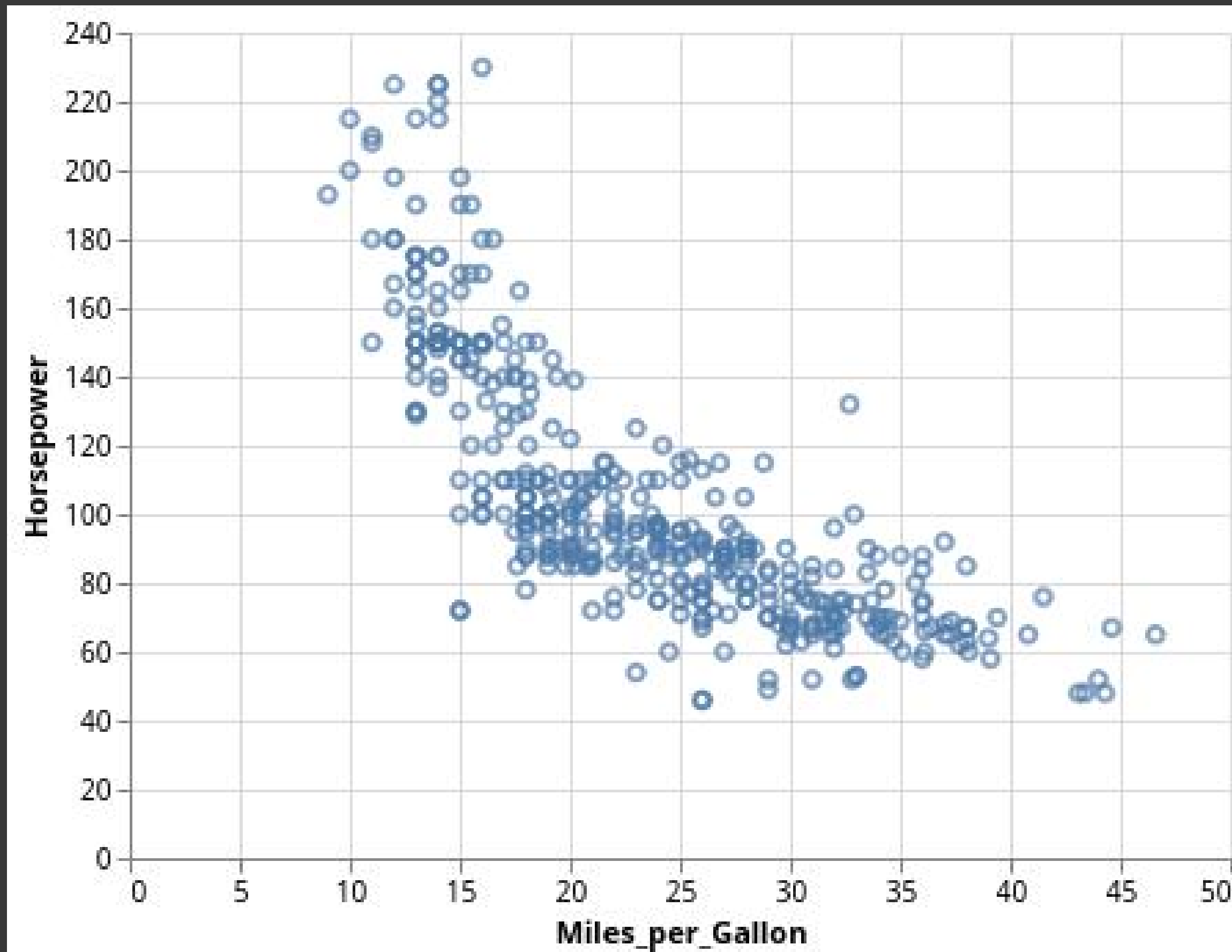
```
alt.Chart(cars)  
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    x='Miles_per_Gallon',  
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altair Code



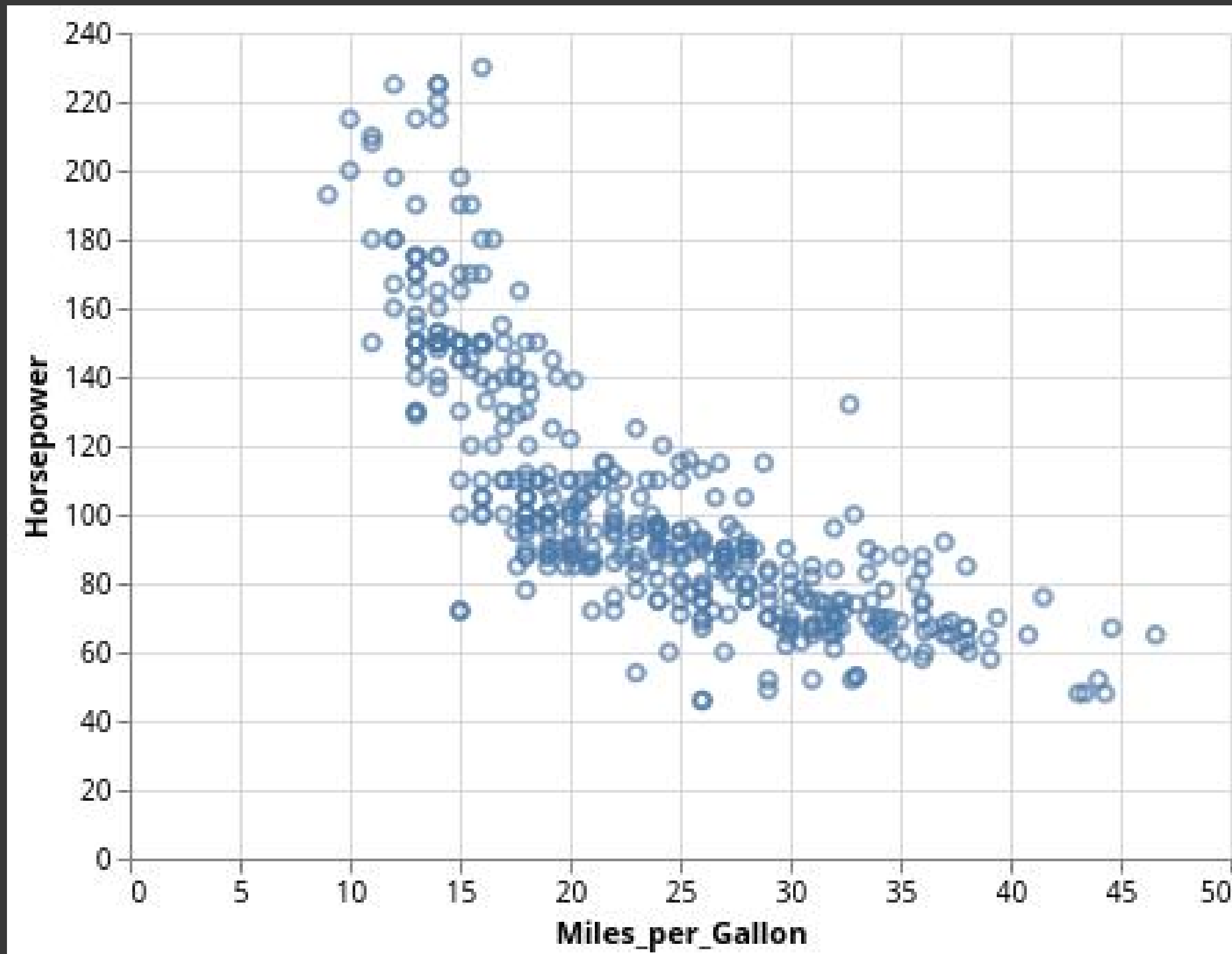
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altair Code



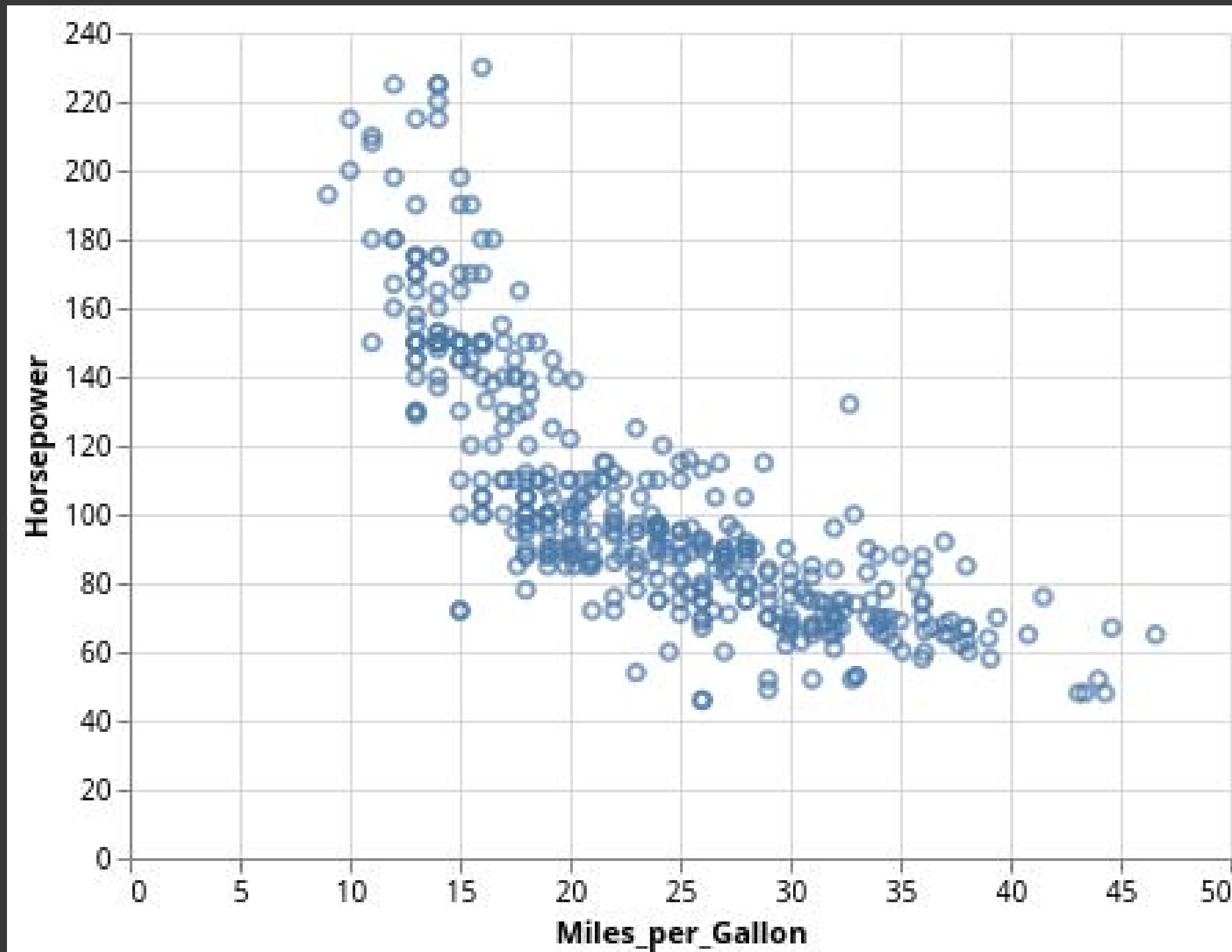
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altair Code



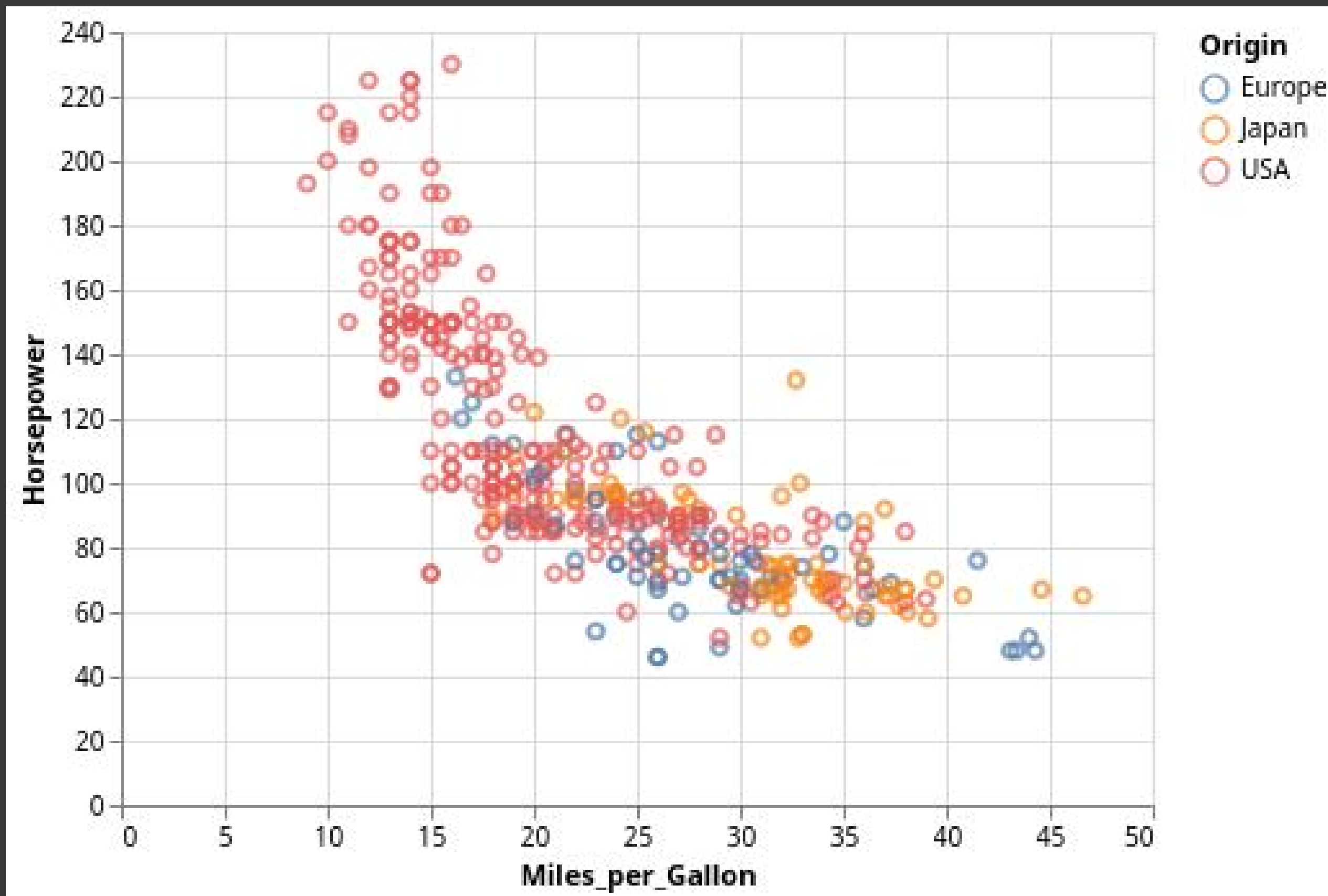
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altair Code



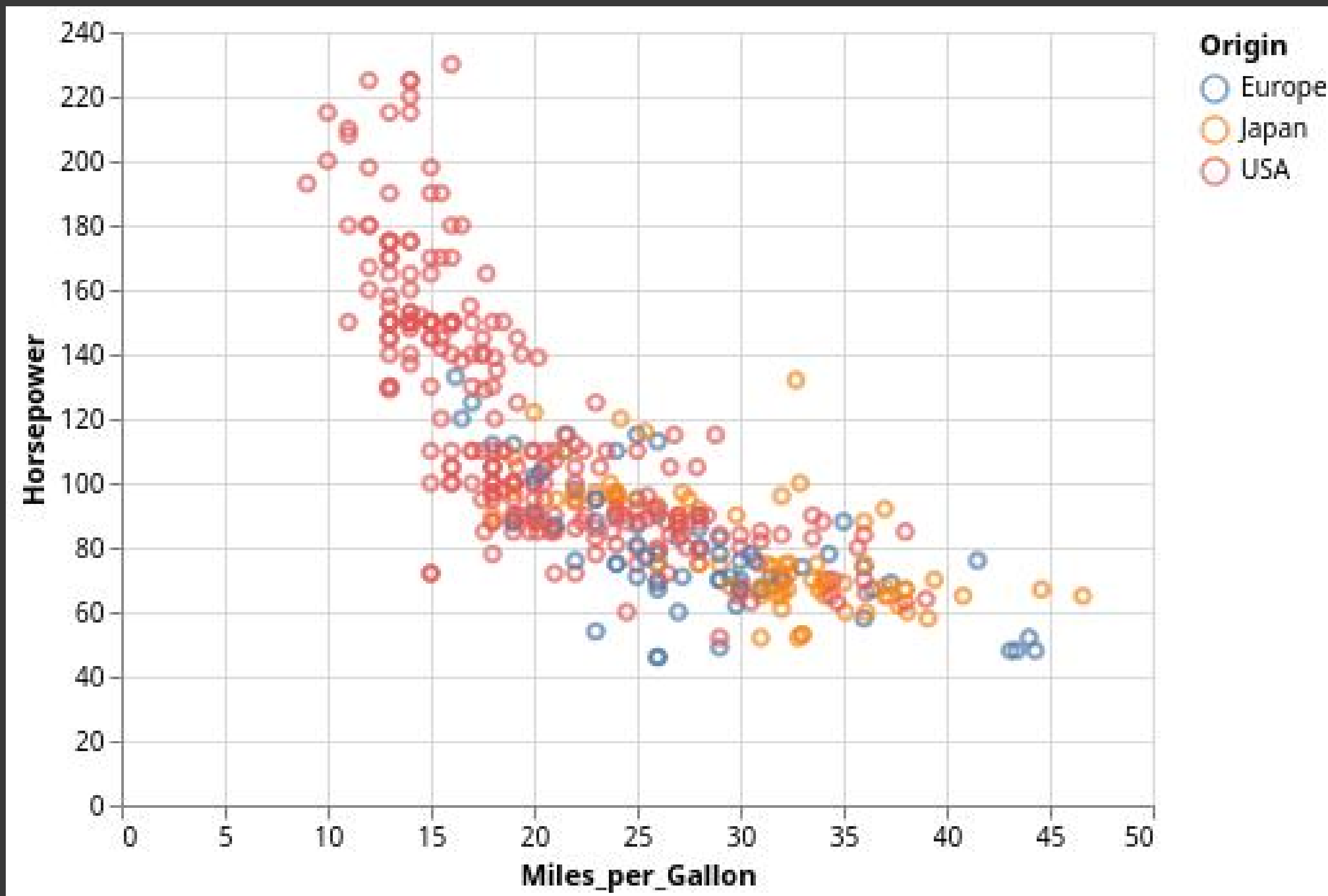
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altair Code



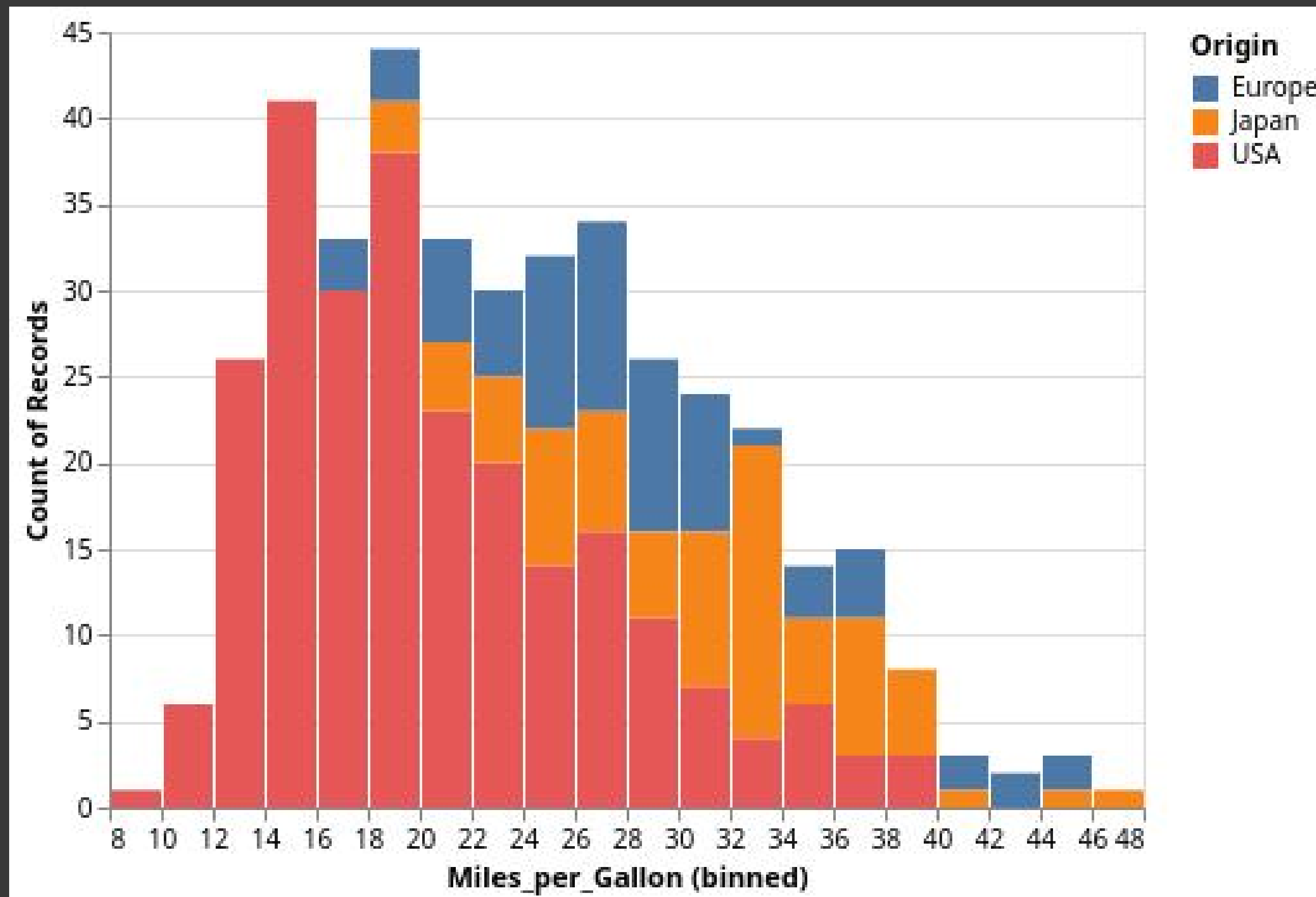
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    y='Horsepower'
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altair Code



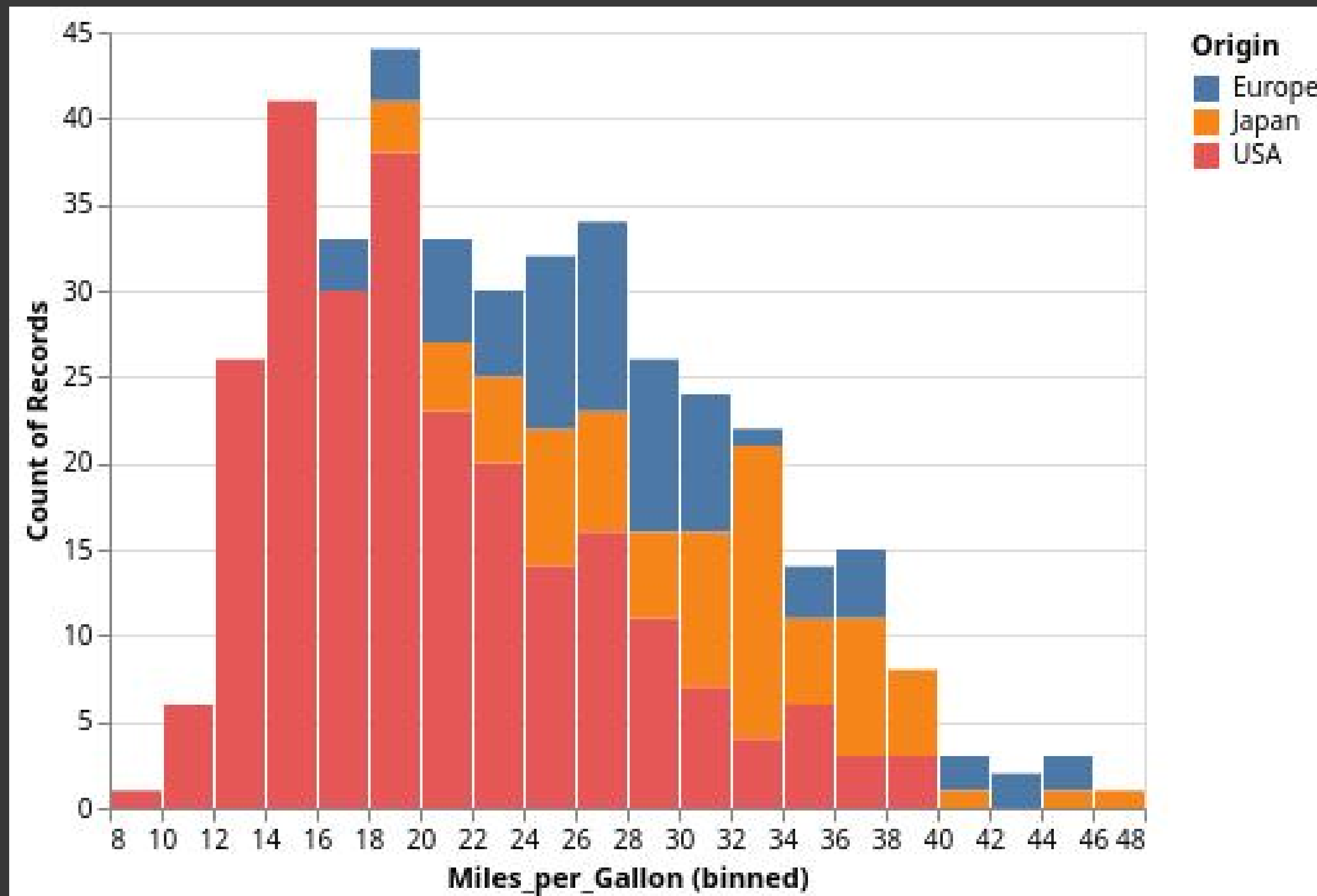
```
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    color='Origin'
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```

altair Code



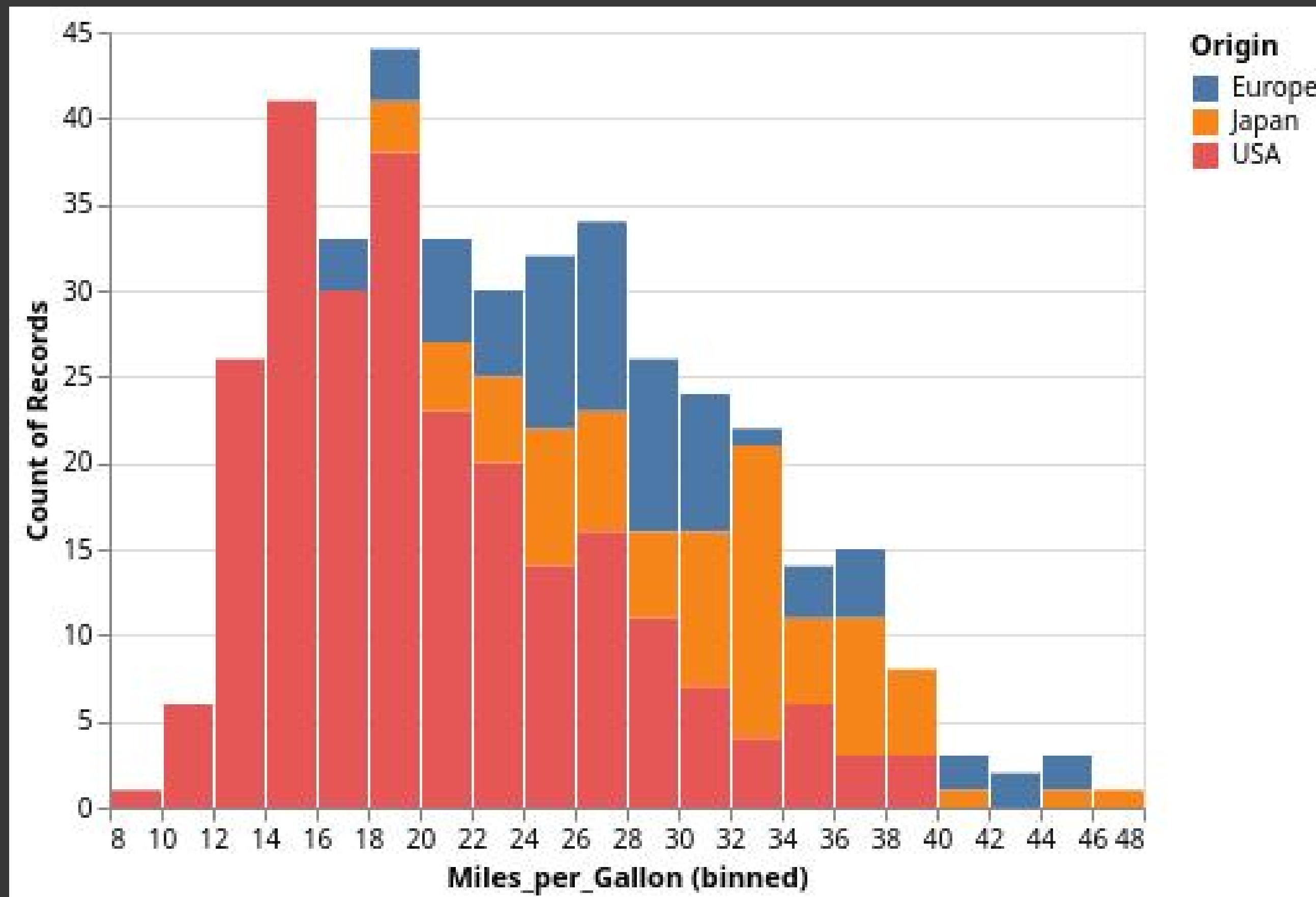
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    color='Origin')
```

altair Code



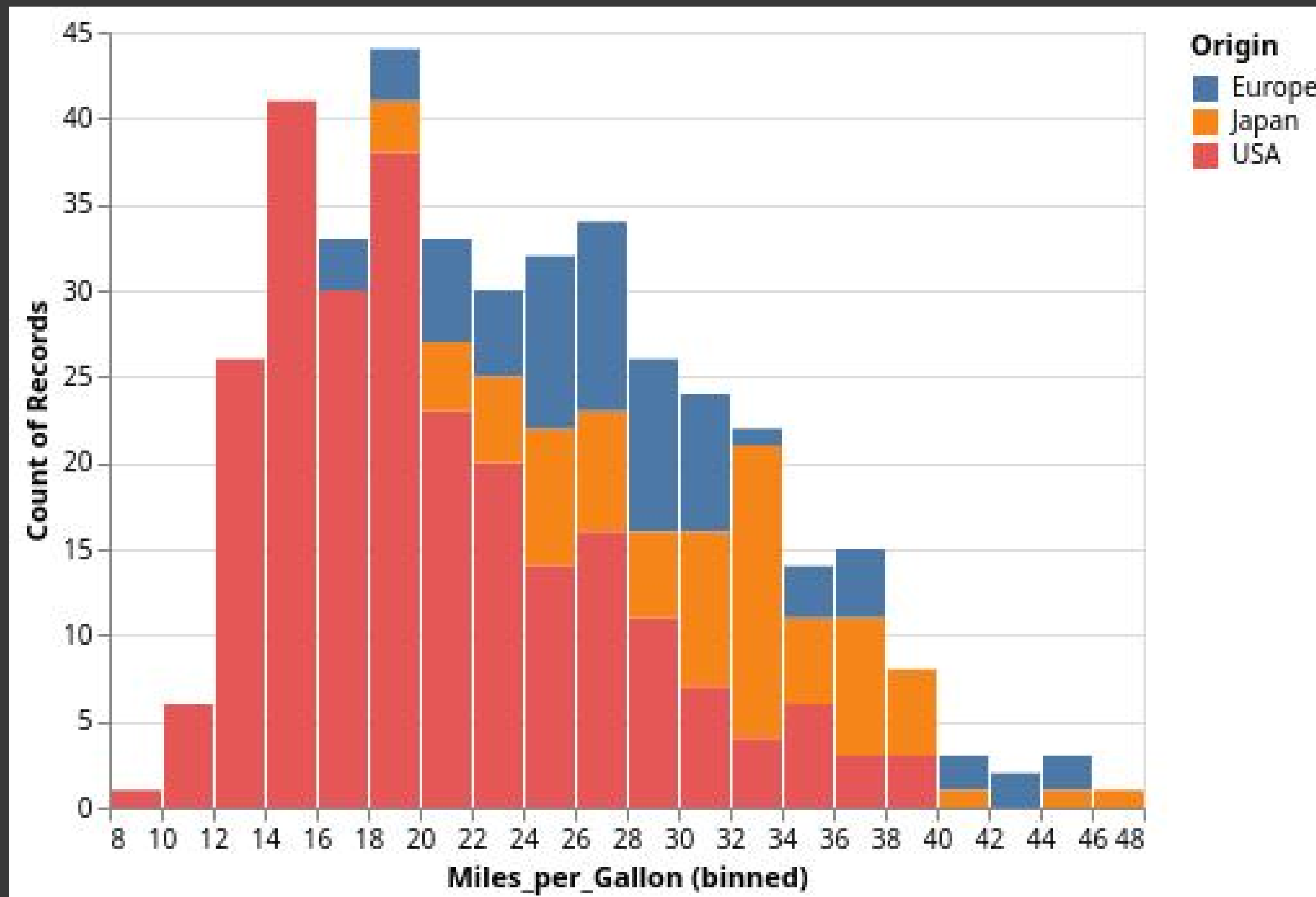
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altair Code

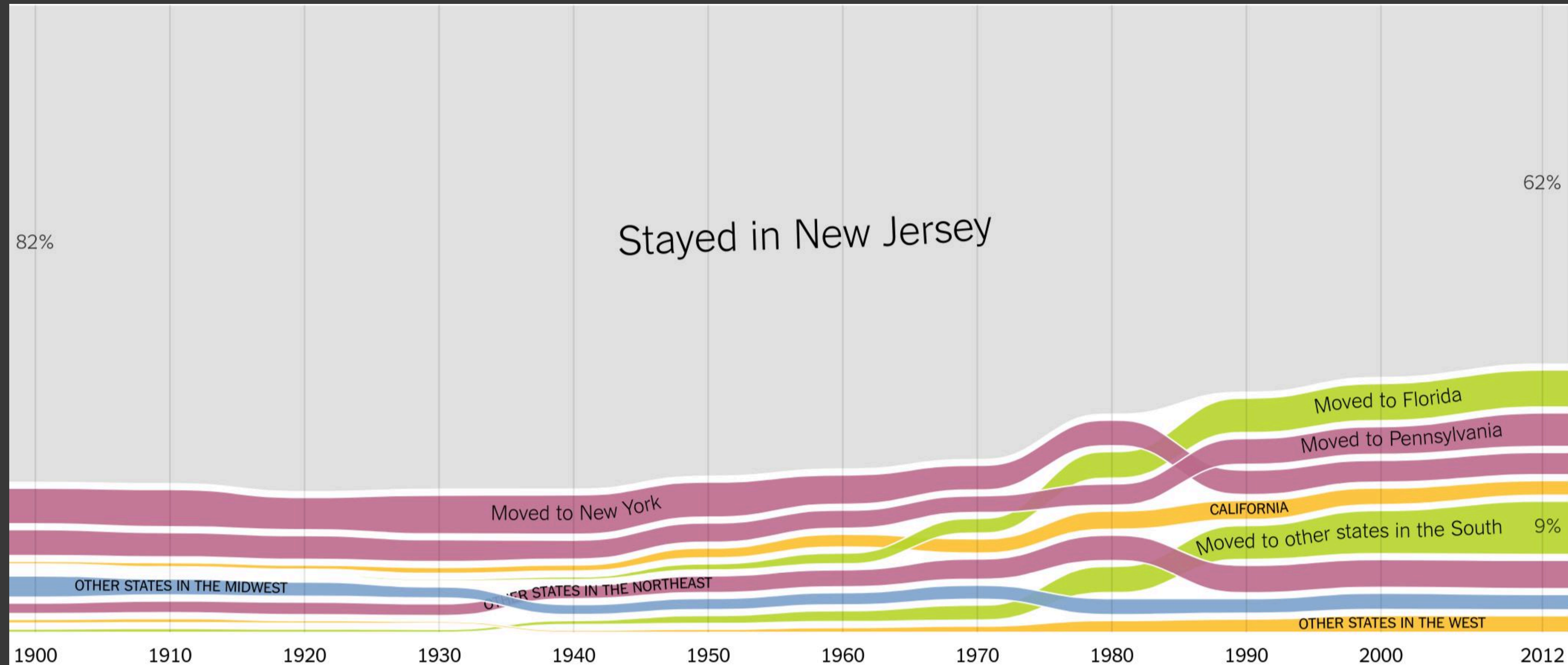


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altair Code

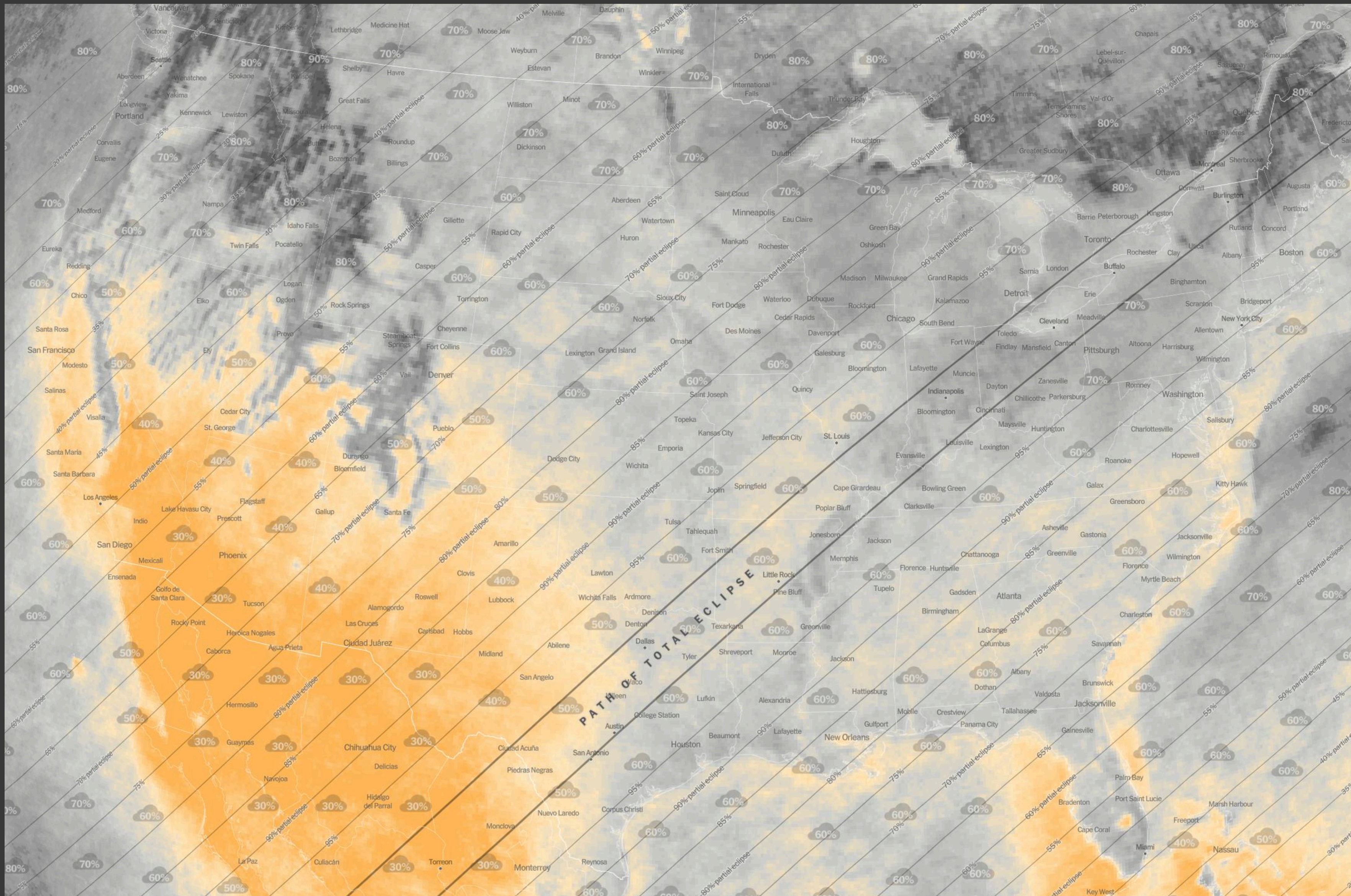


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    y='count()',
    color='Origin')
```



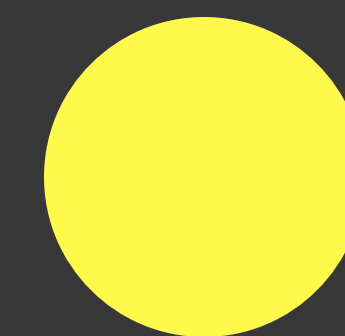
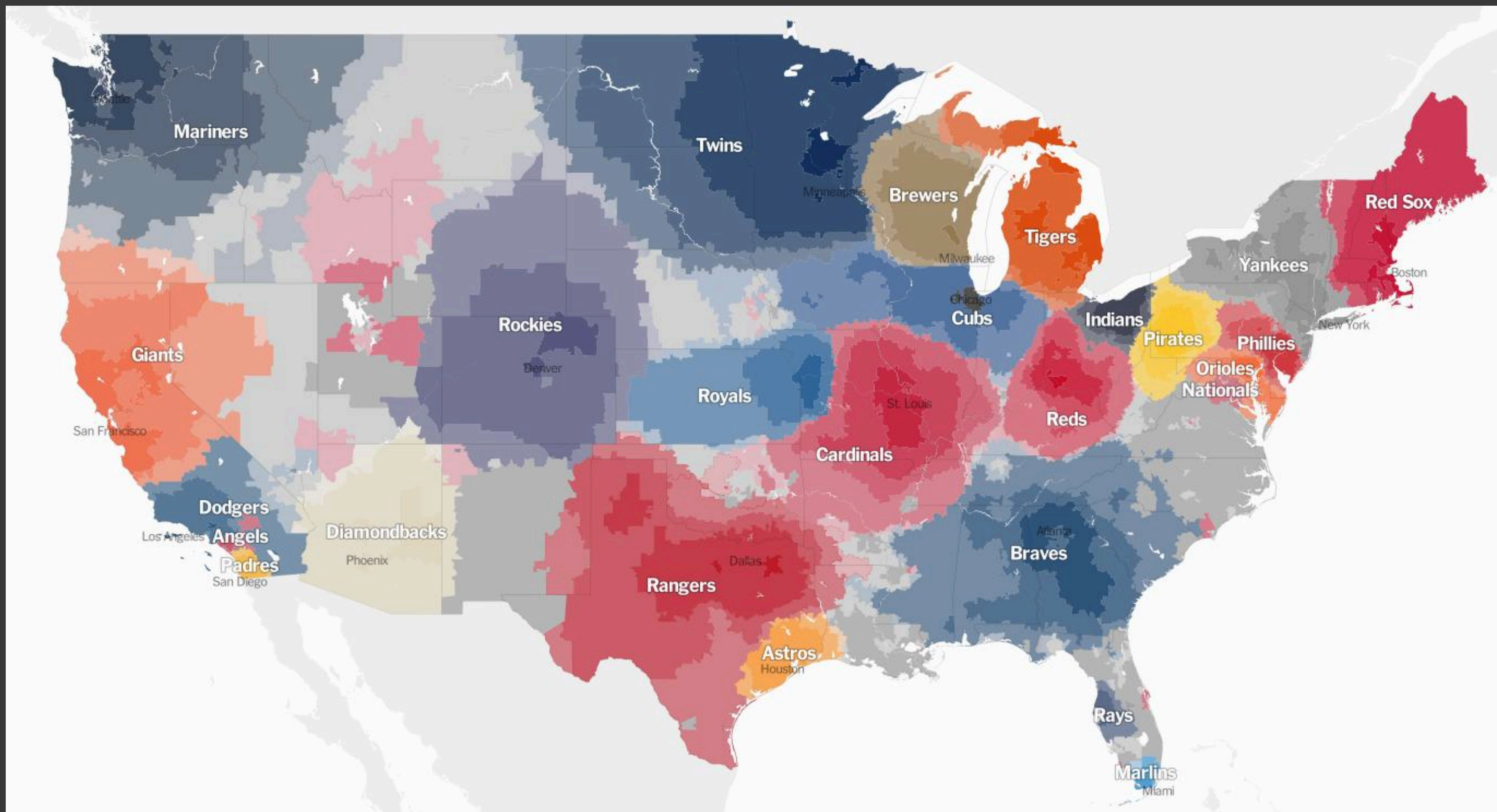
Color: Categorical

Source: NYT



Color: Quantitative

Source: NYT



Color: Both?

Source: NYT

Title → **Change since 2003 in average time spent at home**

+120 minutes per day

Gridlines →

+90

+60

+30

Bars
Marks?

Axes
Encodings?

2004

2010

2015

2020

2022

Tick marks

Among U.S. adults • Source: Analysis of the U.S. Census Bureau's American Time Use Survey by Patrick Sharkey, Princeton University • The New York Times

Caption

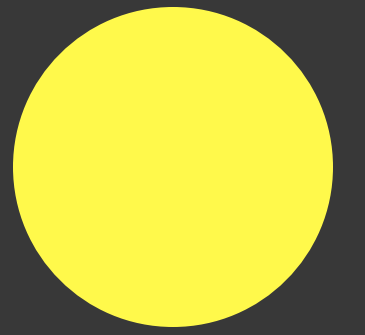
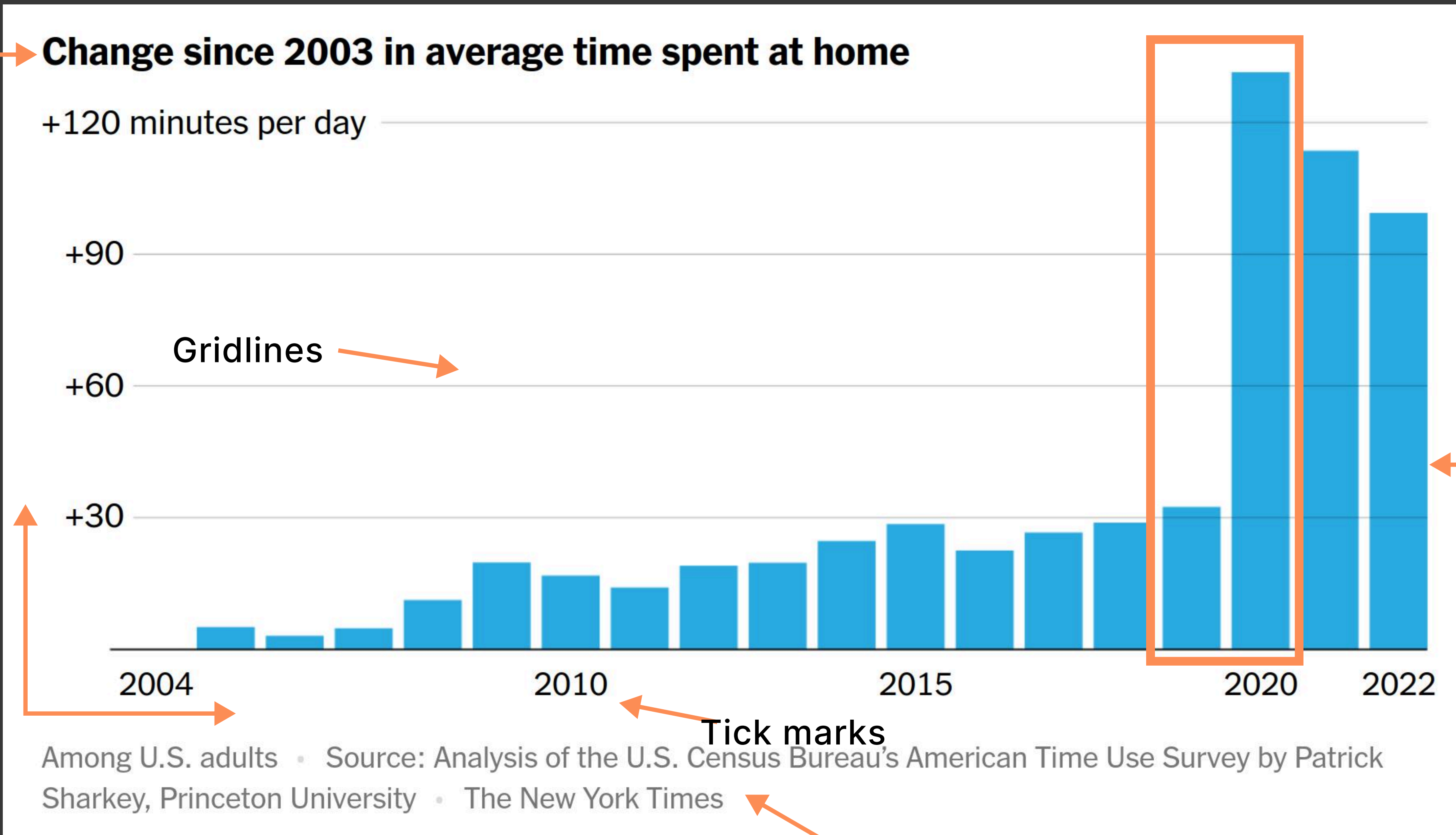
Anatomy of a Chart

Source: NYT

Title

Gridlines

Axes Encodings?



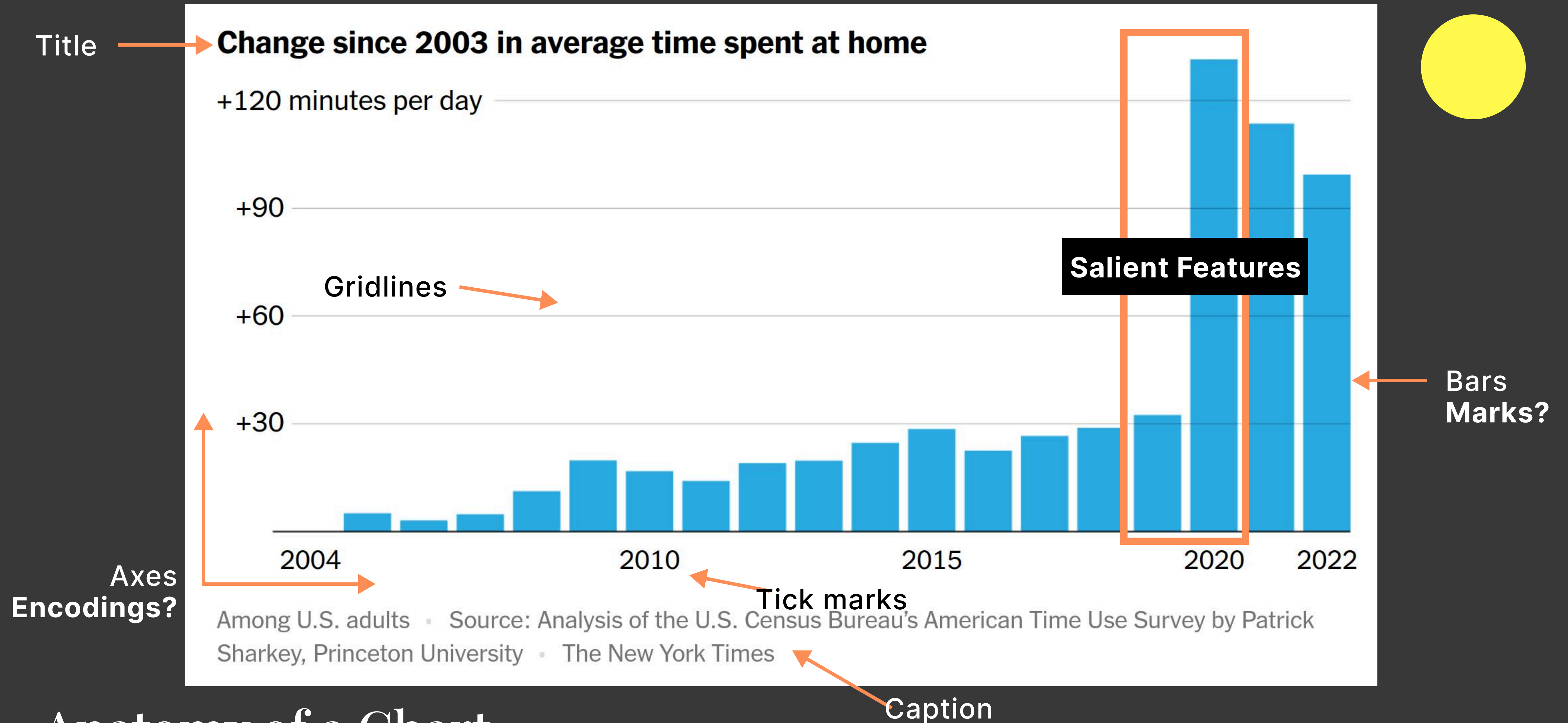
Bars Marks?

Tick marks

Caption

Anatomy of a Chart

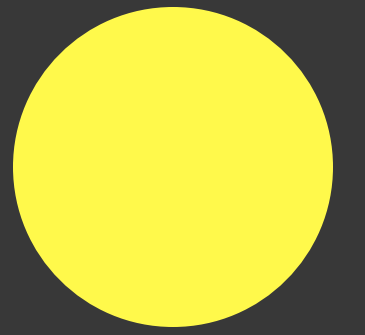
Source: NYT

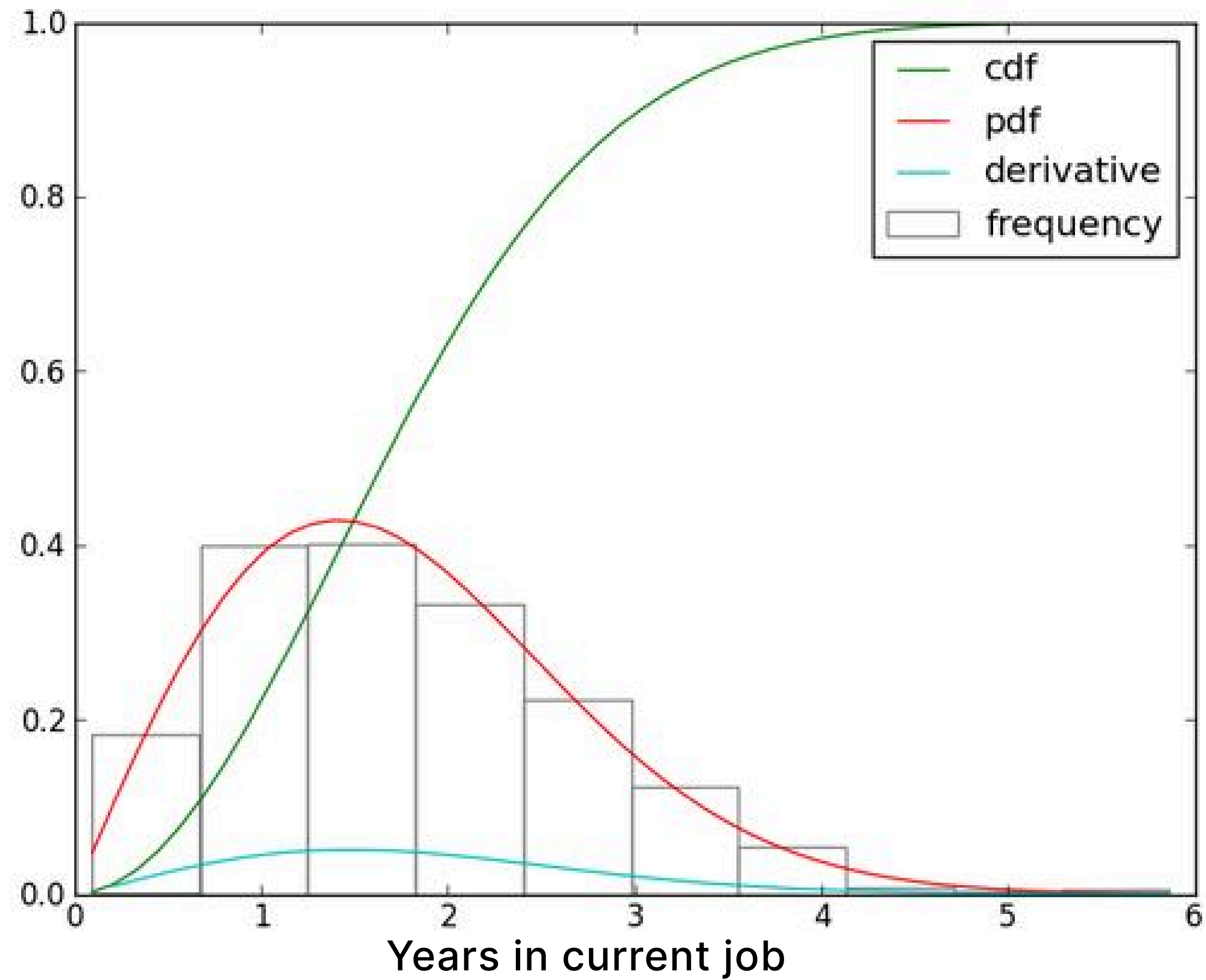


Anatomy of a Chart

Source: NYT

Visualizing Distributions

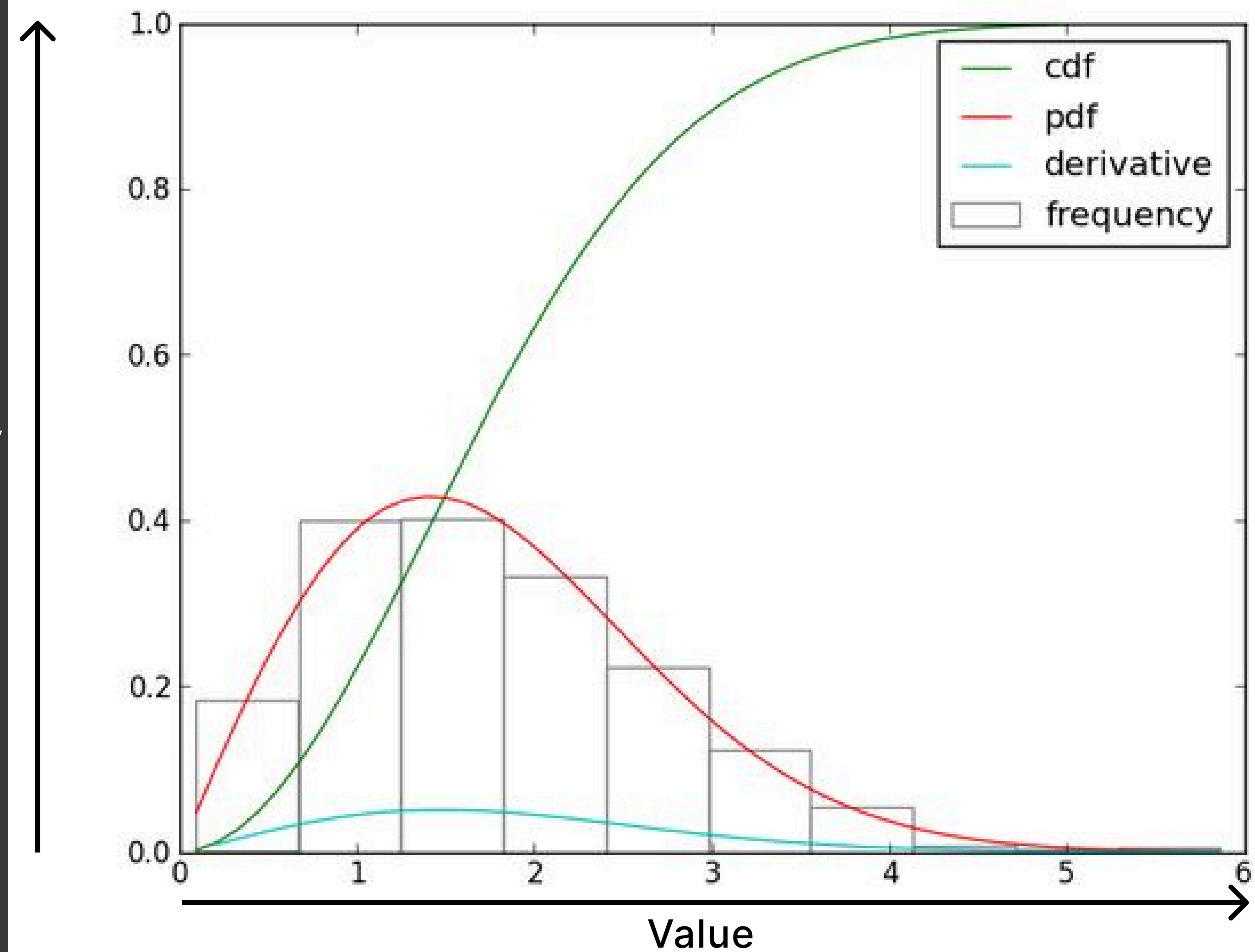




Visualizing Distributions

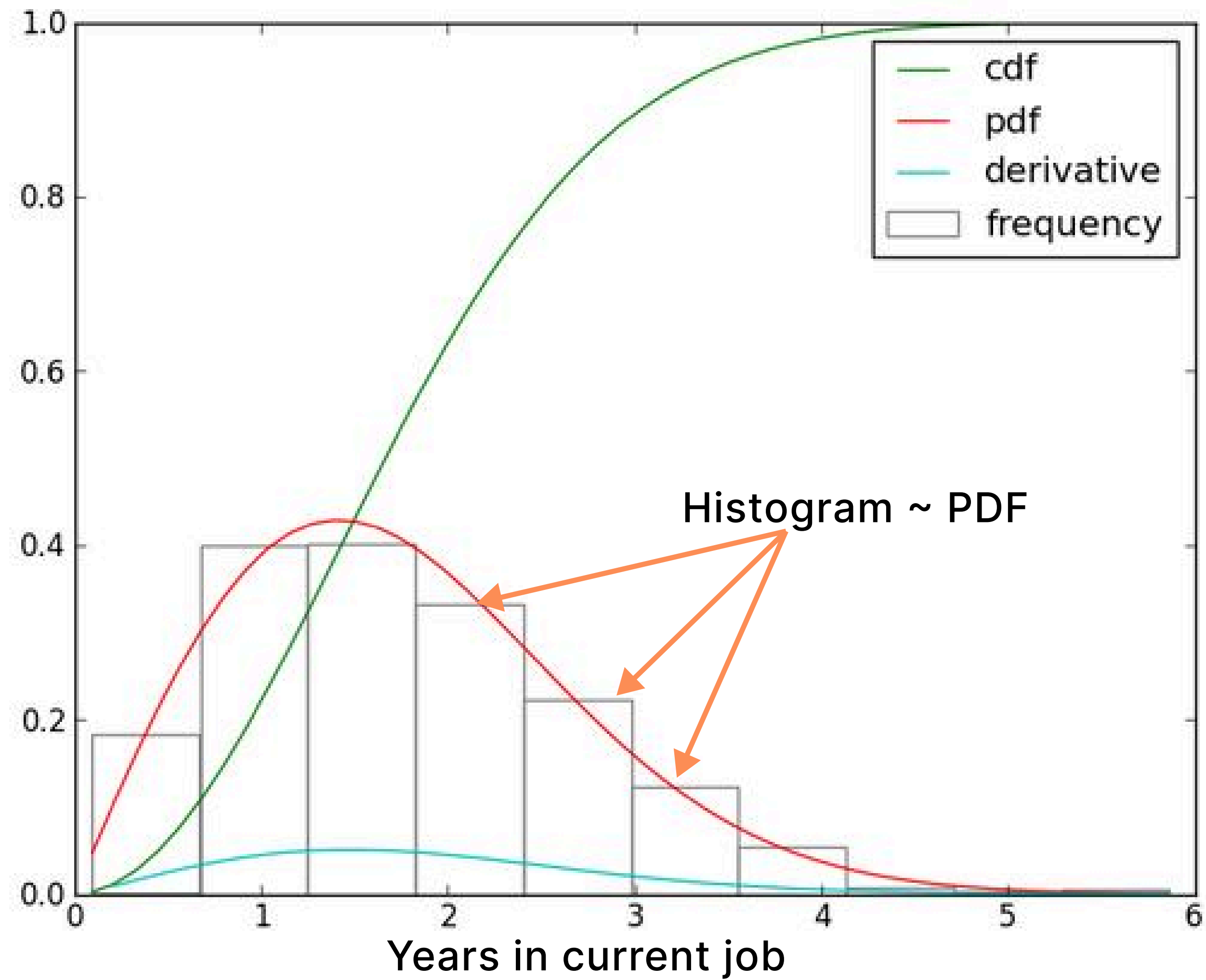
Source

Frequency



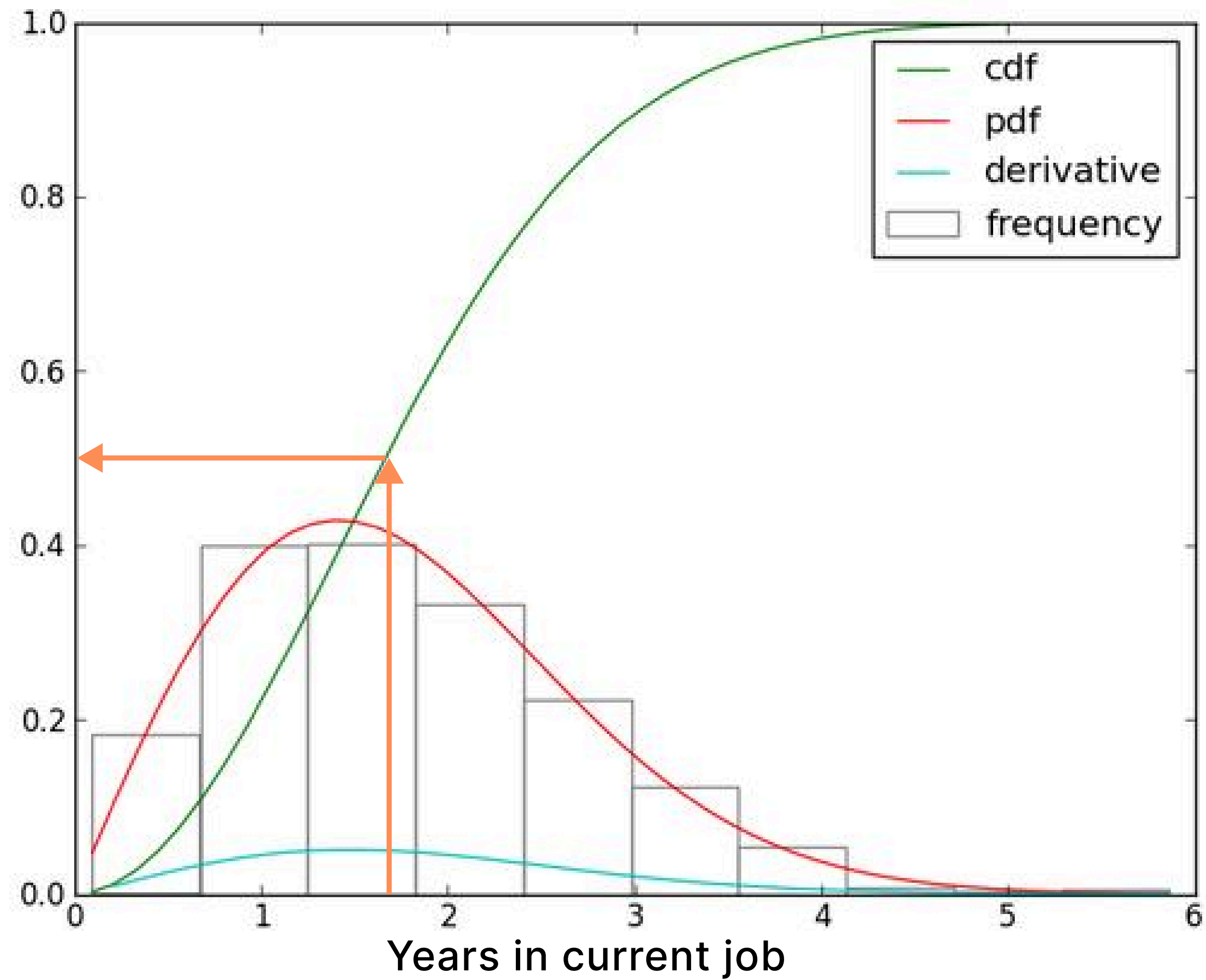
Visualizing Distributions

Source



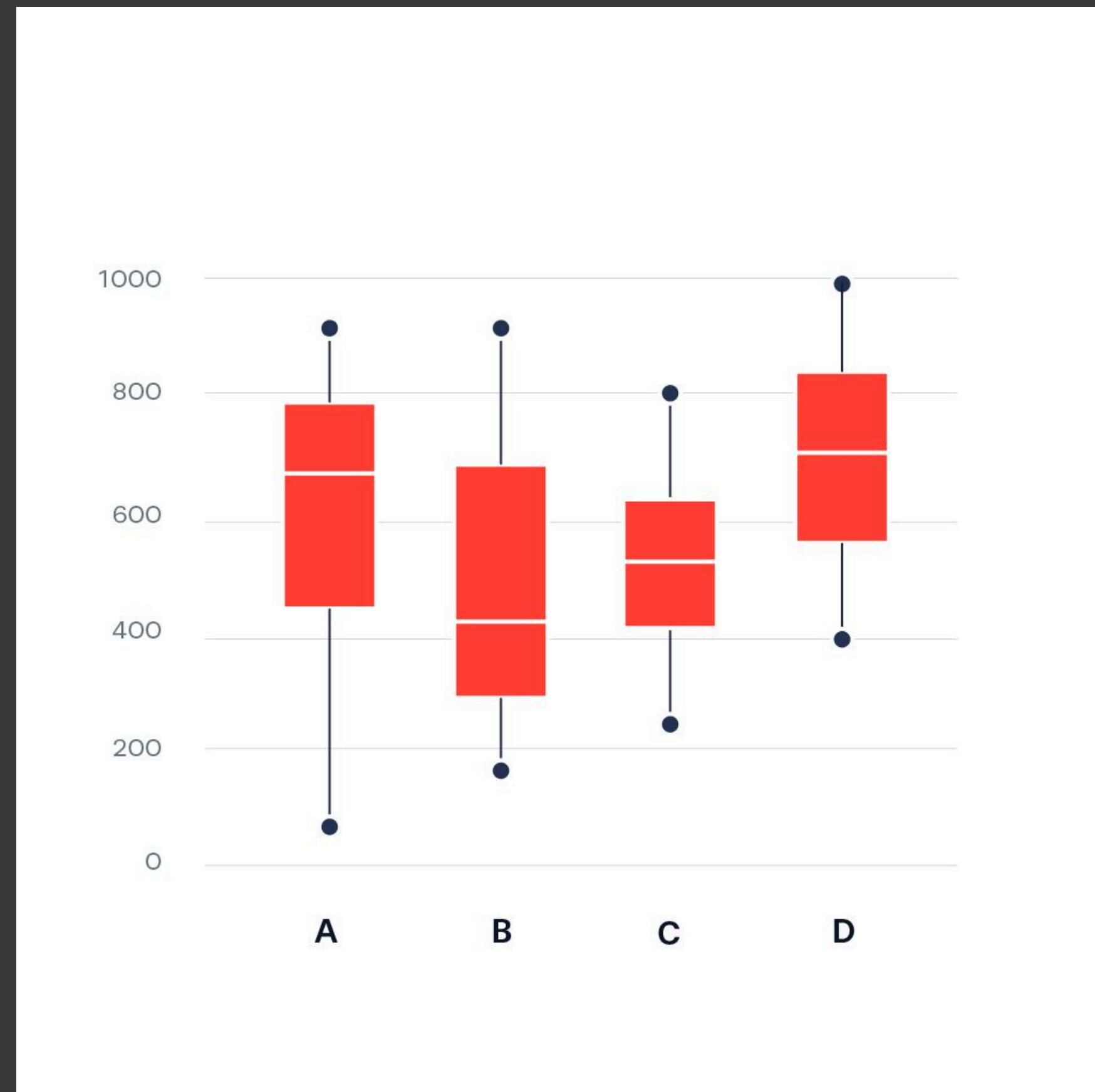
Visualizing Distributions

Source



Visualizing Distributions

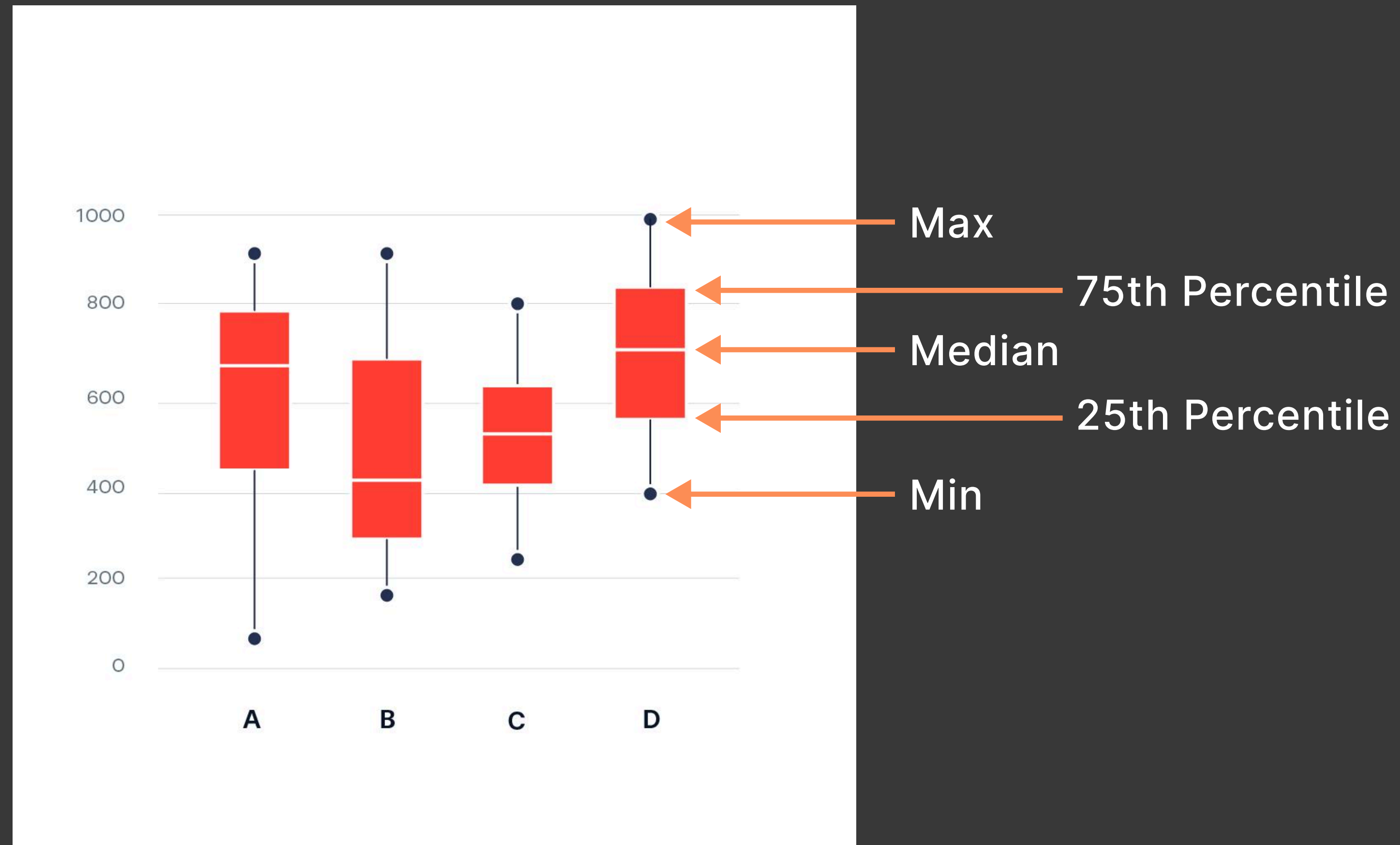
Source



Box Plot

Visualizing Distributions

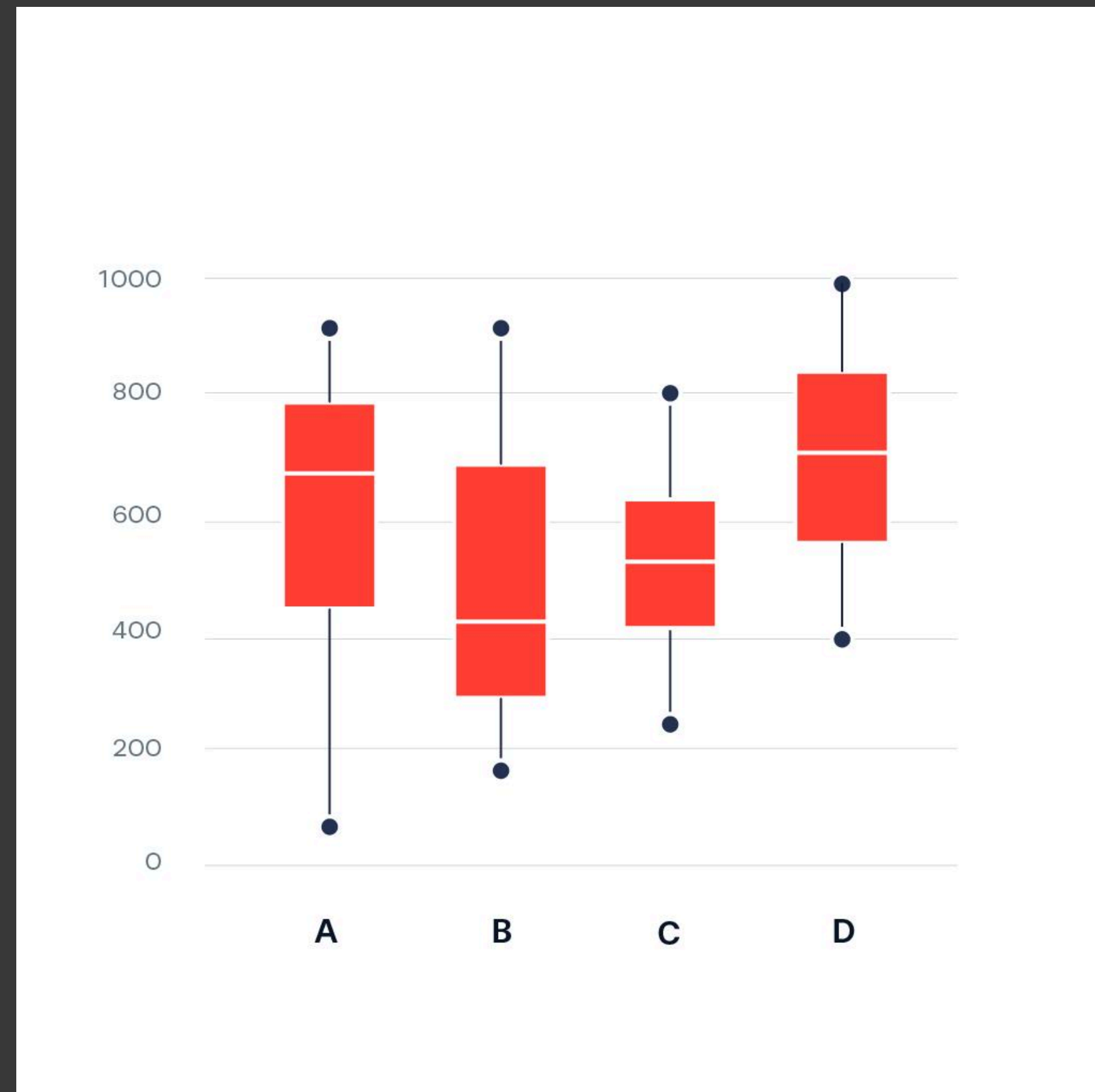
Source: DVP



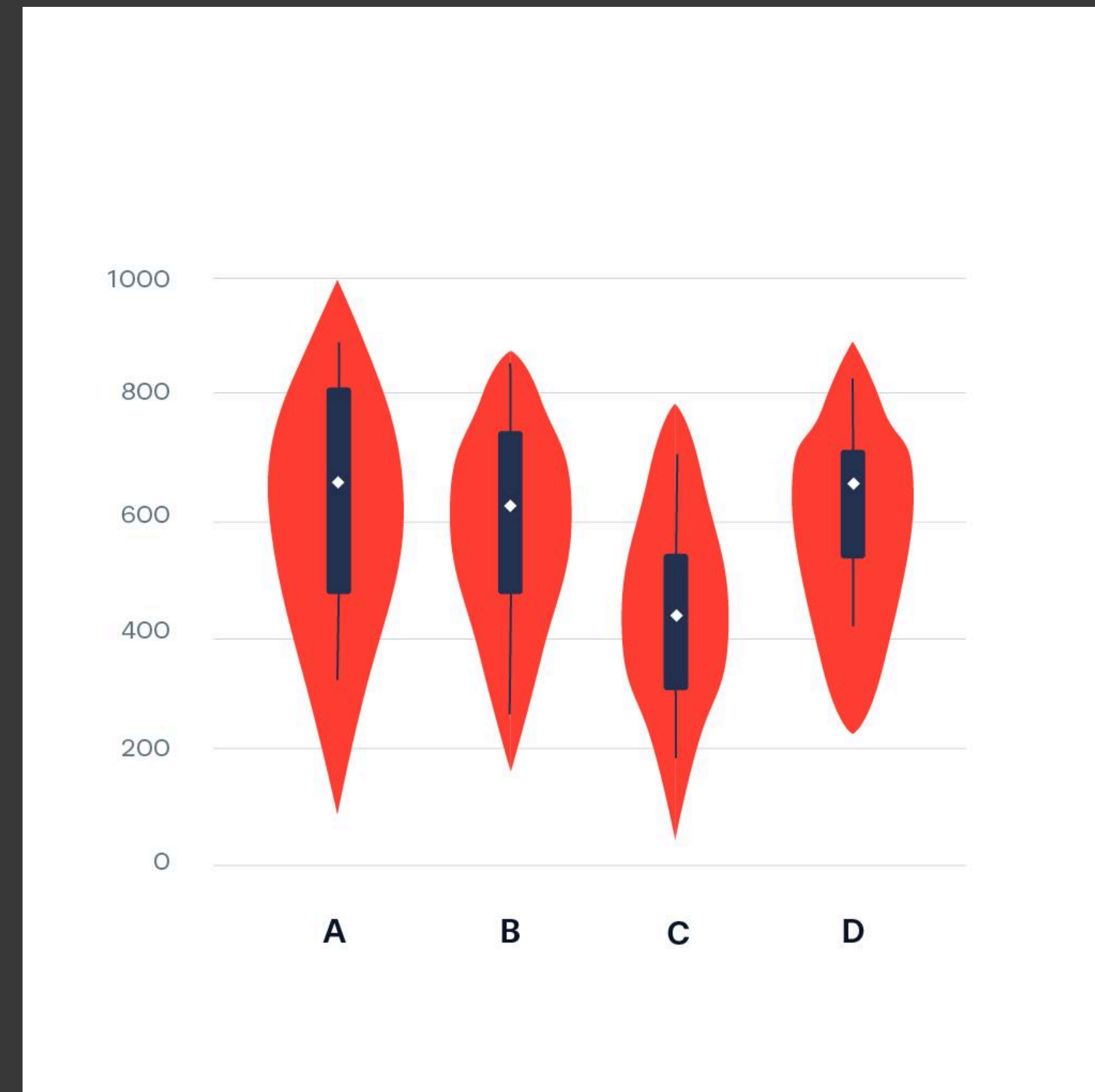
Box Plot

Visualizing Distributions

Source: DVP



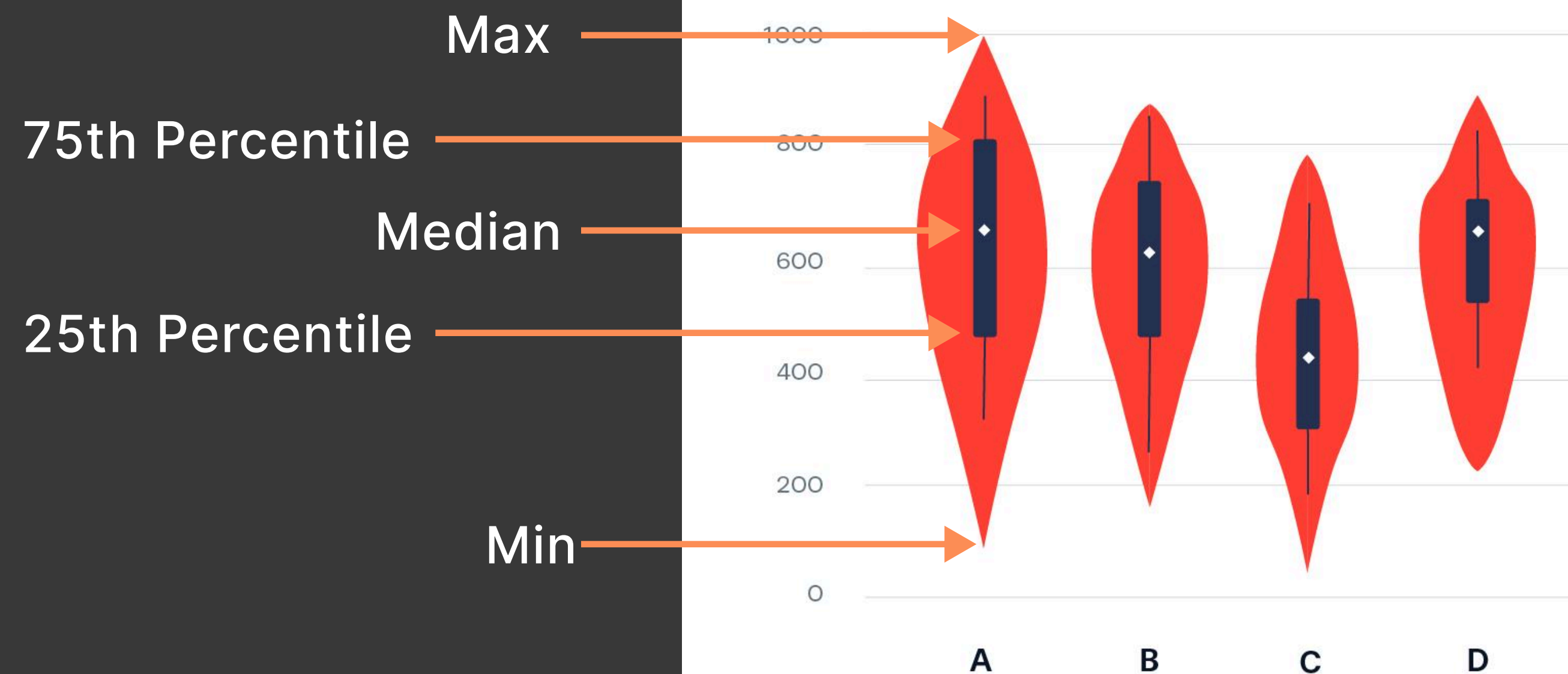
Box Plot



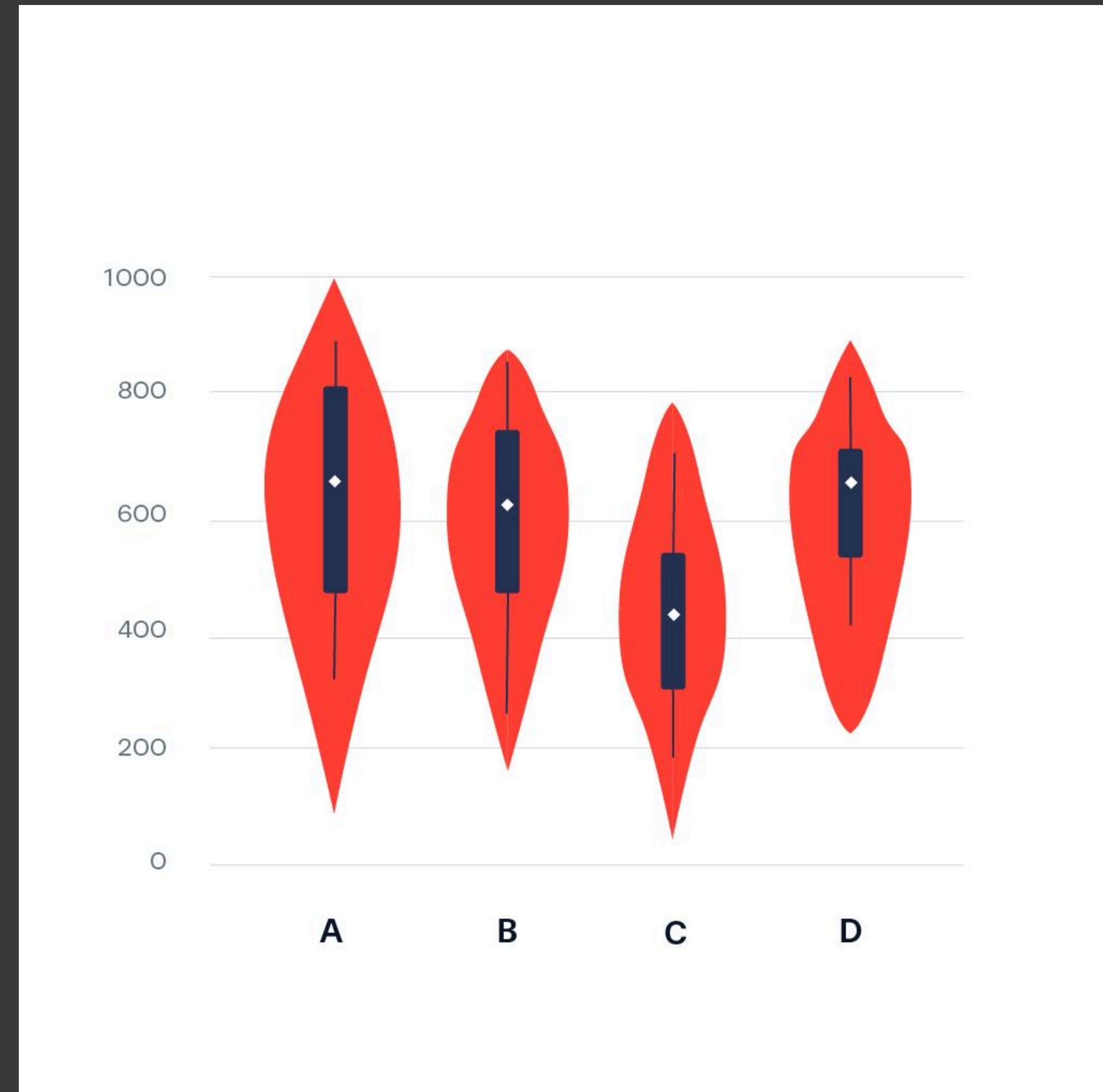
Violin Plot

Visualizing Distributions

Source: DVP



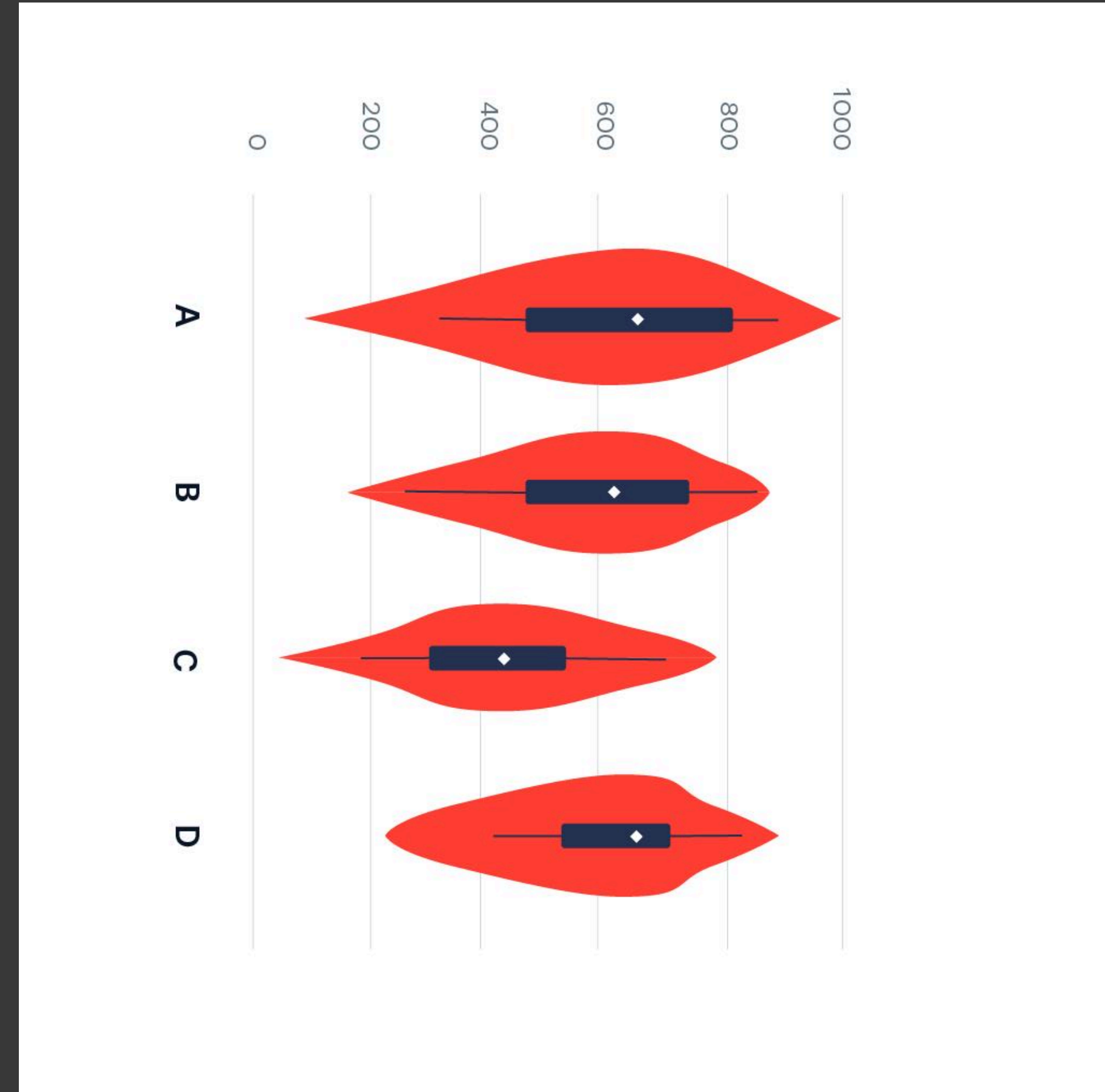
Violin Plot



Violin Plot

Visualizing Distributions

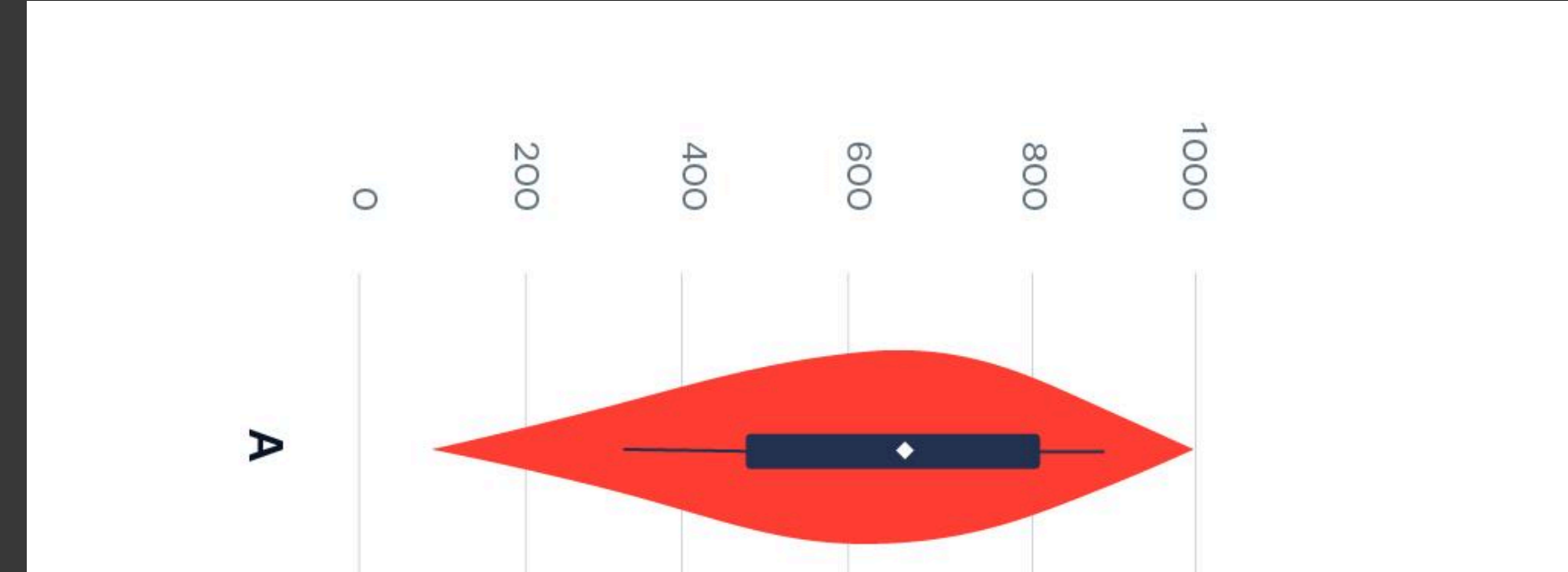
Source: DVP



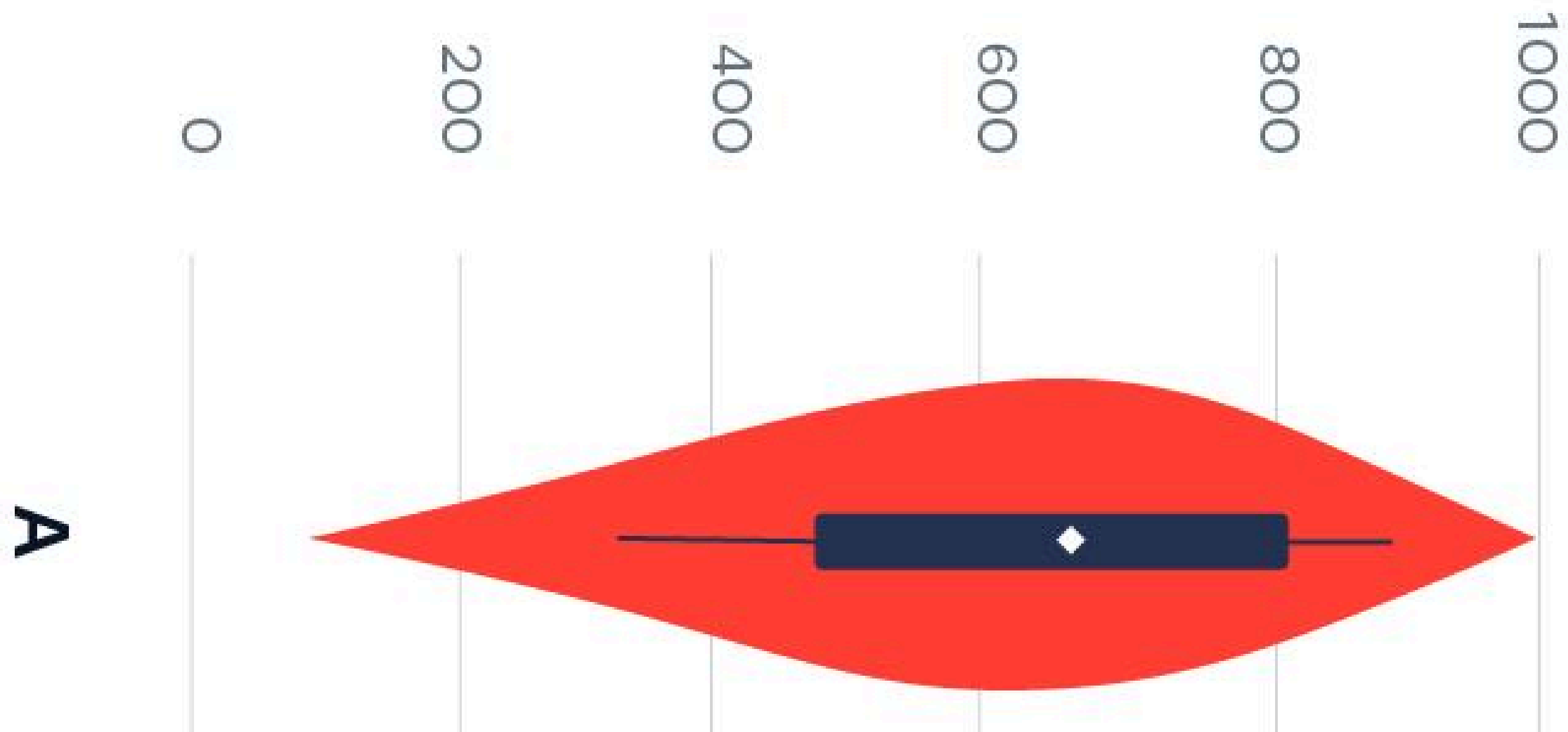
Violin Plot

Visualizing Distributions

Source: DVP

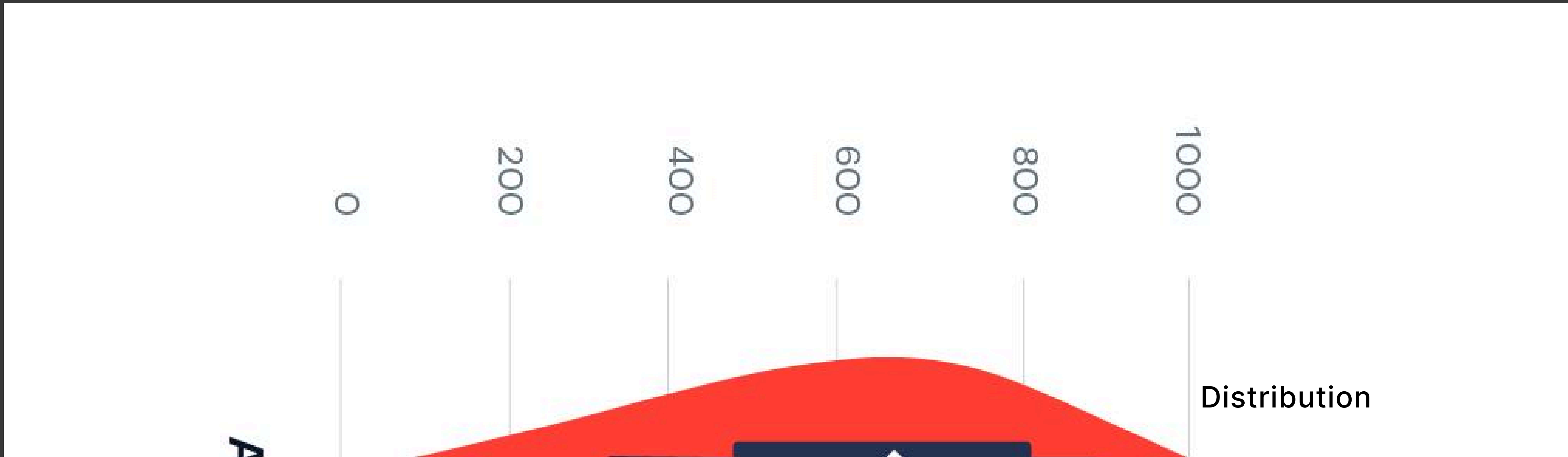


Violin Plot



Visualizing Distributions

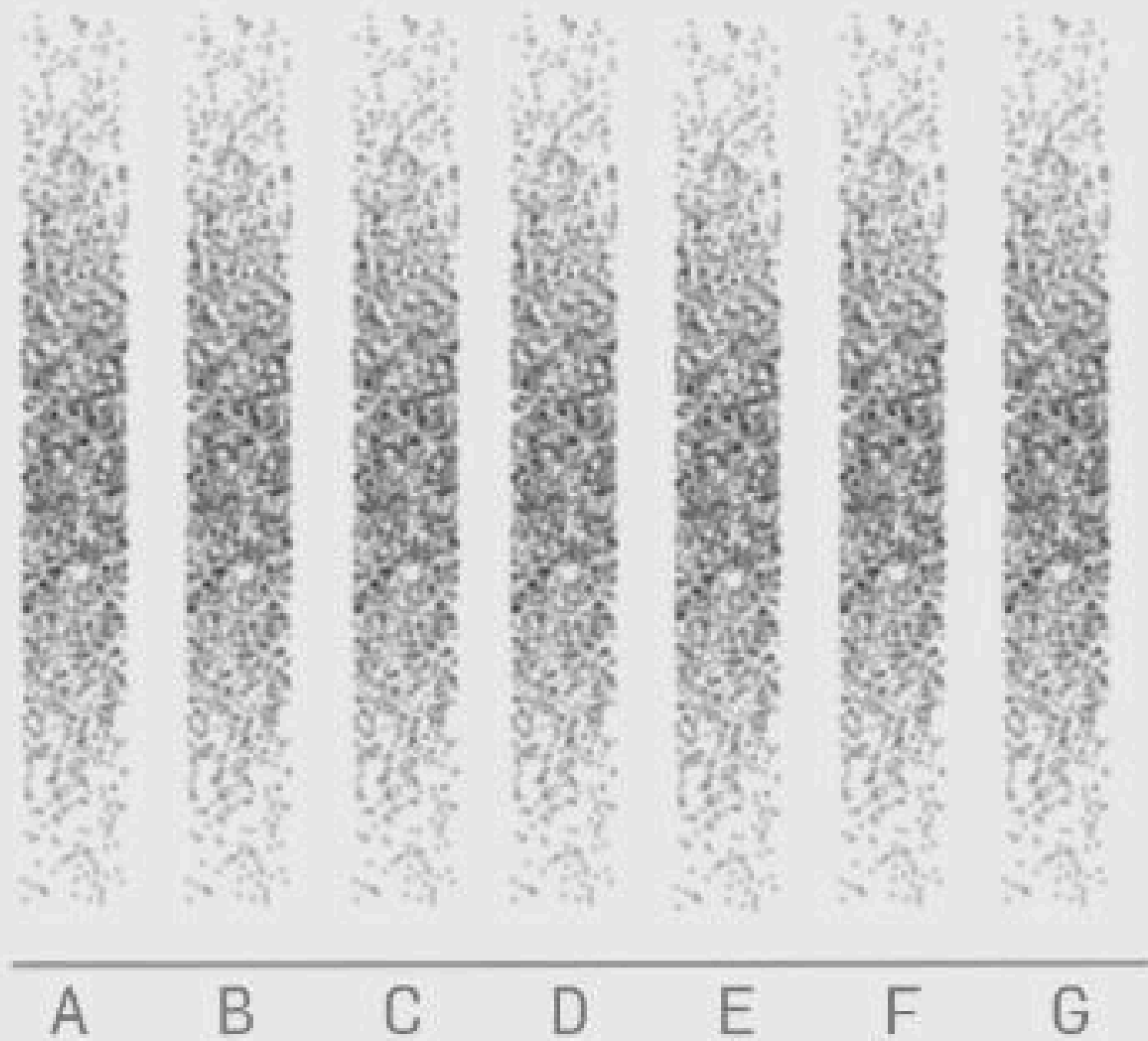
Source: DVP



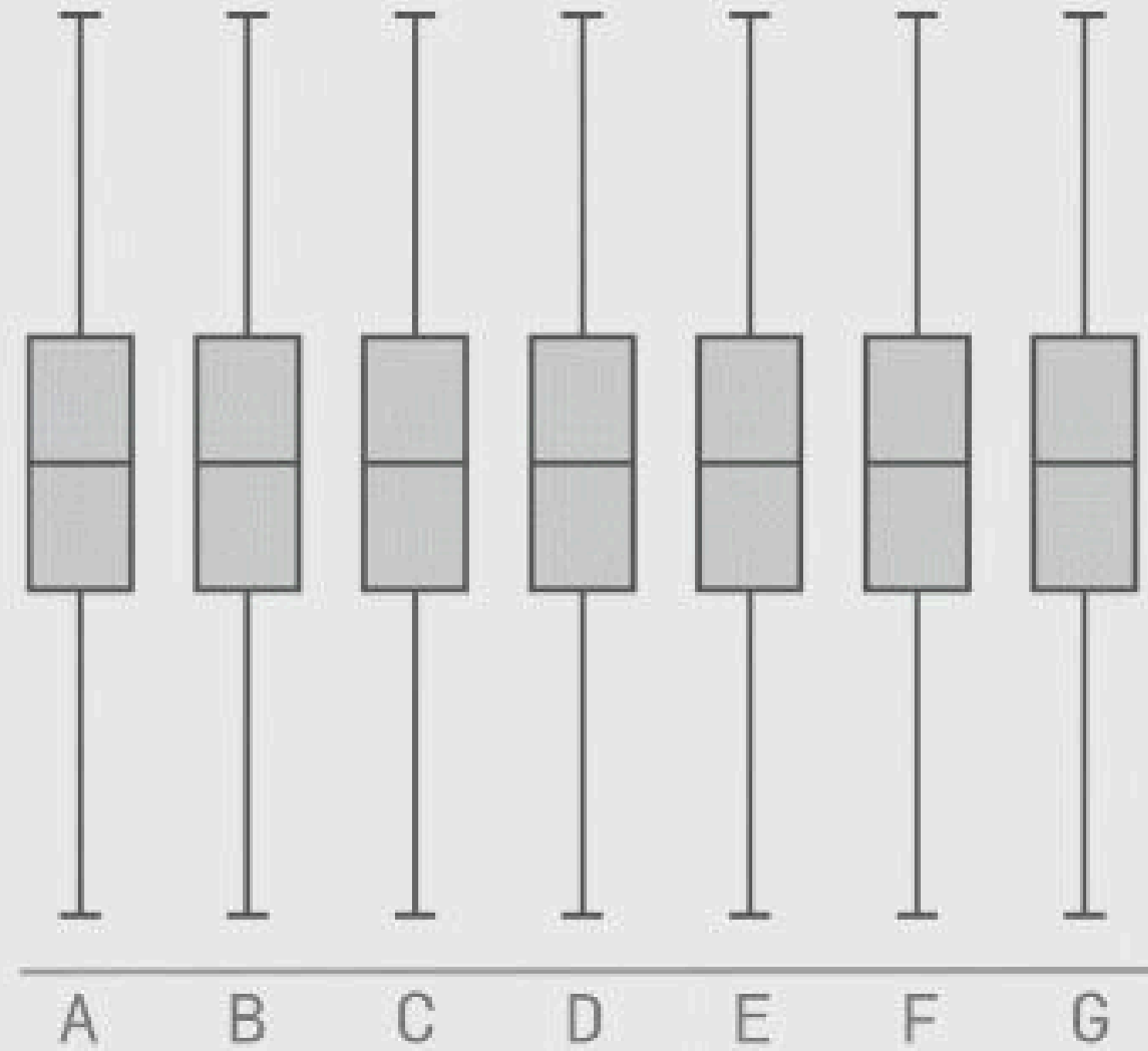
Visualizing Distributions

Source: DVP

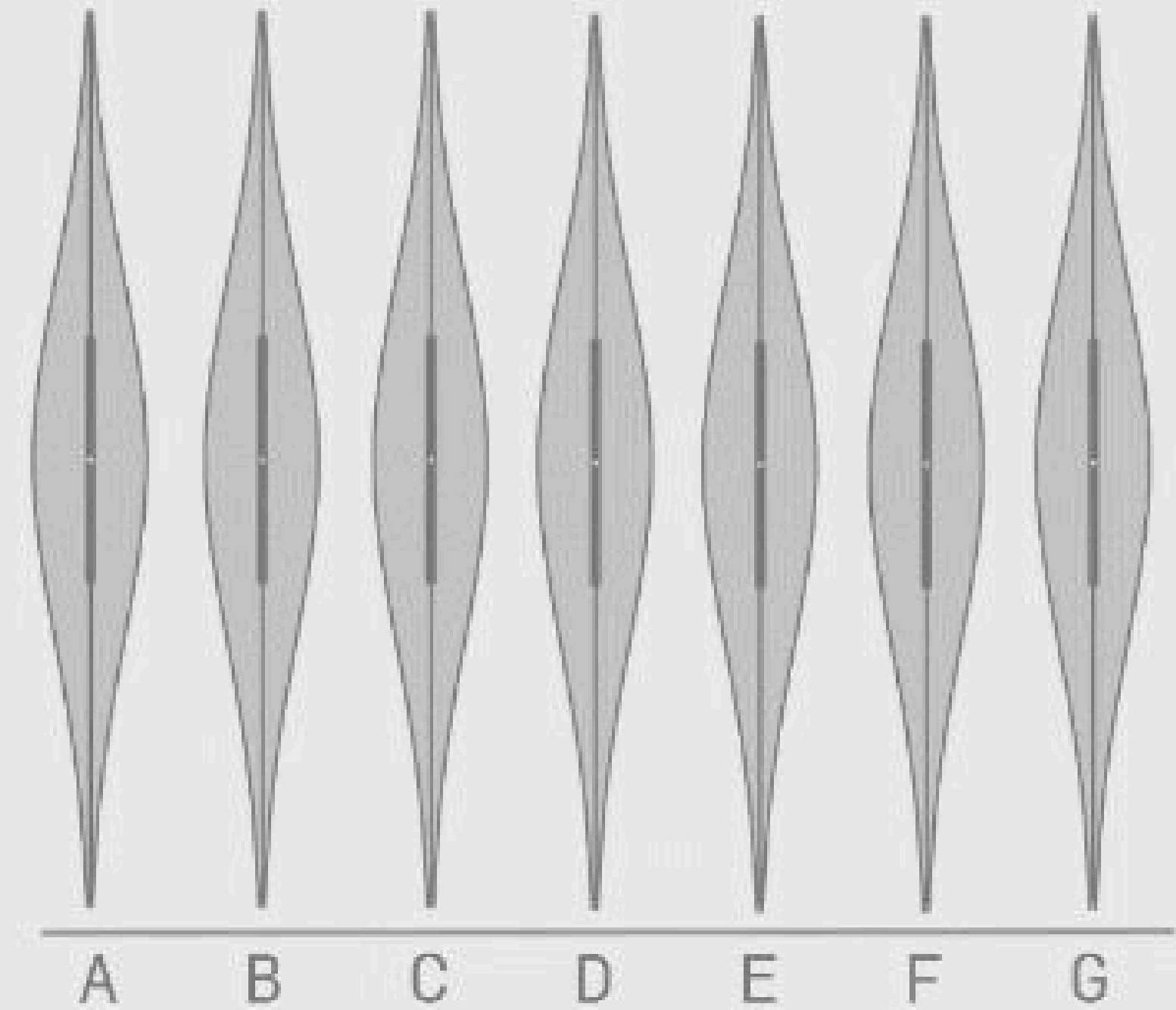
Raw Data



Box-plot of the Data



Violin-plot of the Data



Visualizing Distributions

Source: Autodesk



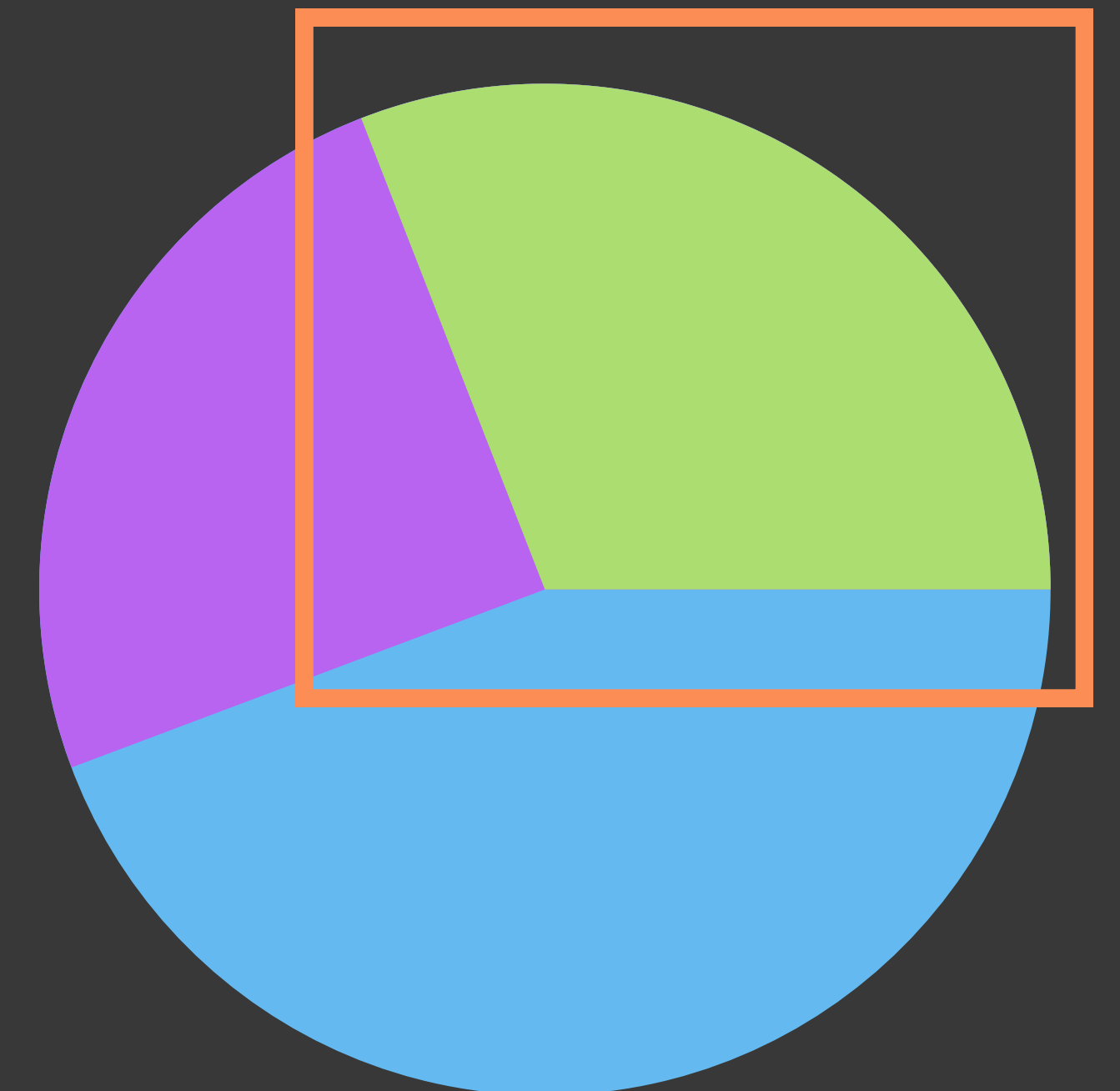
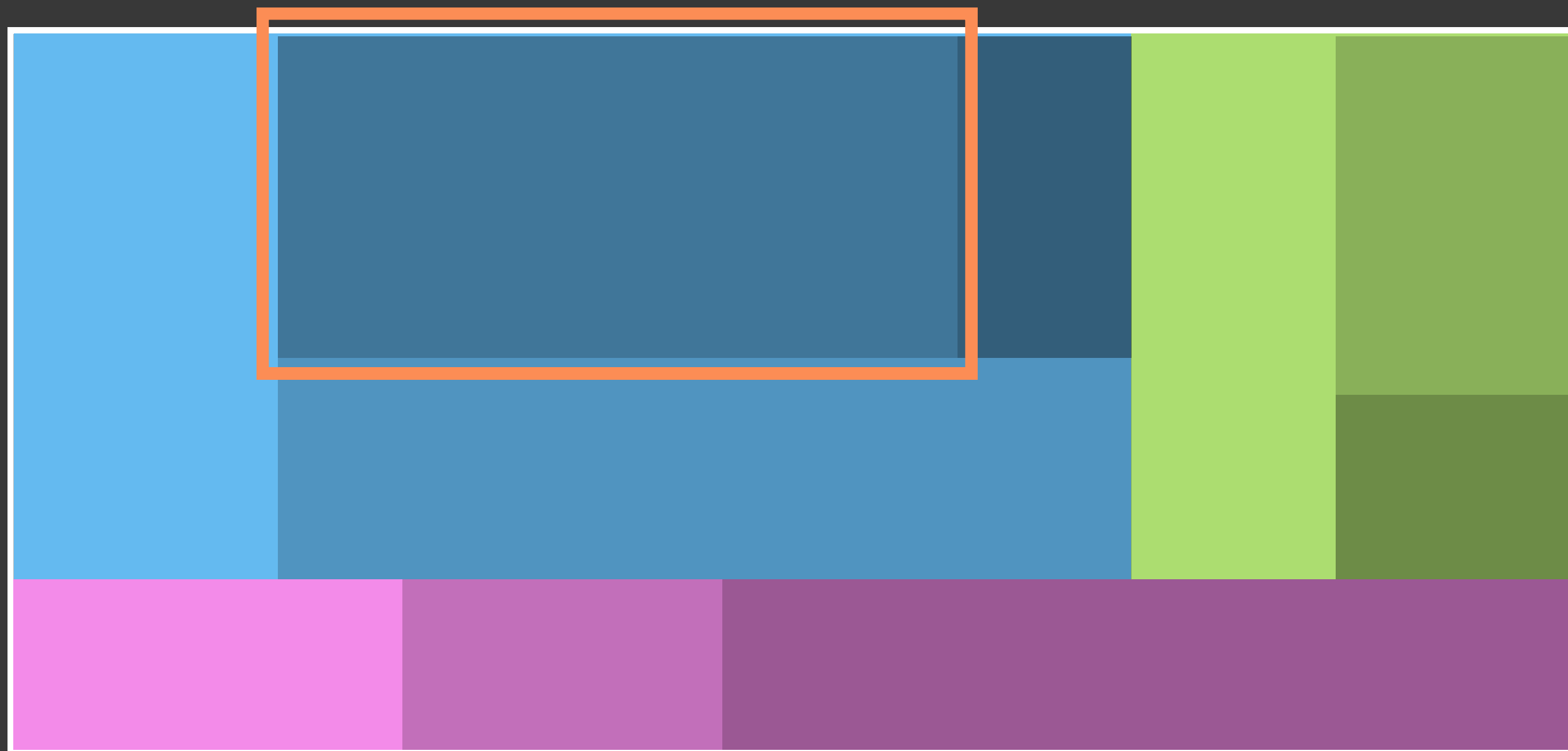
Viz Crimes

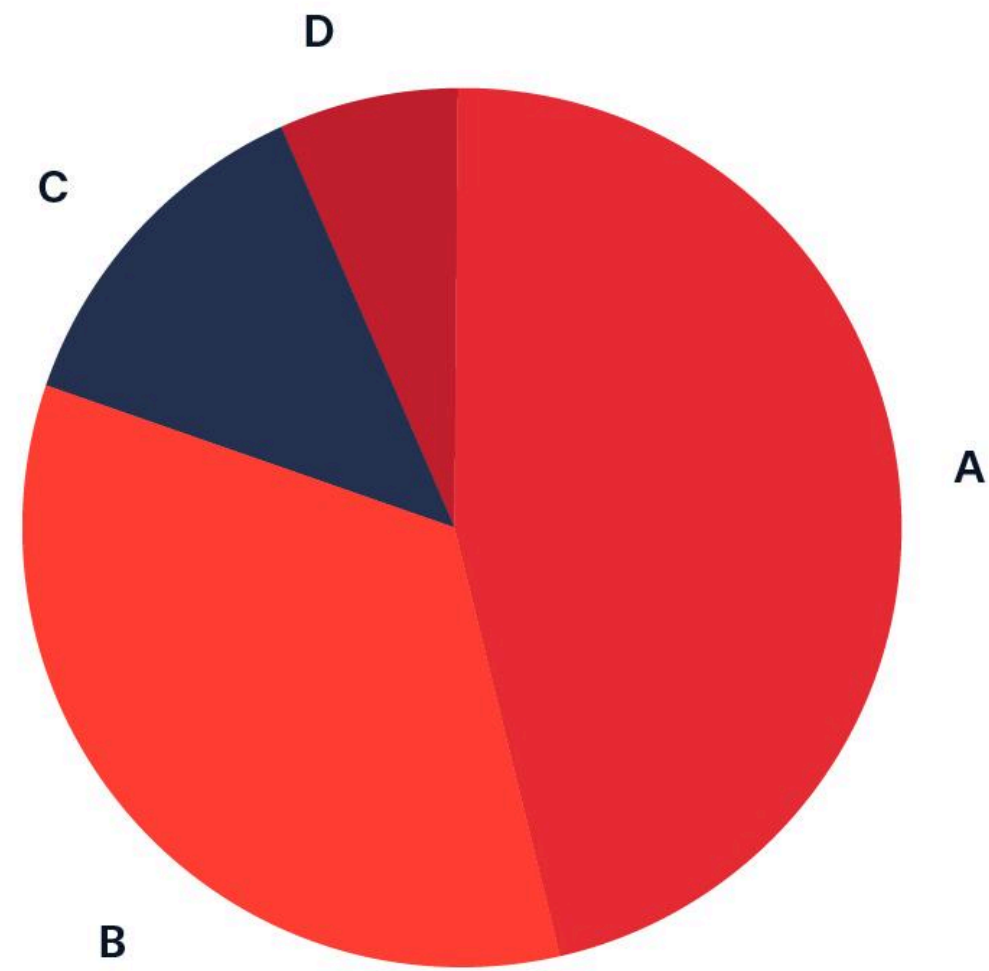
(i.e., what *not* to do)

Encodings

How data maps to perceptual “channels”

Area: Quantitative

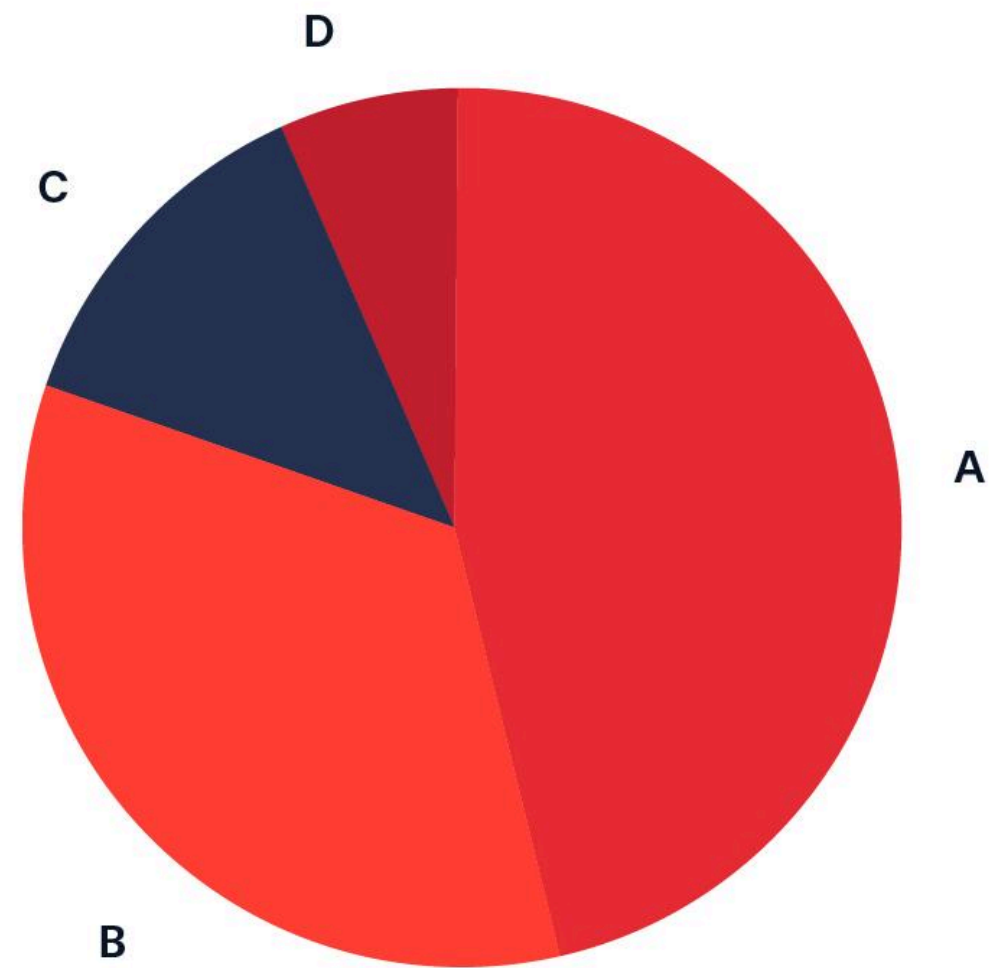




No Pie Charts!

The eye tends to perceive *area* instead of *angle*, making large segments look much larger than small segments.

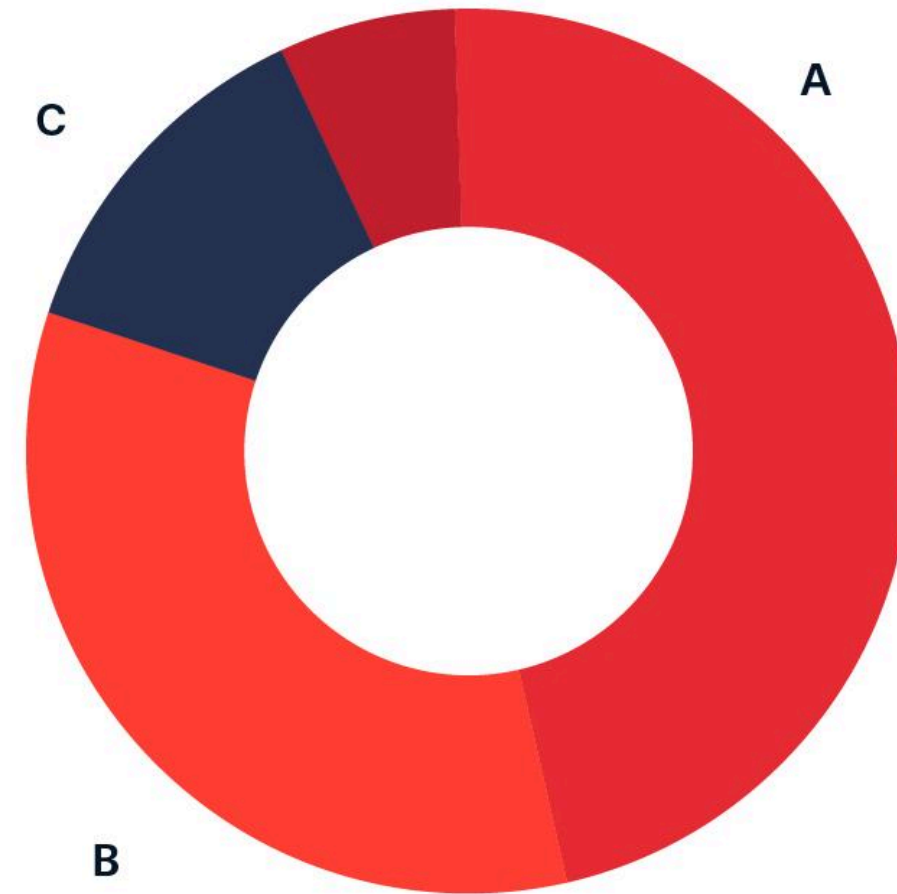
Source: DVP



No Pie Charts!

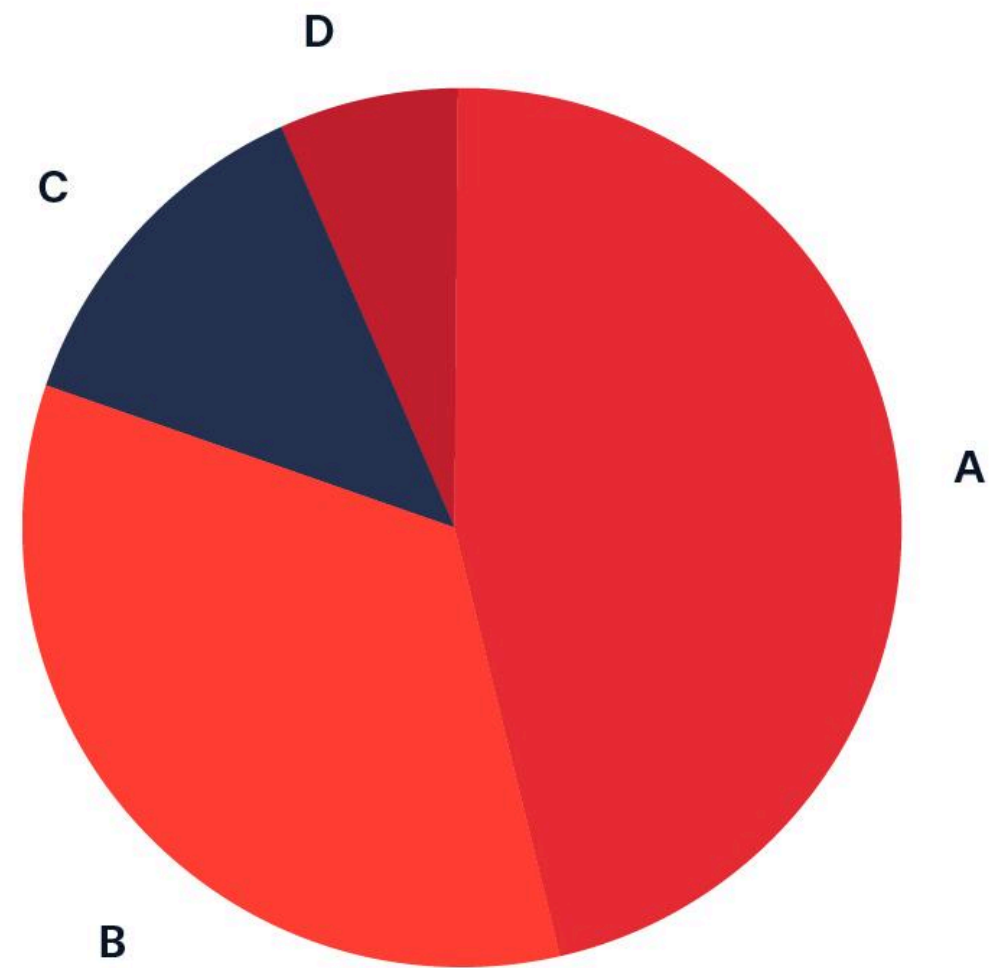
The eye tends to perceive *area* instead of *angle*, making large segments look much larger than small segments.

Source: DVP



Donut Chart

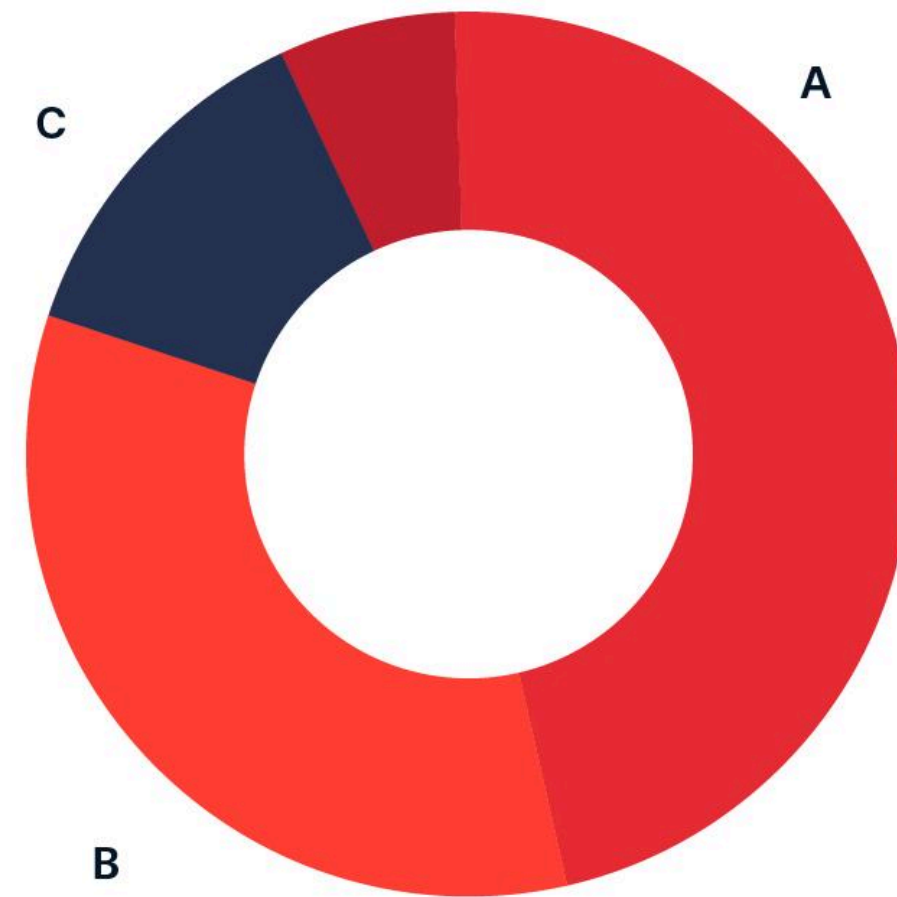
Encourages viewer to perceive *angle* instead of *area*.



No Pie Charts!

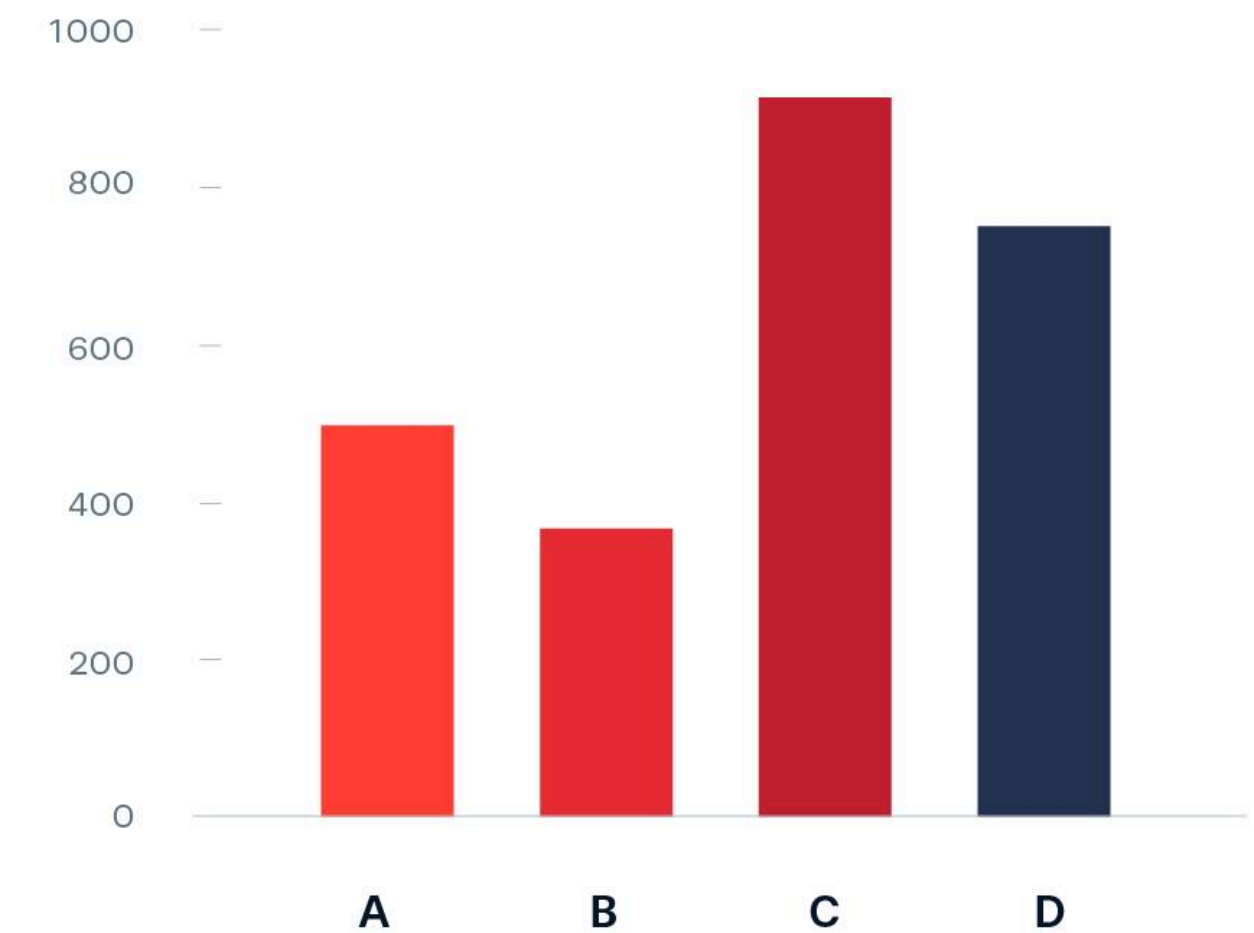
The eye tends to perceive *area* instead of *angle*, making large segments look much larger than small segments.

Source: DVP



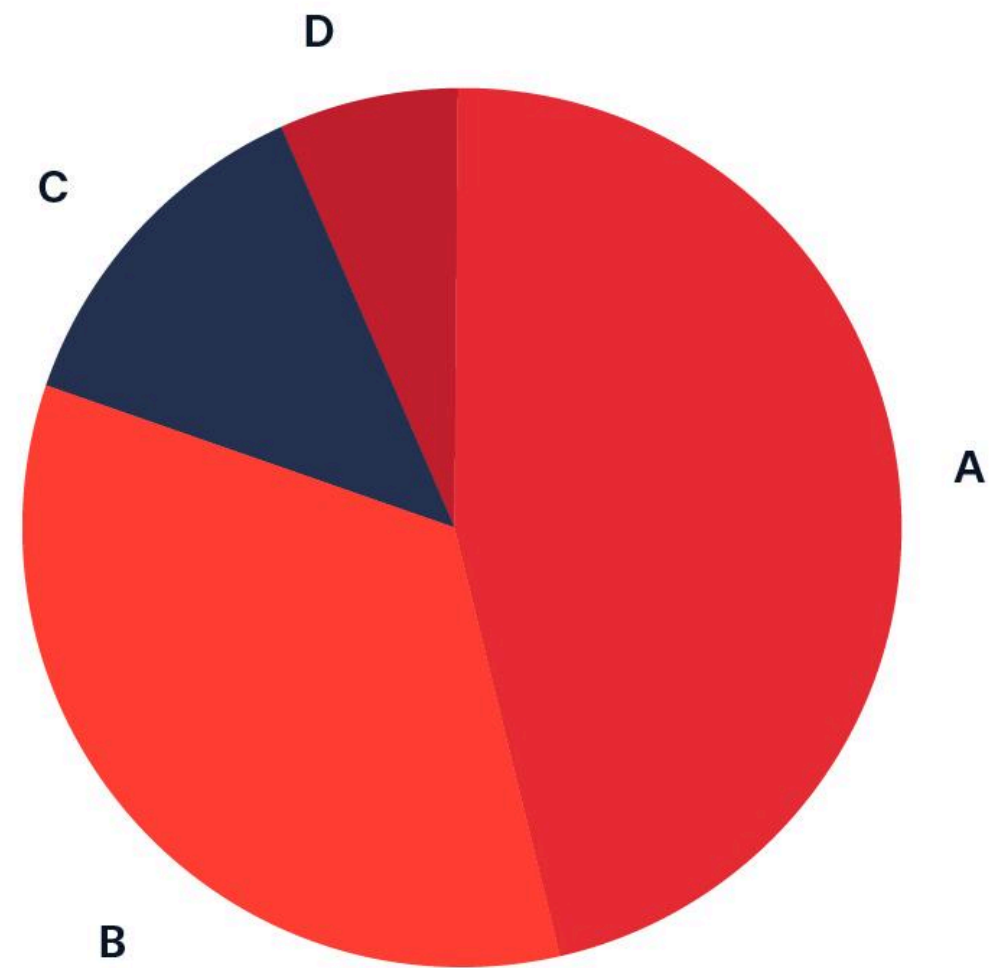
Donut Chart

Encourages viewer to perceive *angle* instead of *area*.



Bar Chart

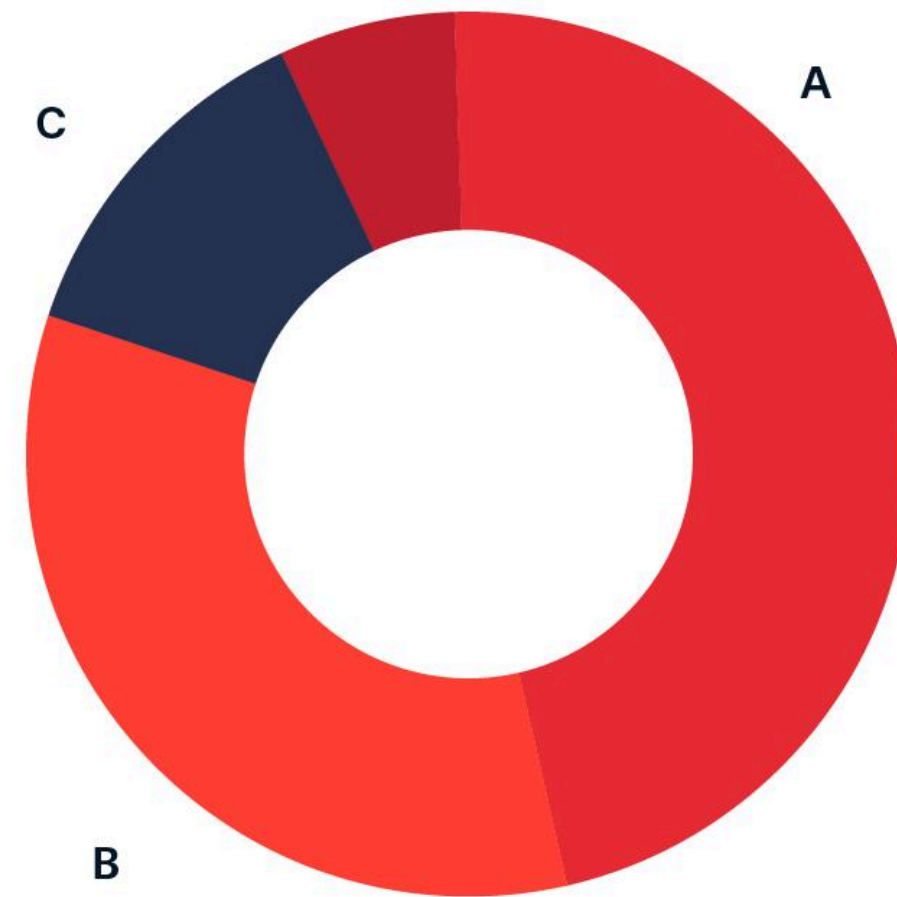
Like a donut chart, but not bendy! Your eye is good at comparing straight lengths!



No Pie Charts!

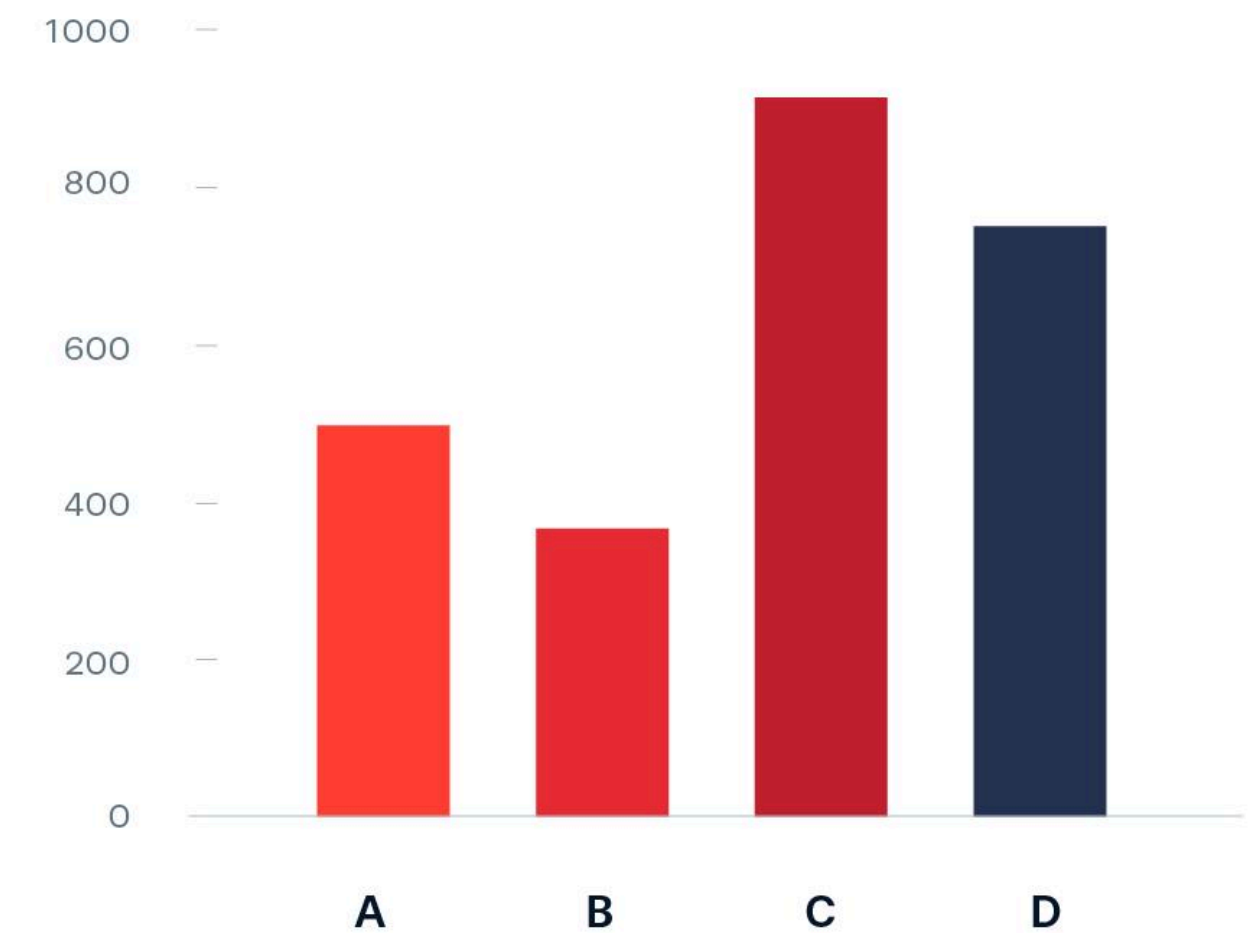
The eye tends to perceive *area* instead of *angle*, making large segments look much larger than small segments.

Source: DVP



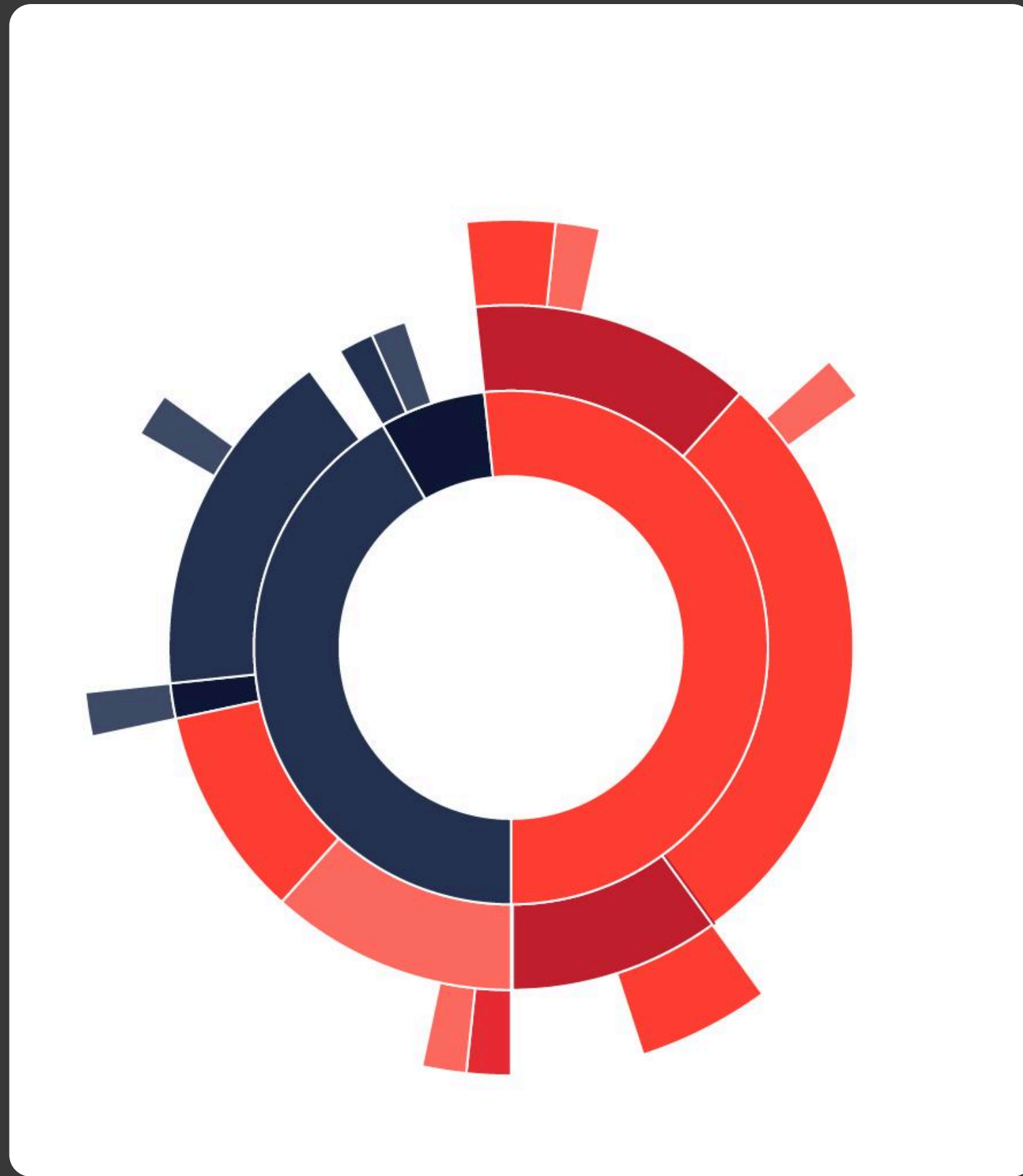
Donut Chart

Encourages viewer to perceive *angle* instead of *area*.



Bar Chart

Like a donut chart, but not bendy! Your eye is good at comparing straight lengths!



Avoid Weird Encodings

Perception tends to work best along straight lines. Viewers more easily understand charts that they've seen before. Don't let aesthetics supersede good visual communication.

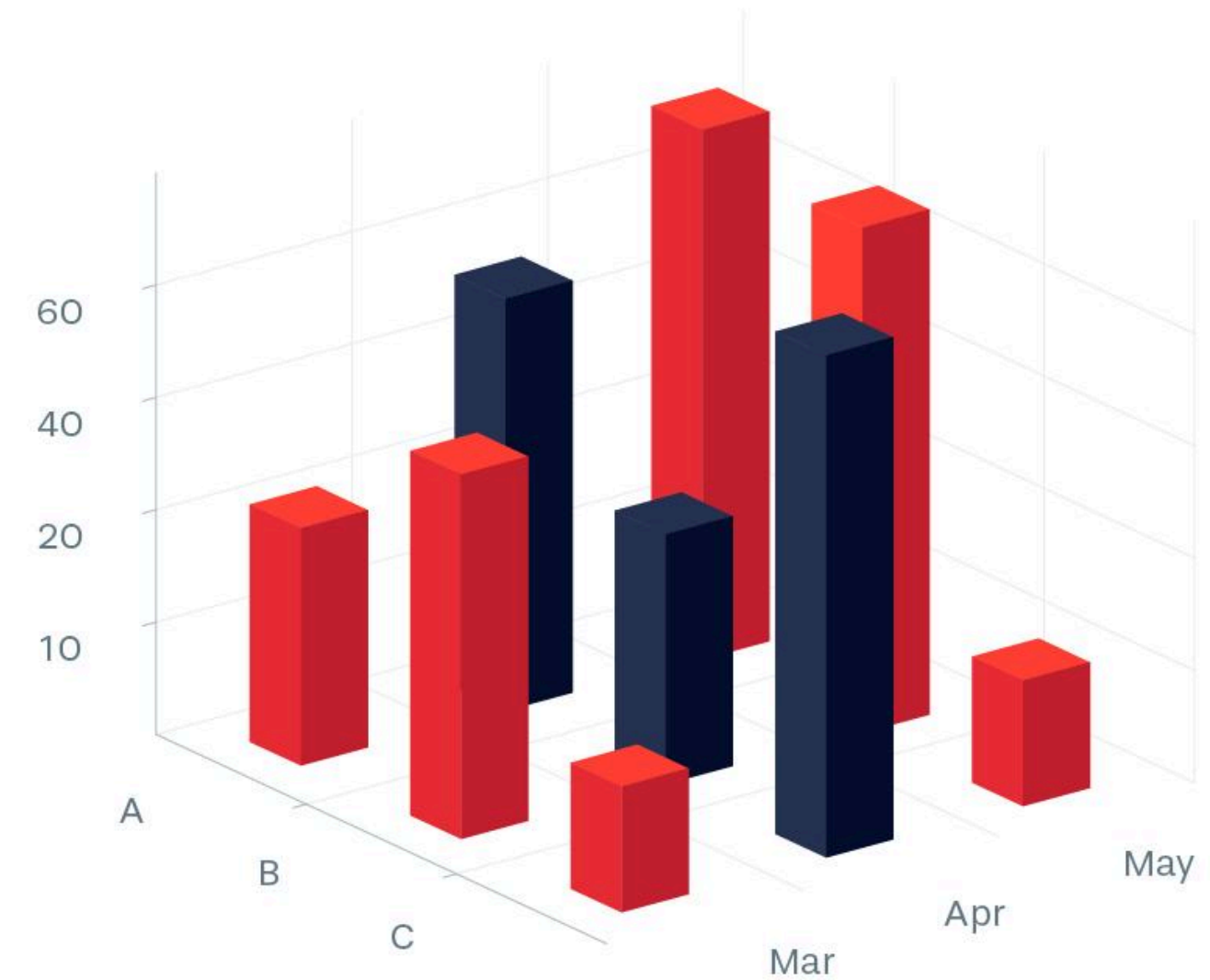
Source: DVP



Avoid Weird Encodings

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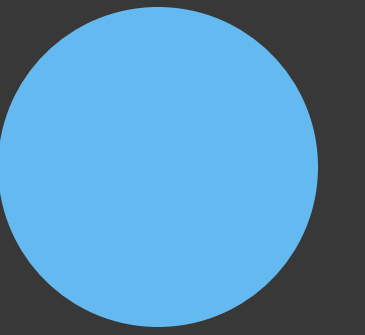
Source: DVP



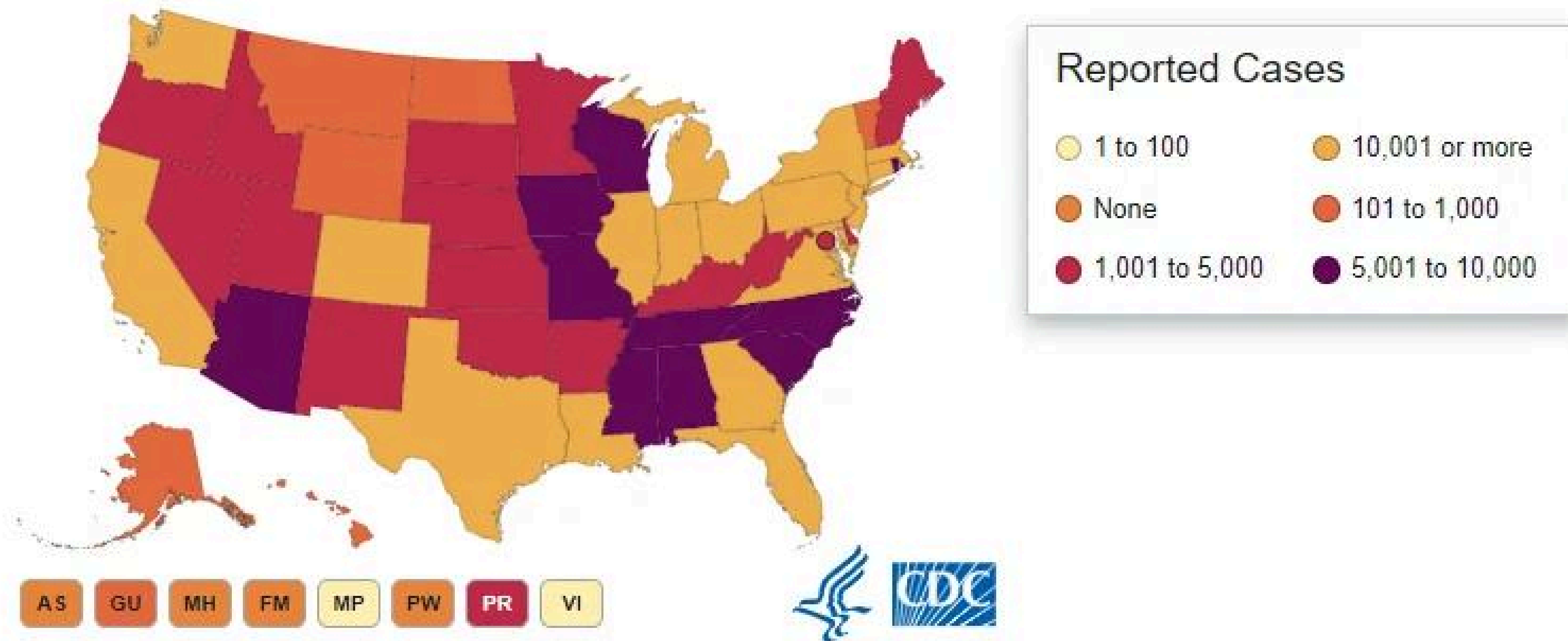
Avoid 3D

There's usually some other encoding you could use. Worst case, make multiple charts for subsets of variables.

Case Studies



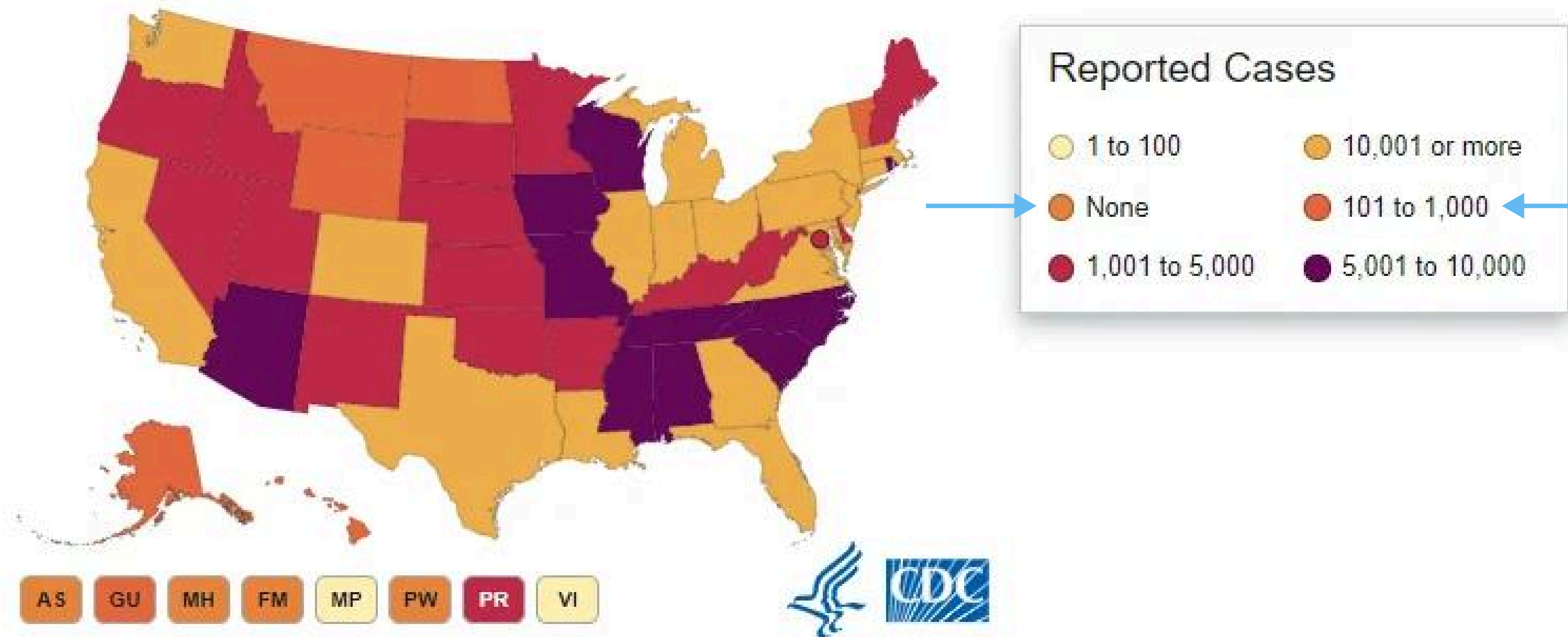
18 states report more than 10,000 cases of COVID-19.



Bad Color Encoding,
Quantitative » Categorical (CDC, 2020)

Source

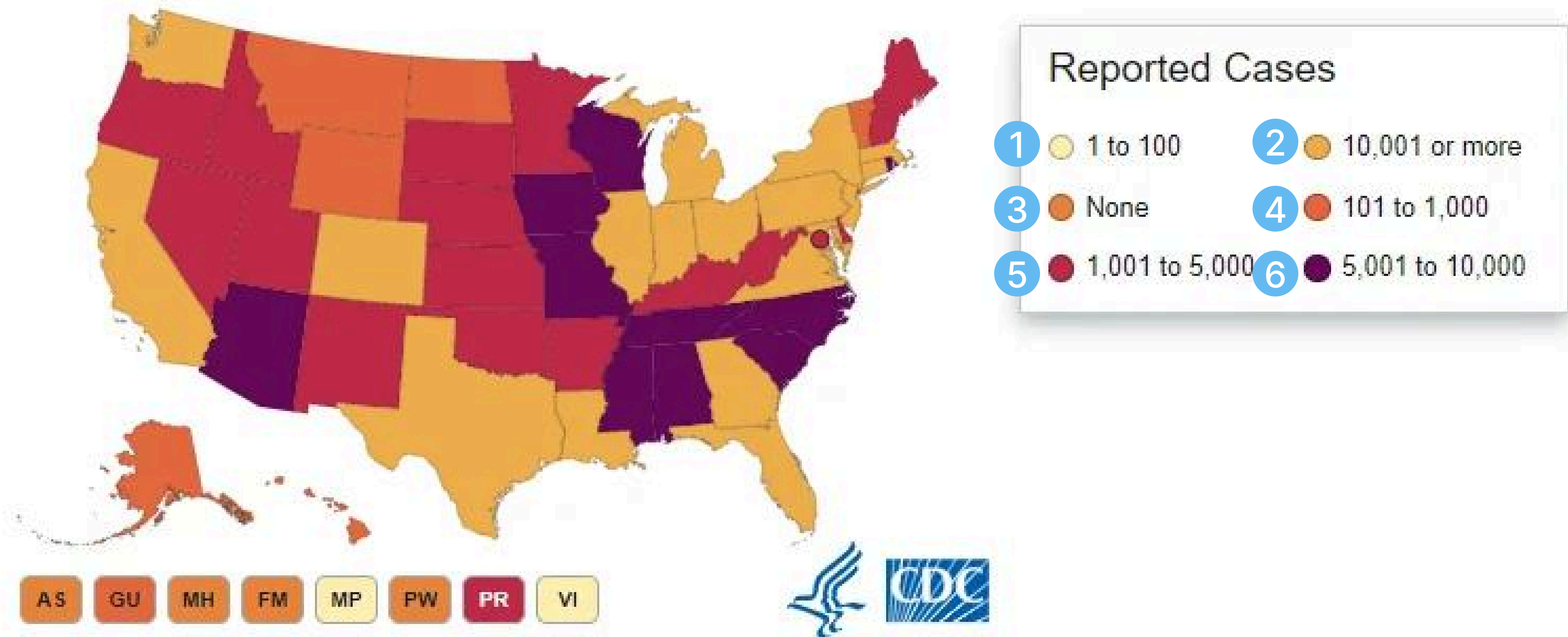
18 states report more than 10,000 cases of COVID-19.



Bad Color Encoding,
Quantitative » Categorical (CDC, 2020)

Source

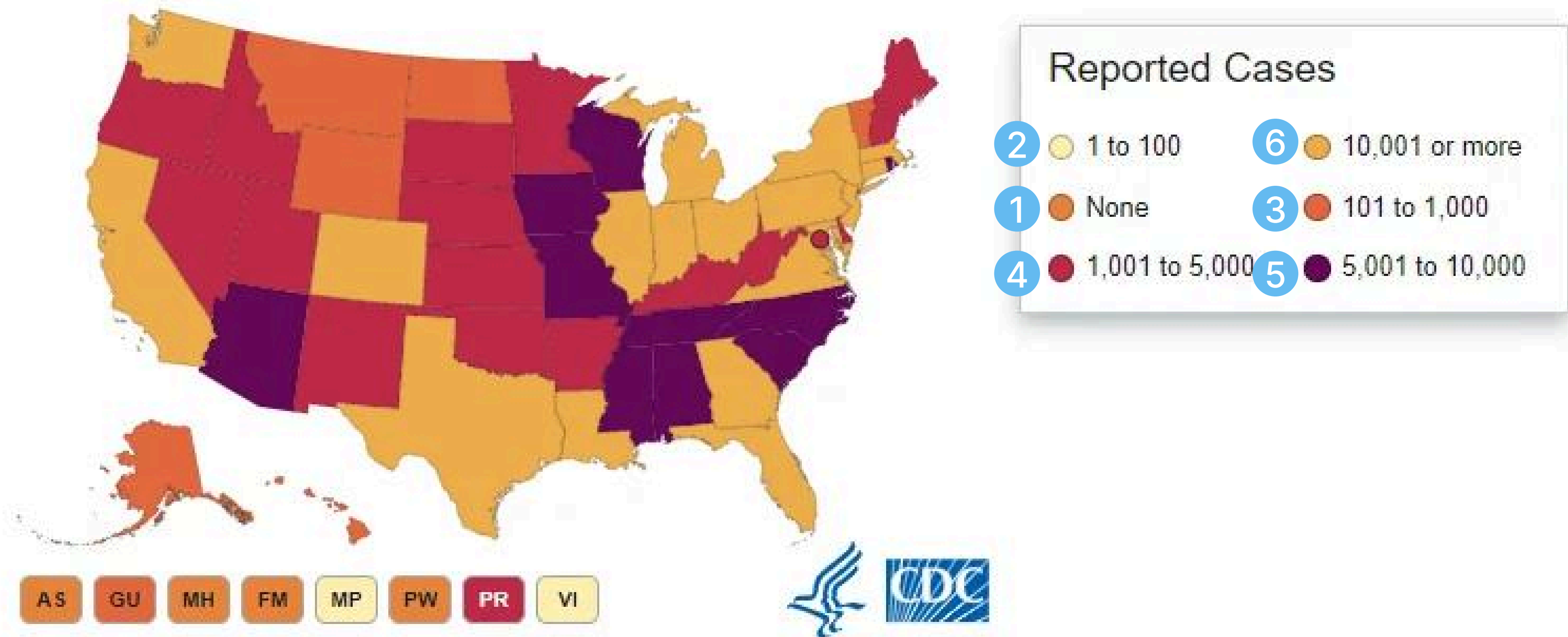
18 states report more than 10,000 cases of COVID-19.



Bad Color Encoding,
Quantitative » Categorical (CDC, 2020)

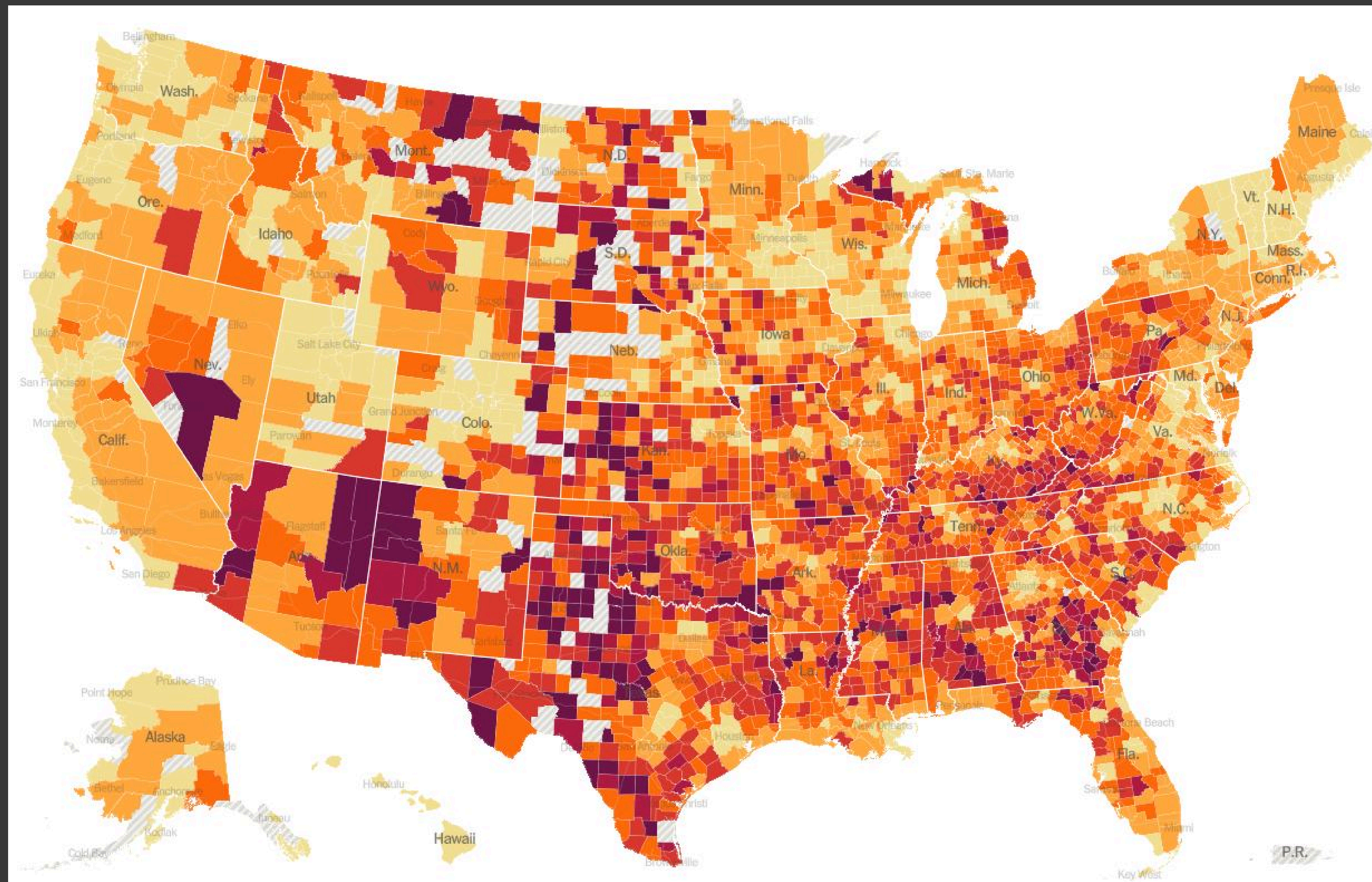
Source

18 states report more than 10,000 cases of COVID-19.



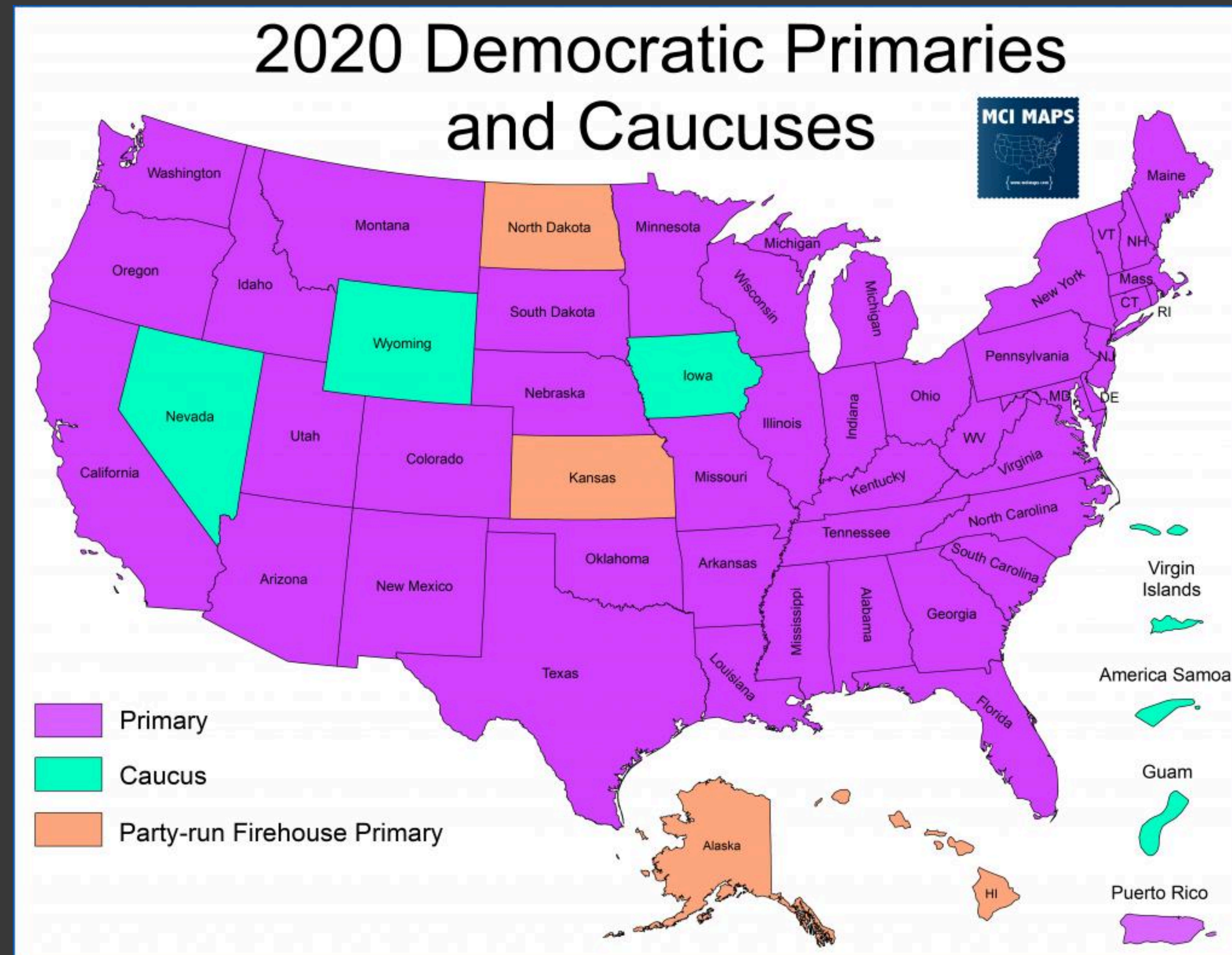
Bad Color Encoding,
Quantitative » Categorical (CDC, 2020)

Source



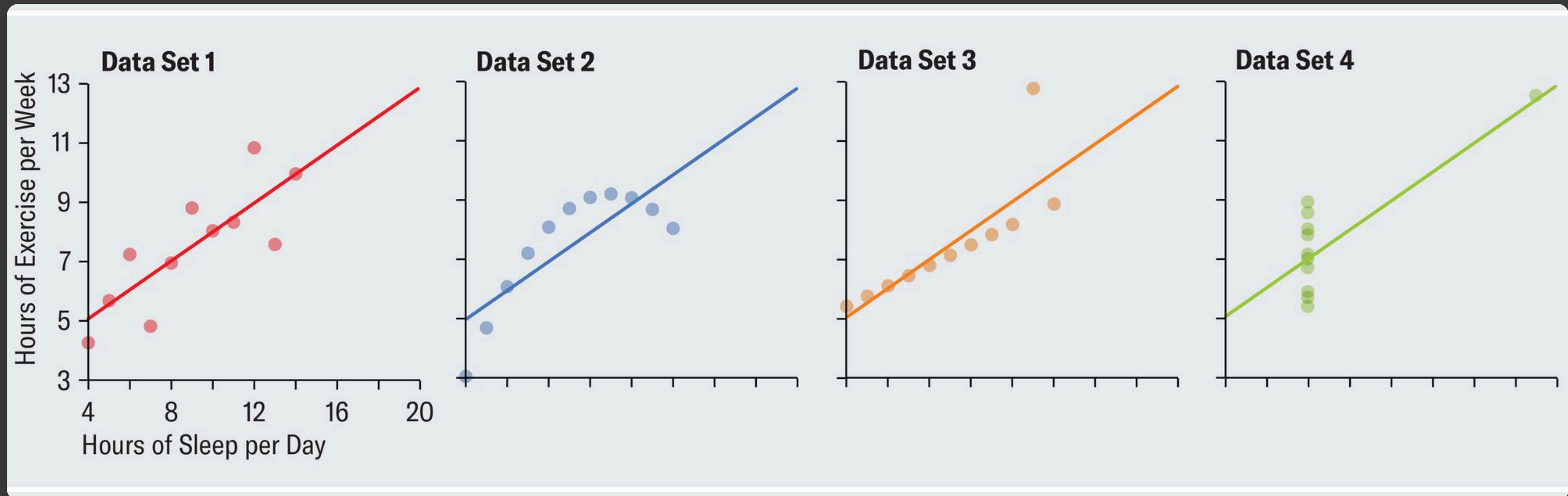
All-Time COVID Deaths (NYTimes, 2023)

Source



Primaries vs. Caucuses (MCI Maps, 2020)

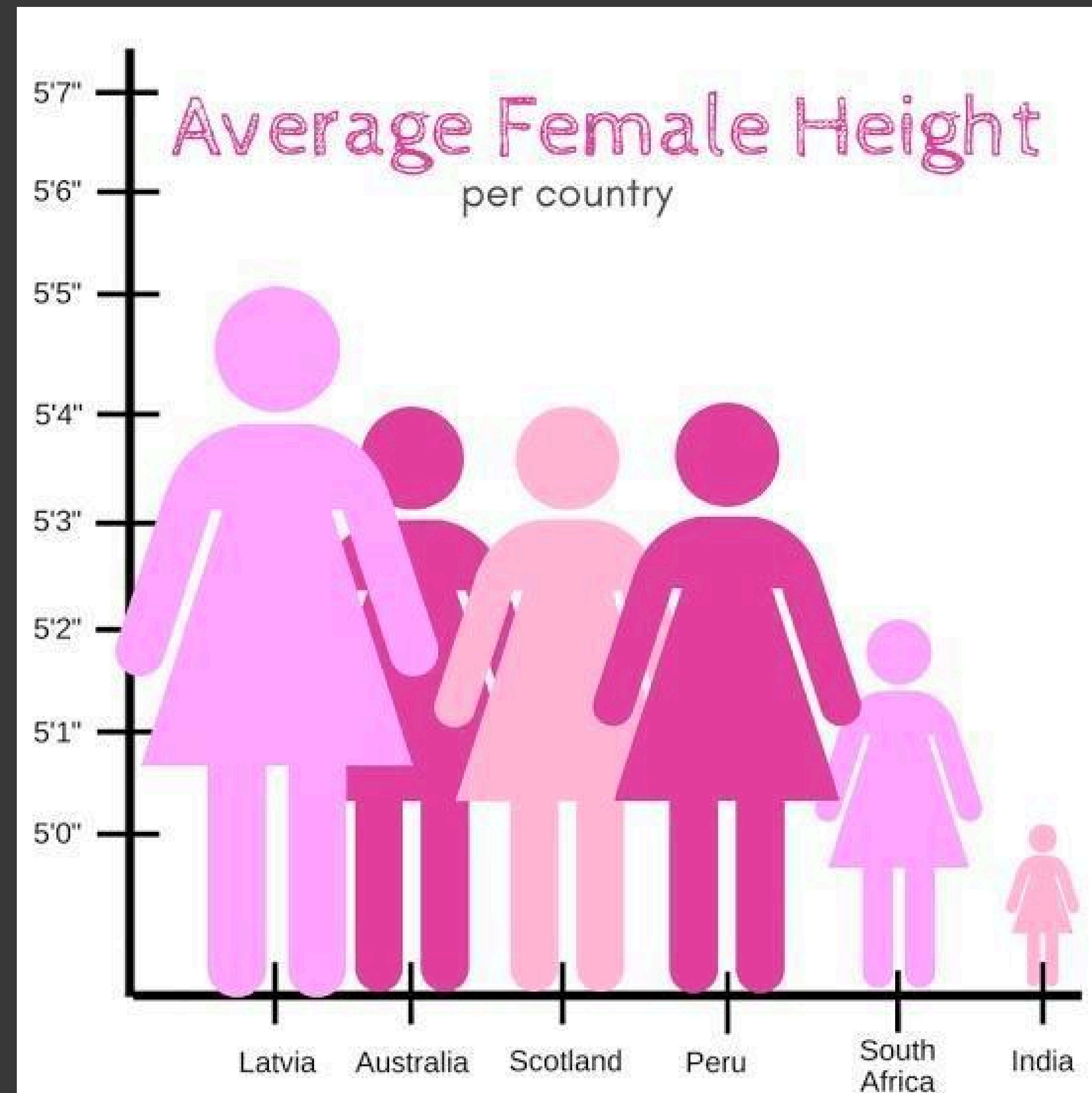
Source



Hours of exercise per week: average, 7.5; standard deviation, 2.03
Hours of sleep per day: average, 9; standard deviation, 3.32
Correlation between the two: 0.816

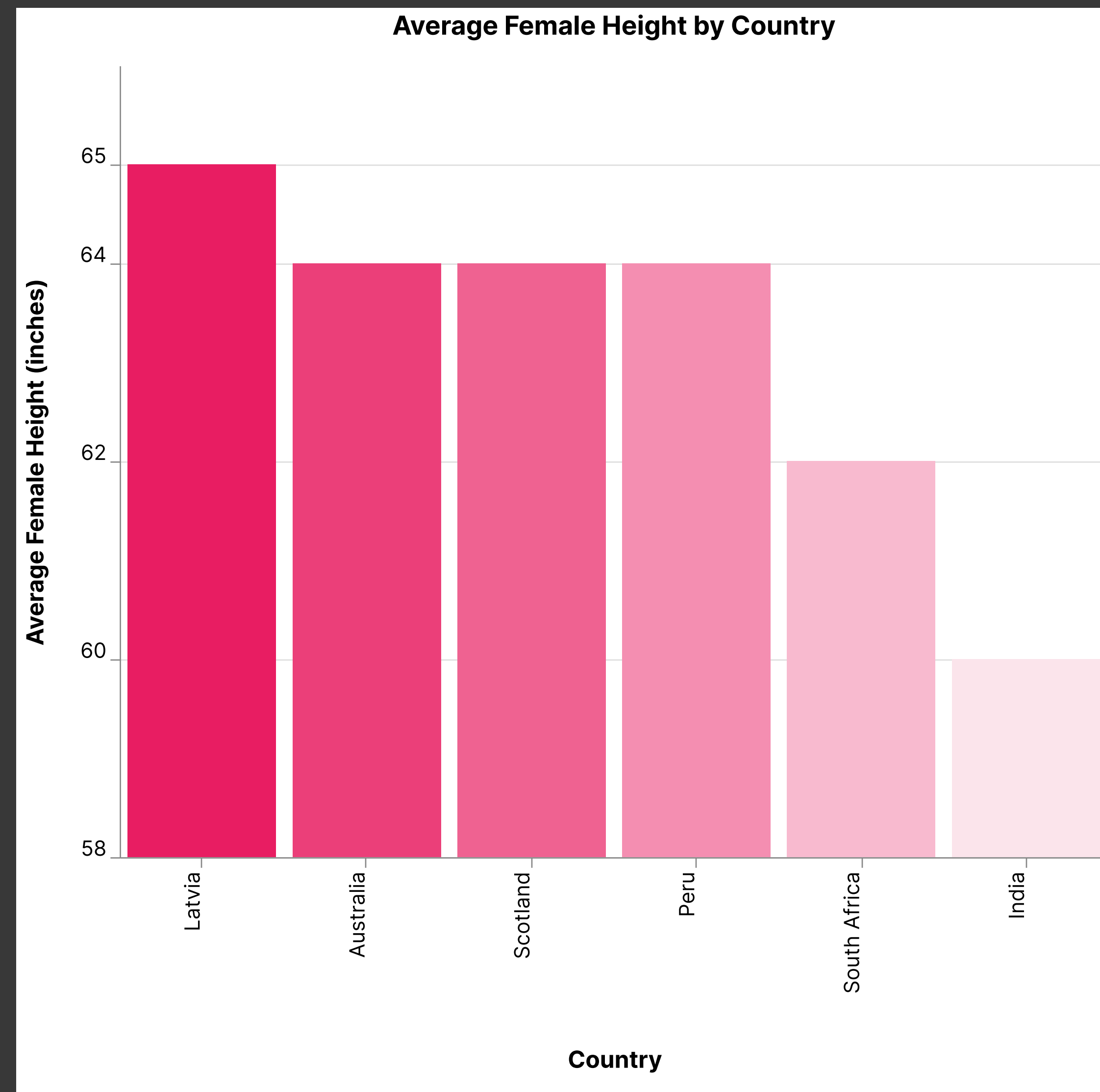
Anscombe's Quartet (Scientific American, 2023)

Source

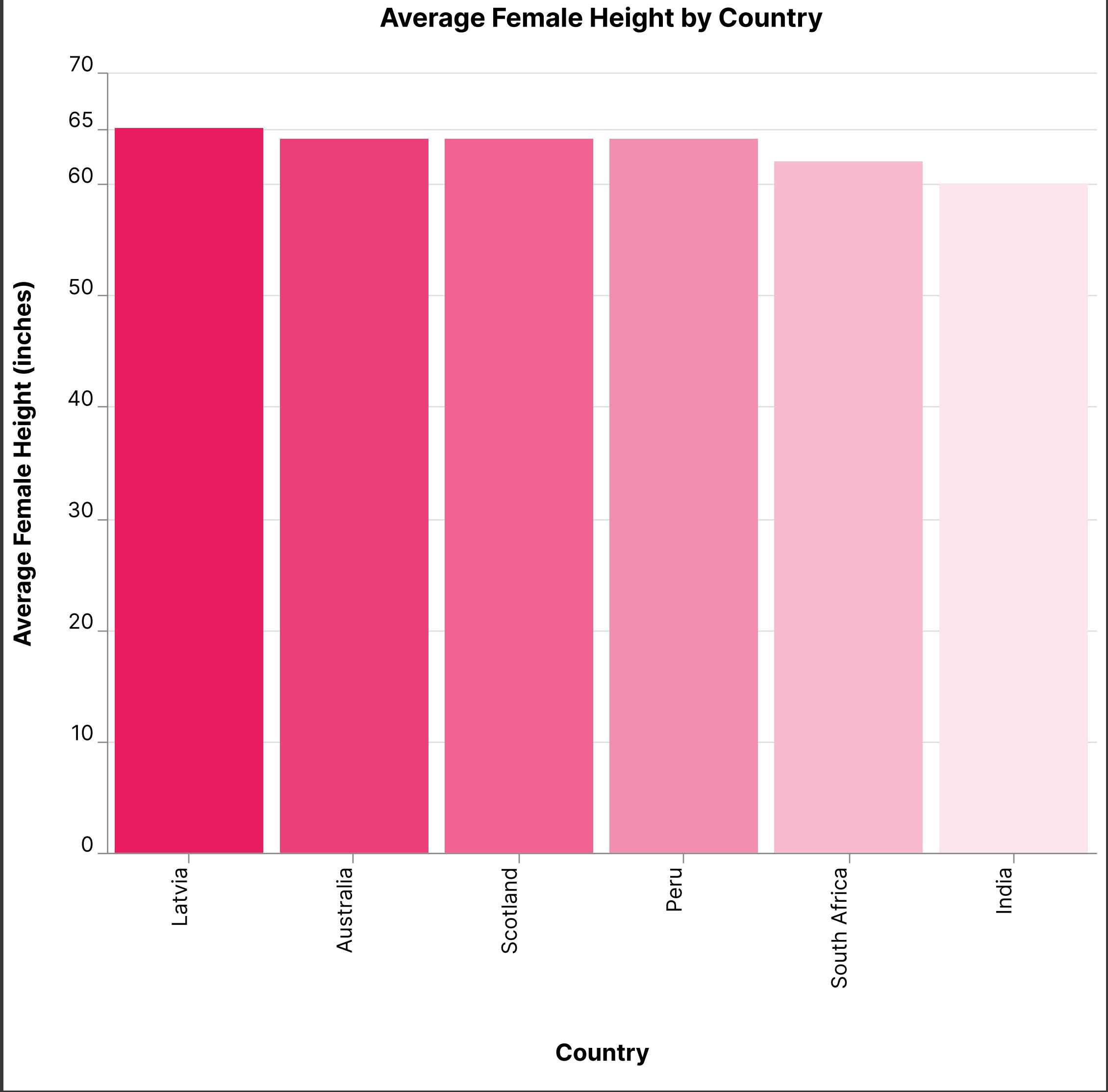


Average Female Height

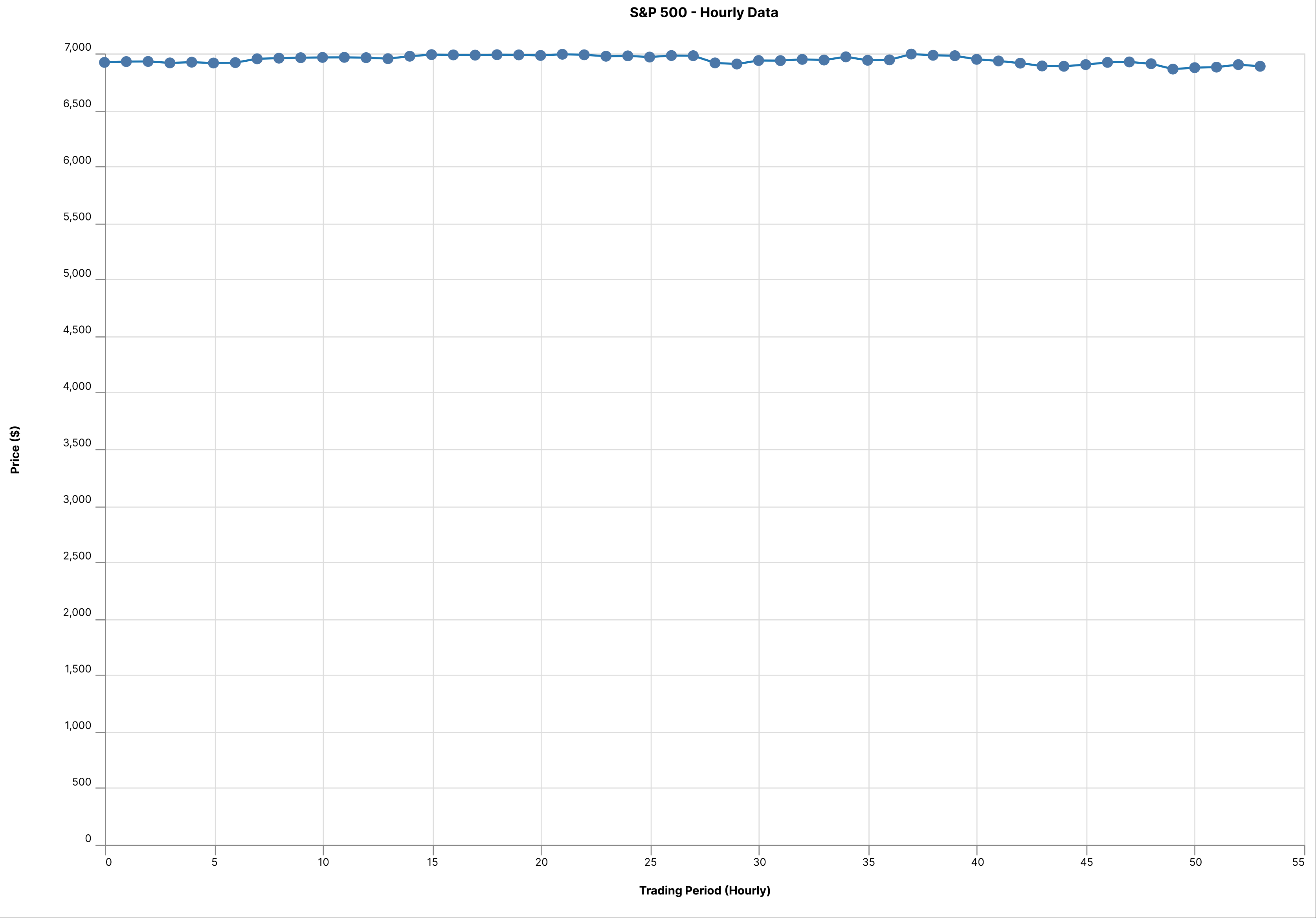
Source



Average Female Height



Average Female Height



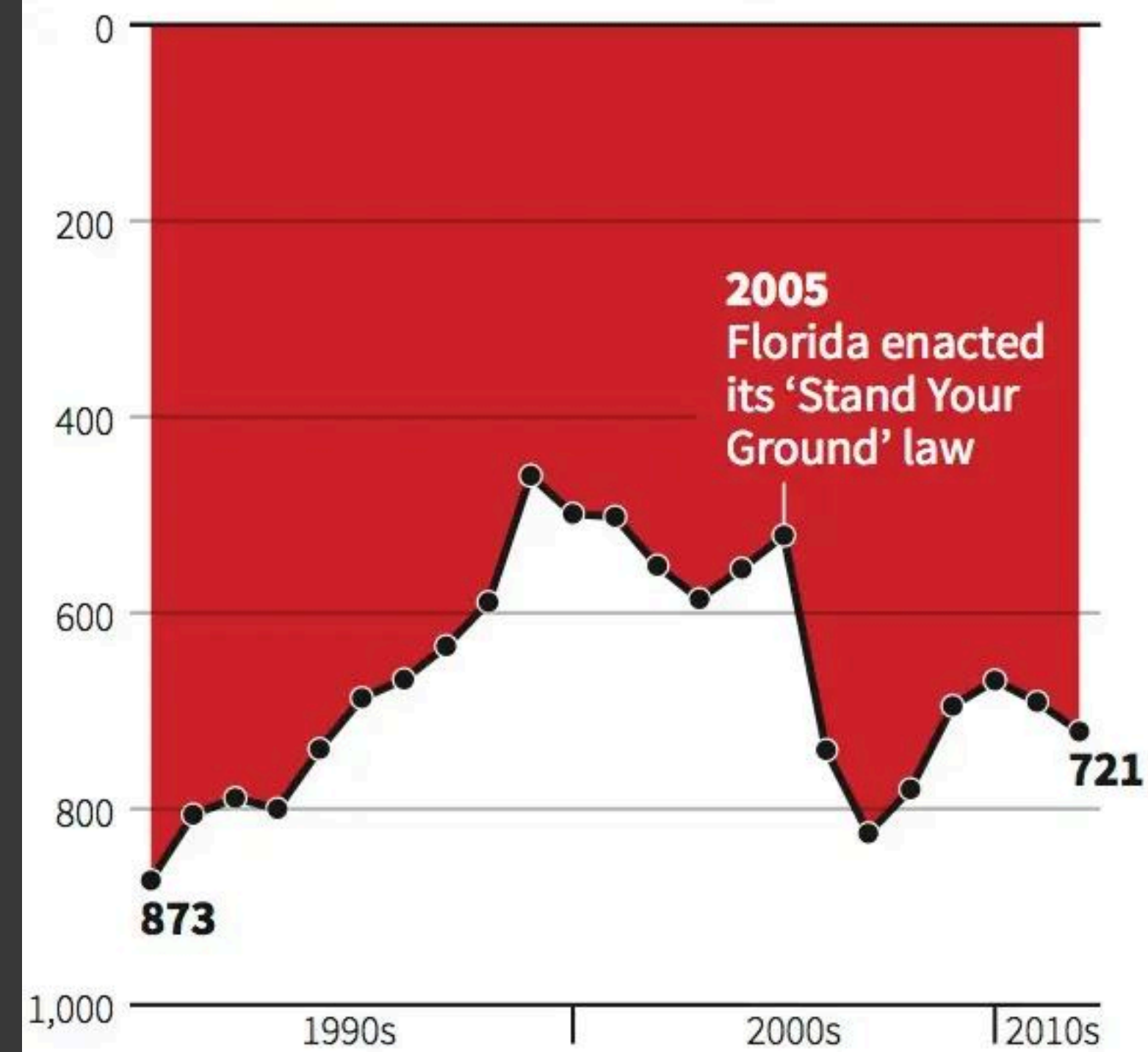
S&P 500



S&P 500

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

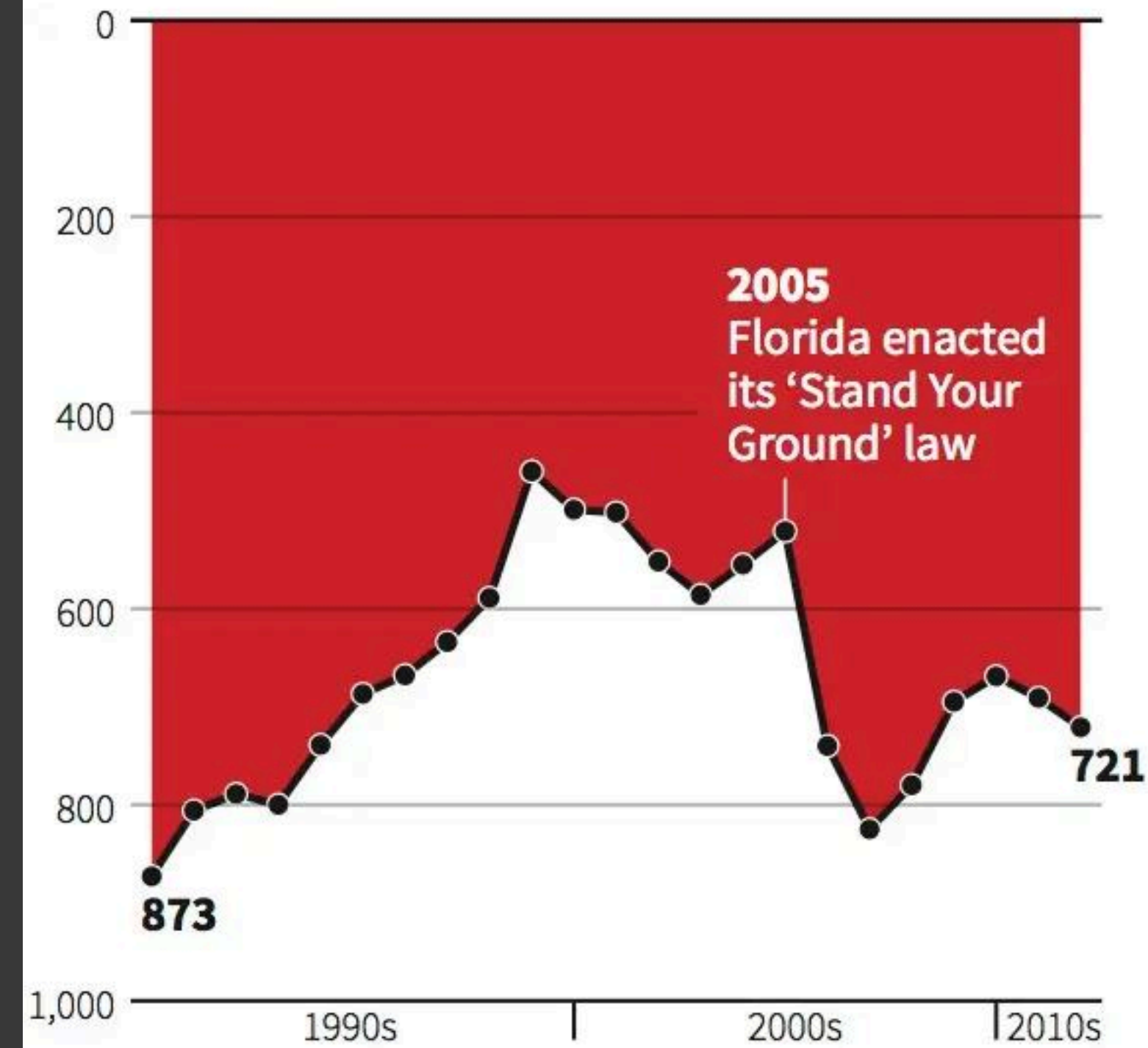
REUTERS

Conventions > Art

Source: Reuters

Gun deaths in Florida

Number of murders committed using firearms

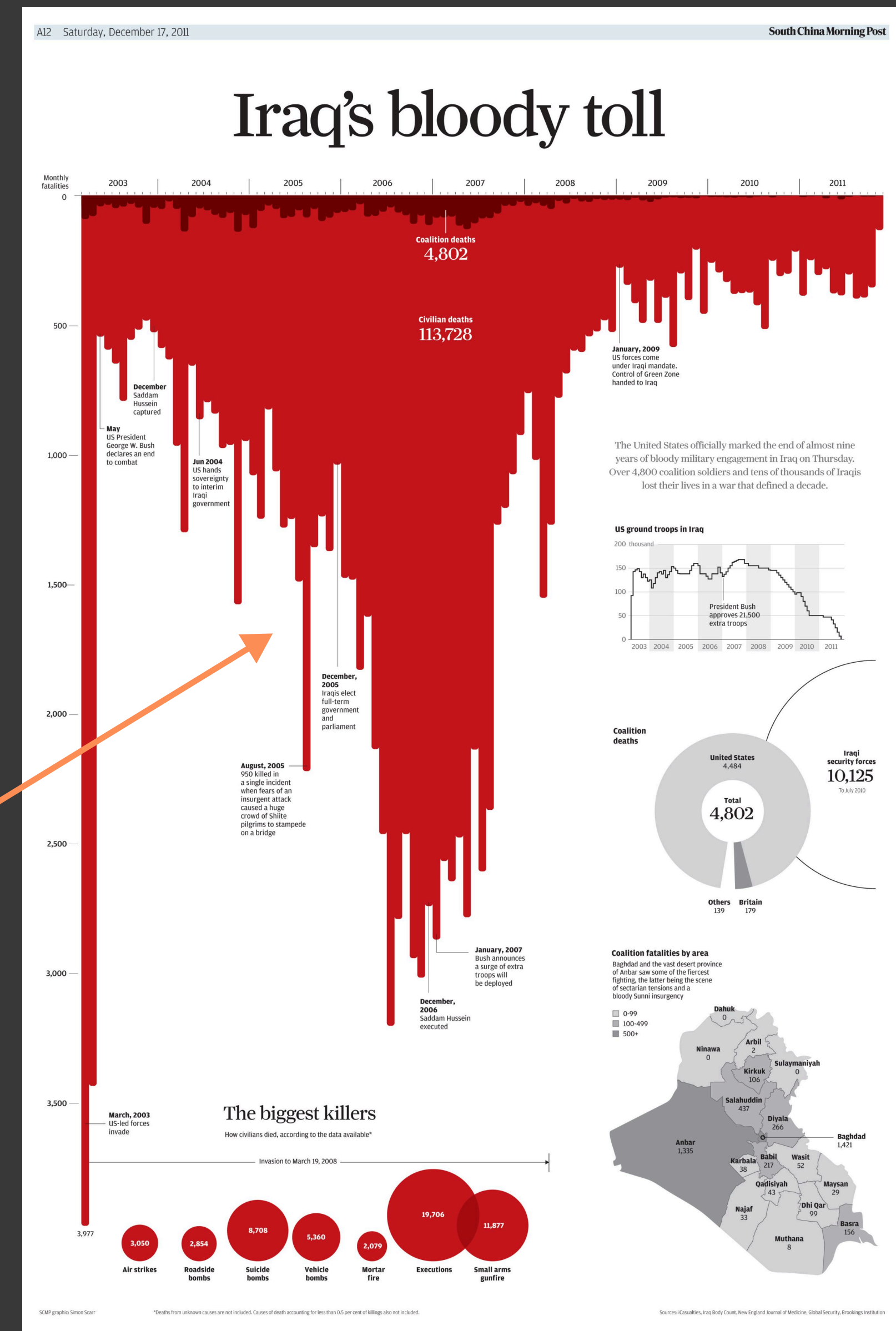


Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

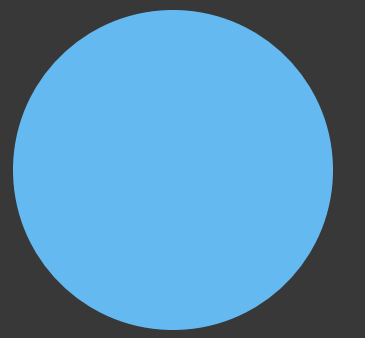
Won multiple awards!



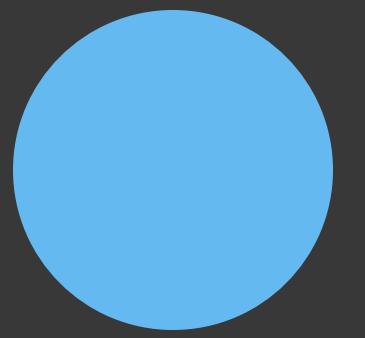
Conventions > Art

Source: Reuters,
SCMP

A Personal Tale



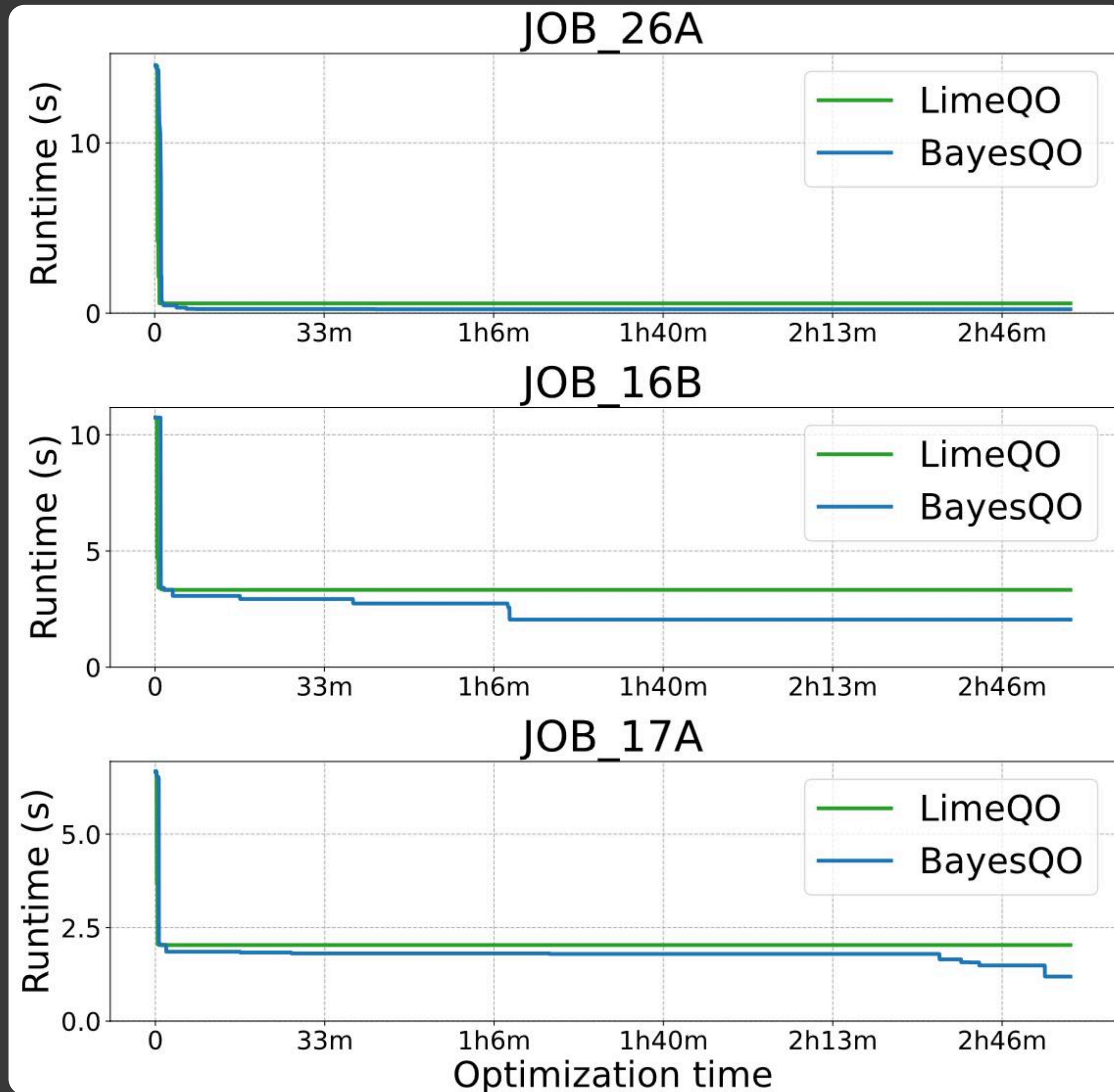
Visualization is *narrative*



Visualization is a form of storytelling.

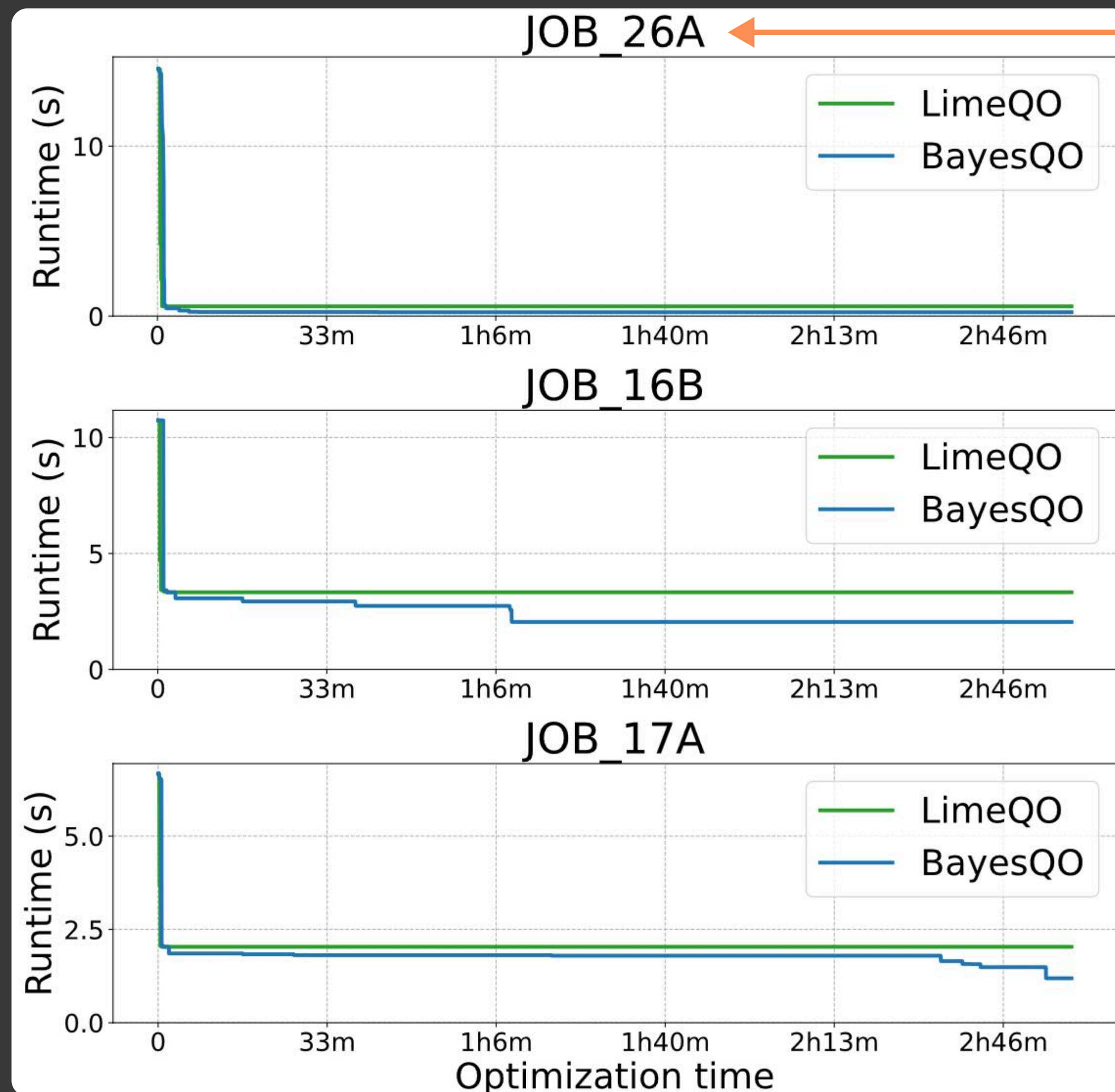
Get the basics right so that the viewer can at least “read” your charts without thinking too hard.

Case Study: My Own Research Paper



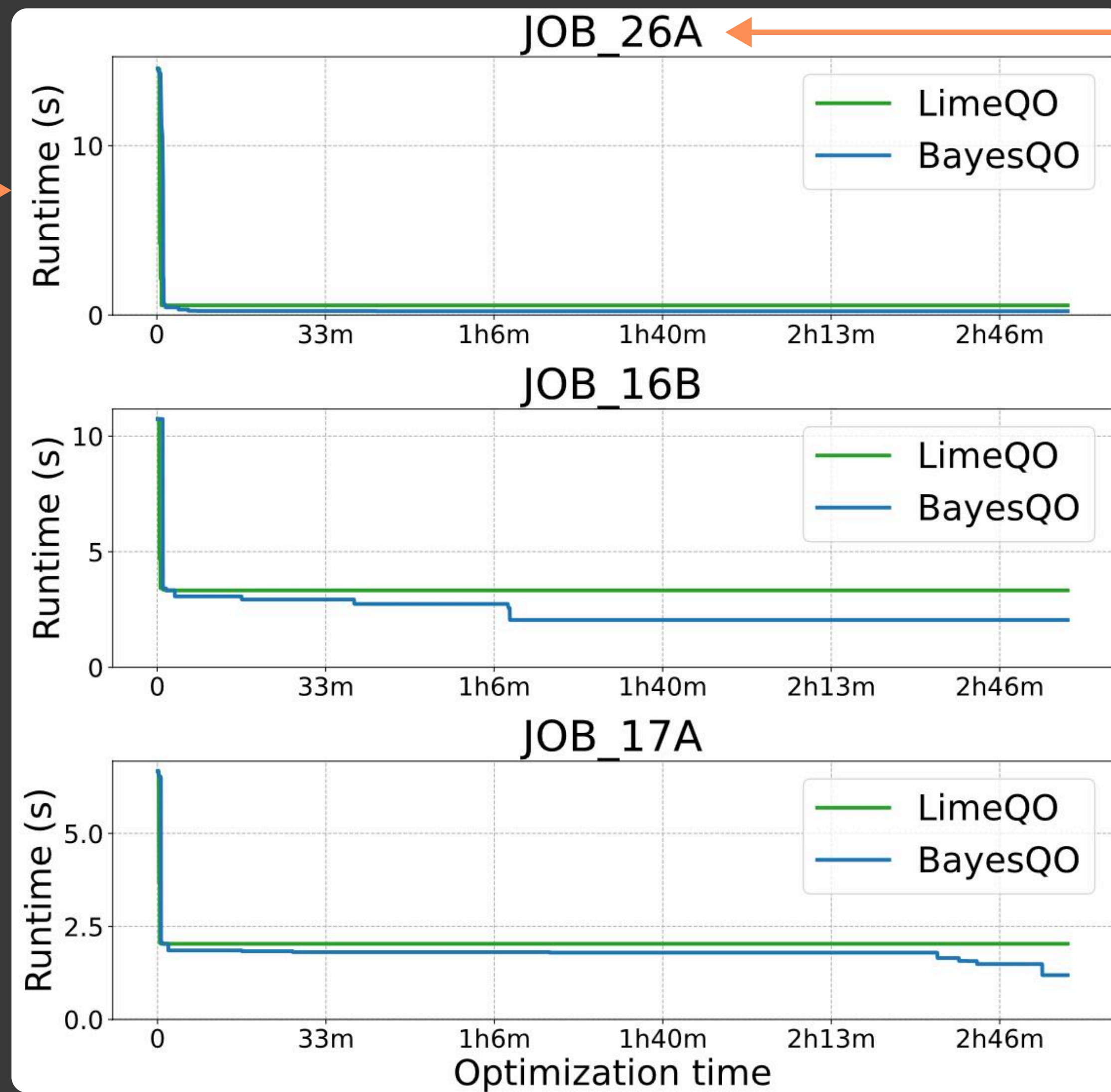
Fastest query runtimes generated by our method (BayesQO) vs. a baseline (LimeQO). Lower is better.

Case Study: My Own Research Paper



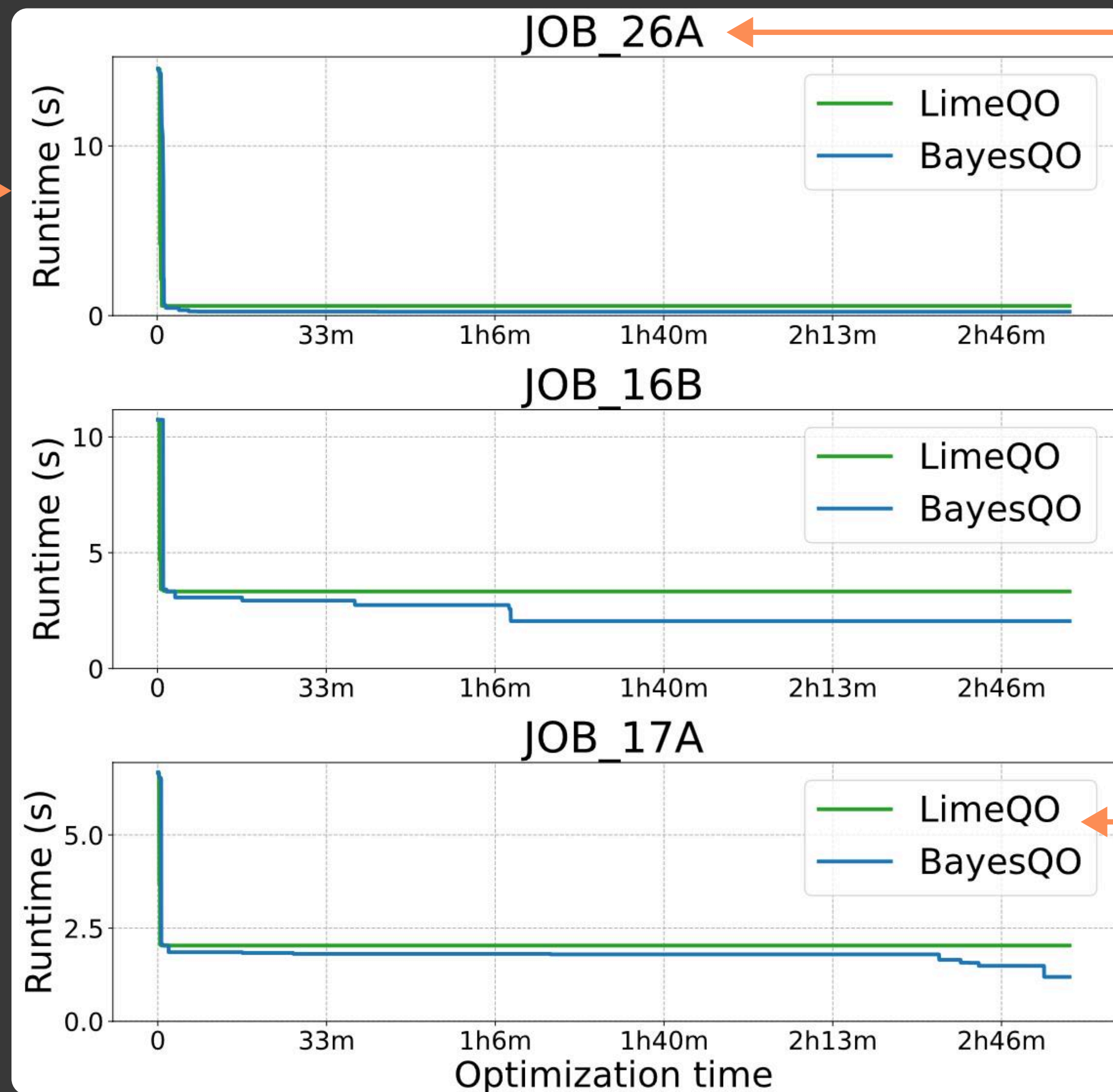
✓ Titles

Case Study: My Own Research Paper



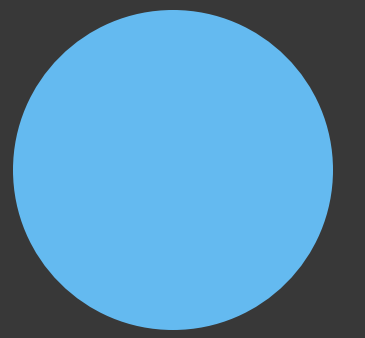
- ✓ Titles
- ✓ Axis labels with units

Case Study: My Own Research Paper



- ✓ Titles
- ✓ Axis labels with units
- ✓ Legend
- ✓ Legible font size

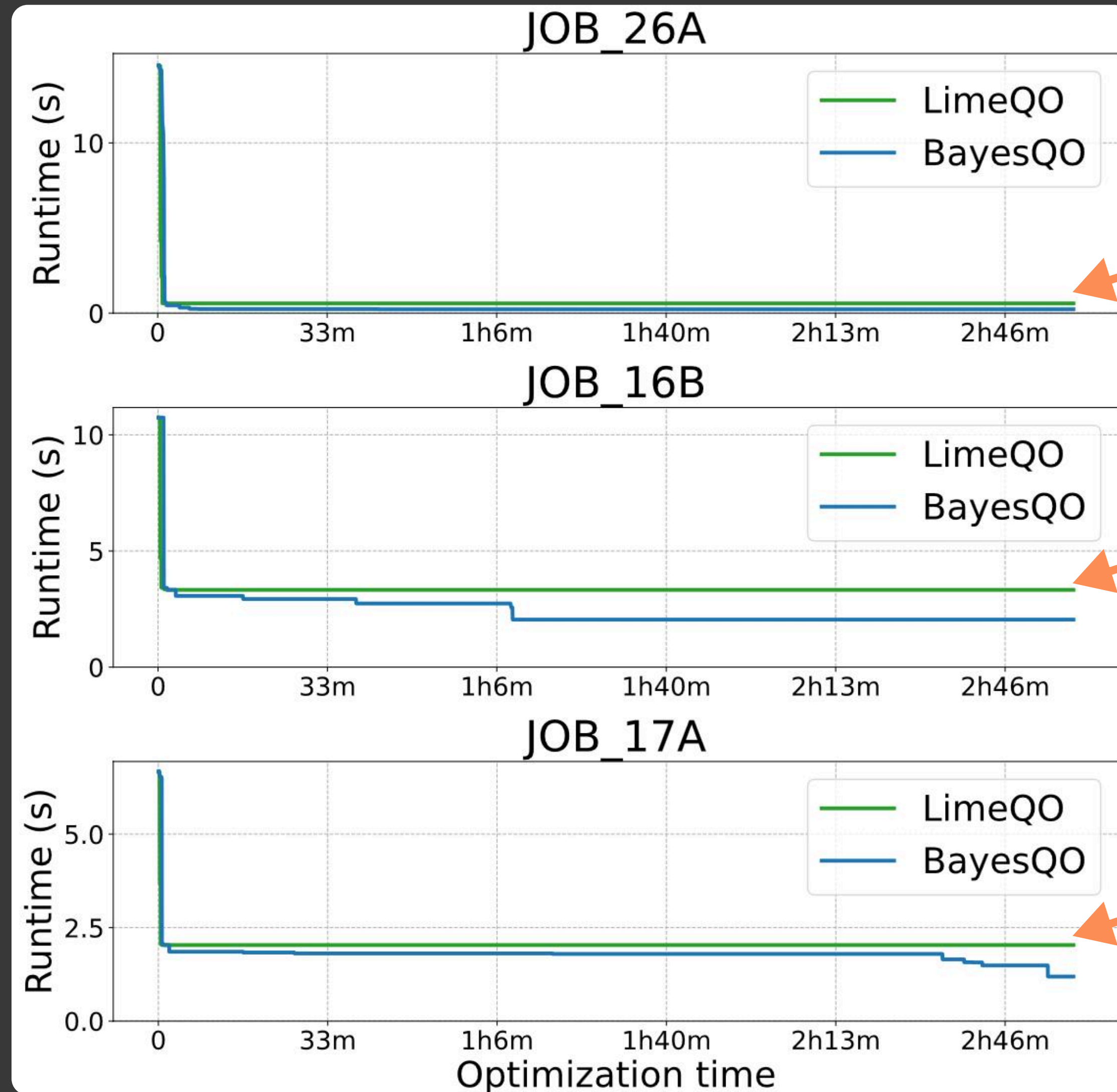
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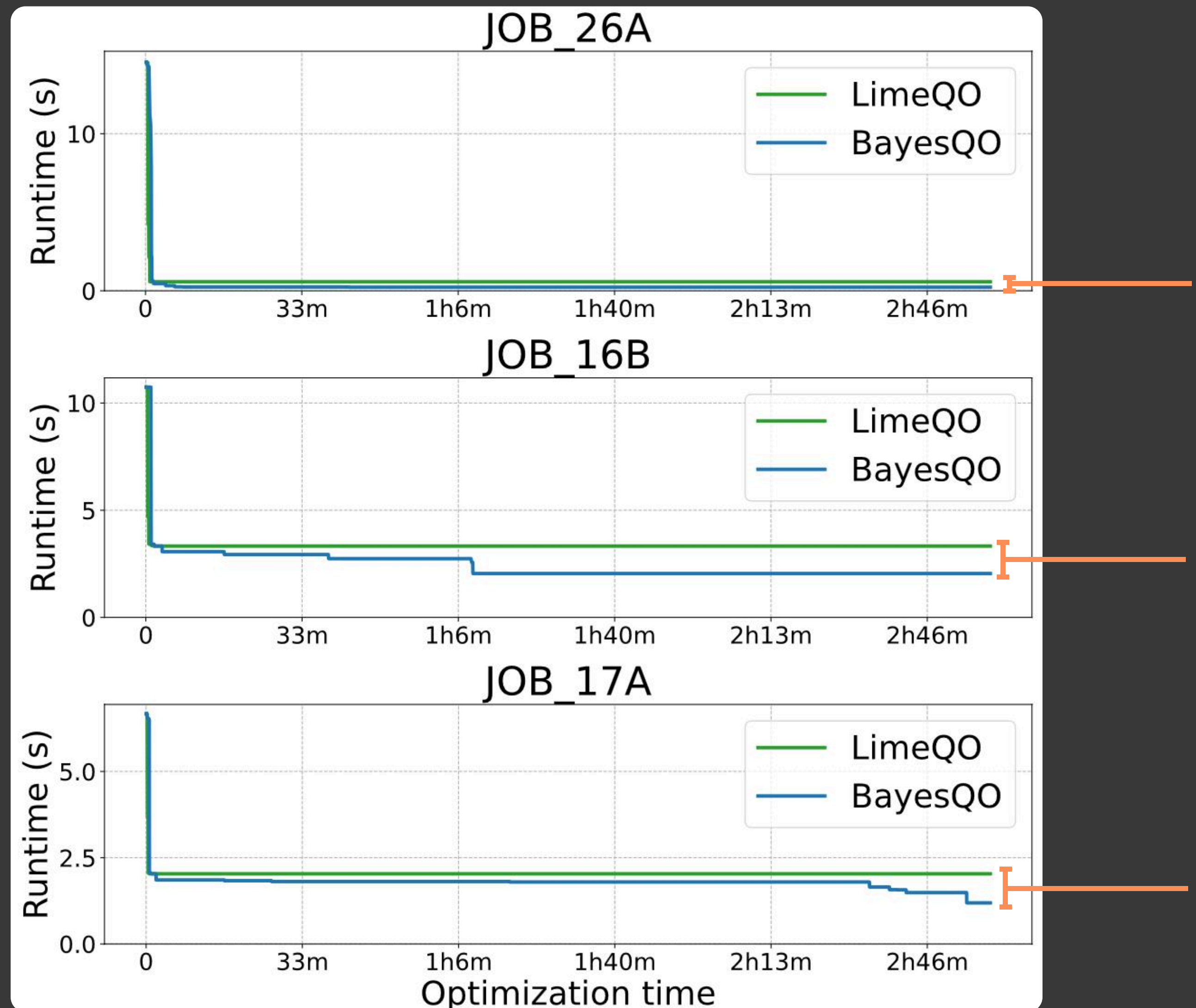
But what you intend and think is salient is not the same
as what the viewer will think is salient!

Case Study: My Own Research Paper



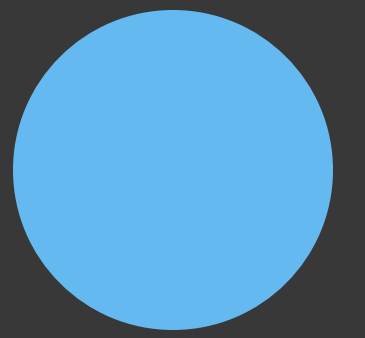
Us: “Look, the green line bottoms out and then doesn’t get any better!”

Case Study: My Own Research Paper



Them: “Wow, your method isn’t that much better!”

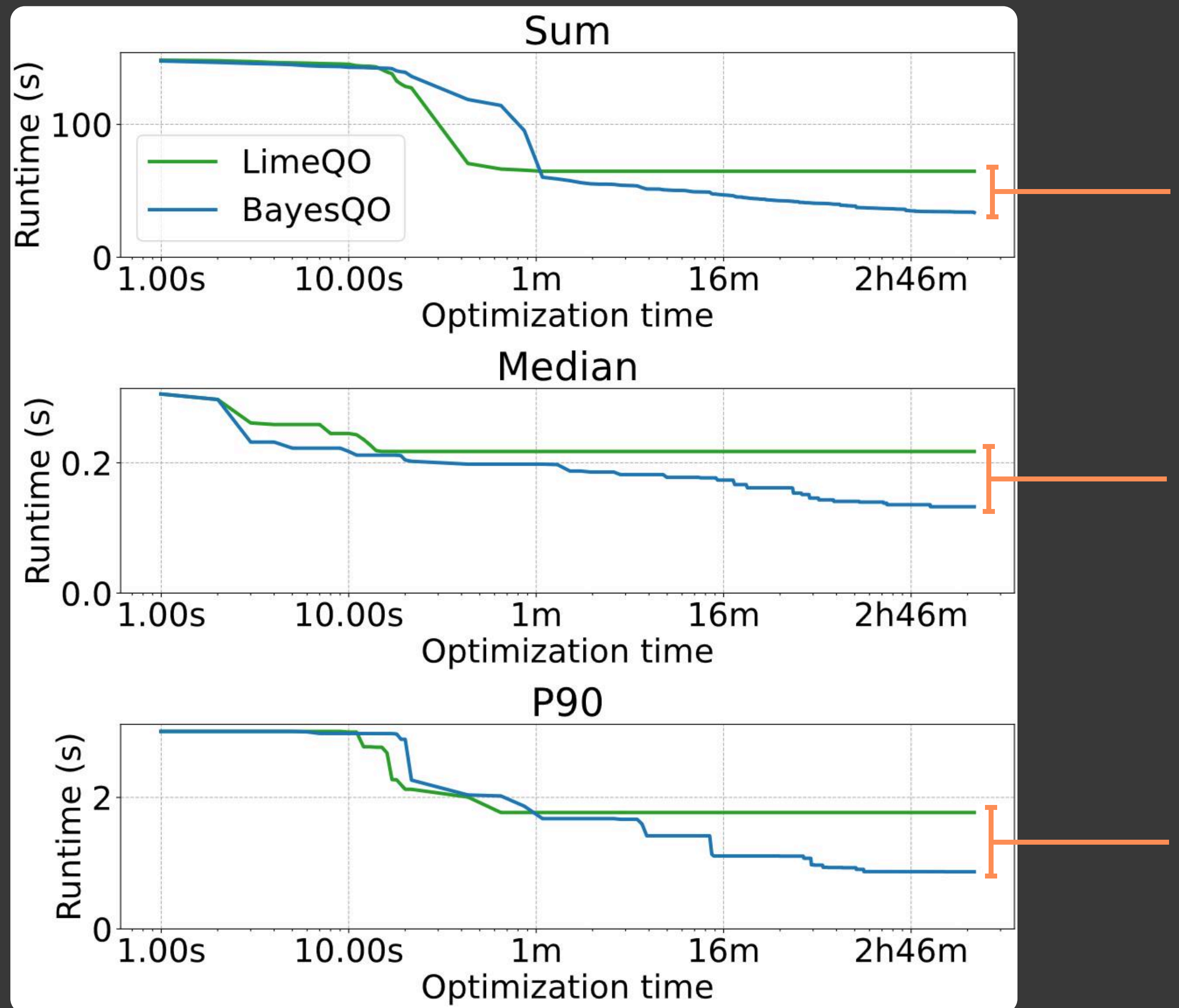
Visualization is *narrative*



Visualization is a form of storytelling.

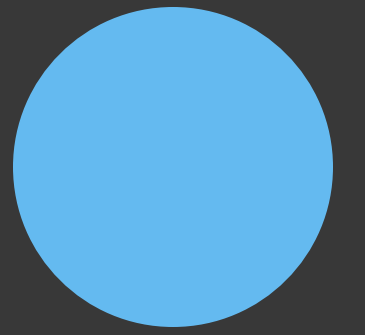
What you *choose to say* is critical.

Case Study: My Own Research Paper



Us: “Look, the gaps are bigger now!”

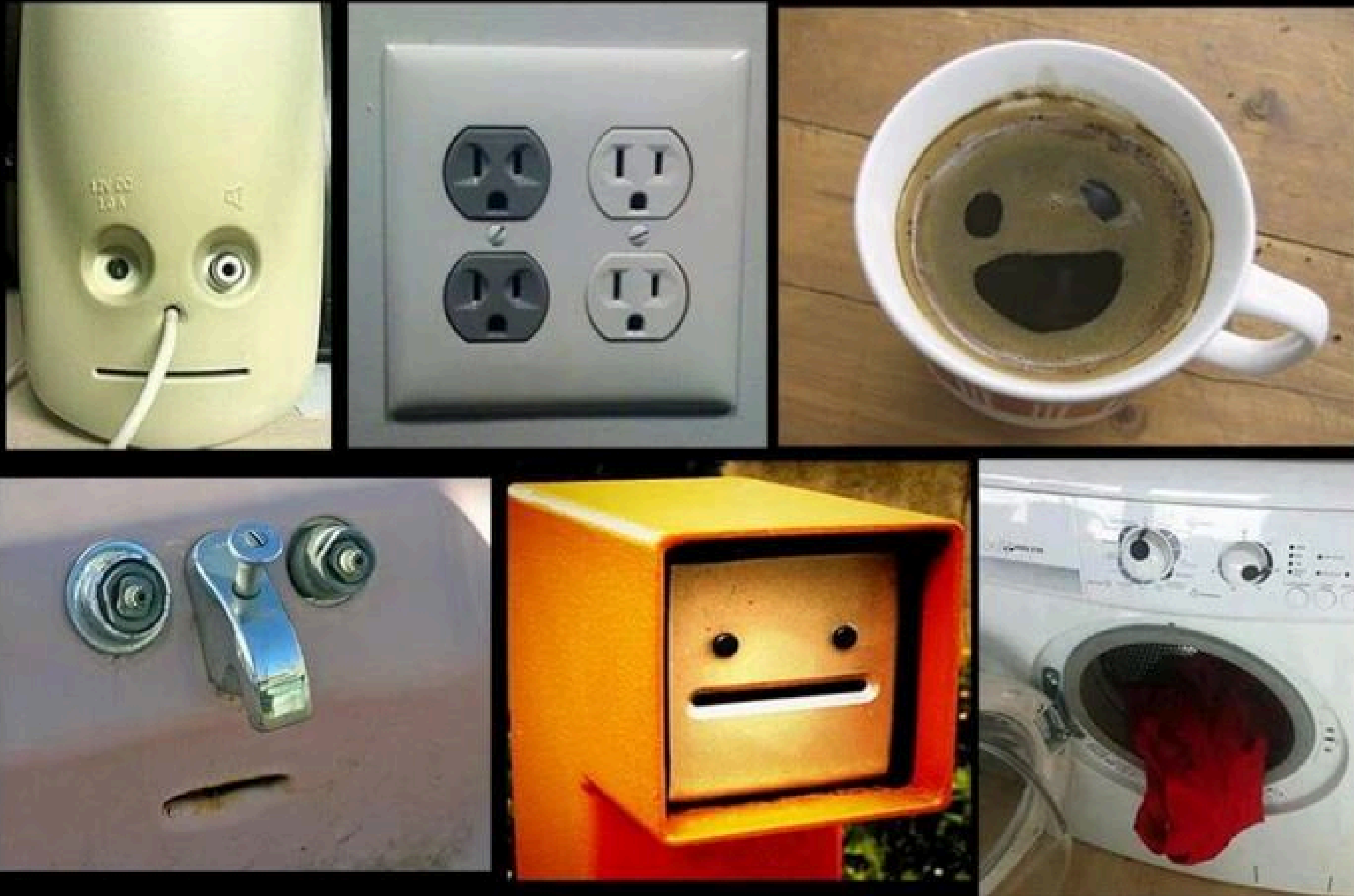
The Takeaway



1. Think carefully about all of the things that you *could say* using the data that you have.
2. Create a visualization adhering to the best practices.
3. Think carefully about *all of the possible interpretations* a viewer could take away from your visualization.

Telling Stories Too well





Pareidolia

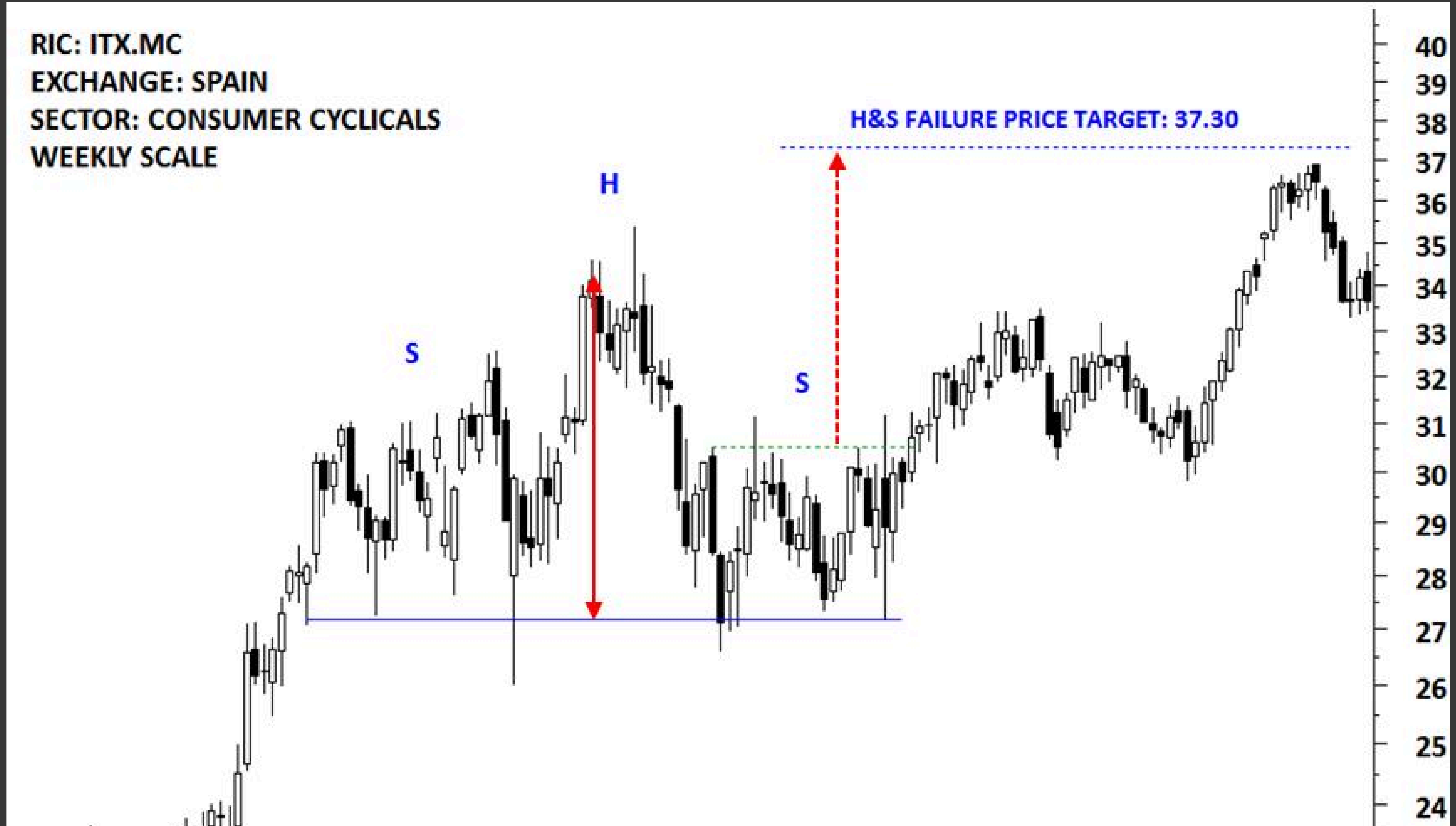
Source



“Technical Analysis”

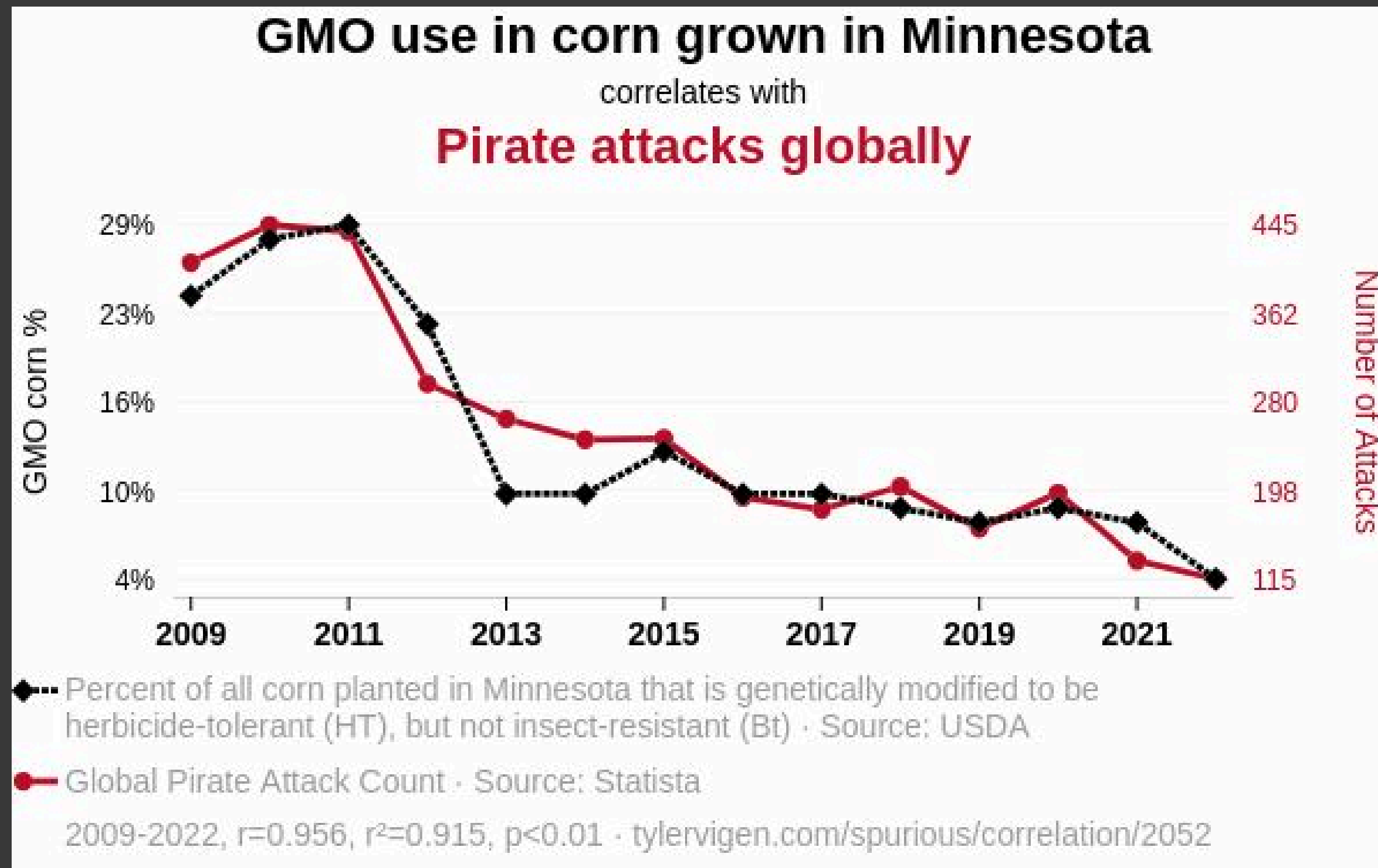
Source: [TrendSpider](#)

RIC: ITX.MC
EXCHANGE: SPAIN
SECTOR: CONSUMER CYCLICALS
WEEKLY SCALE



“Technical Analysis”

Source: TechCharts



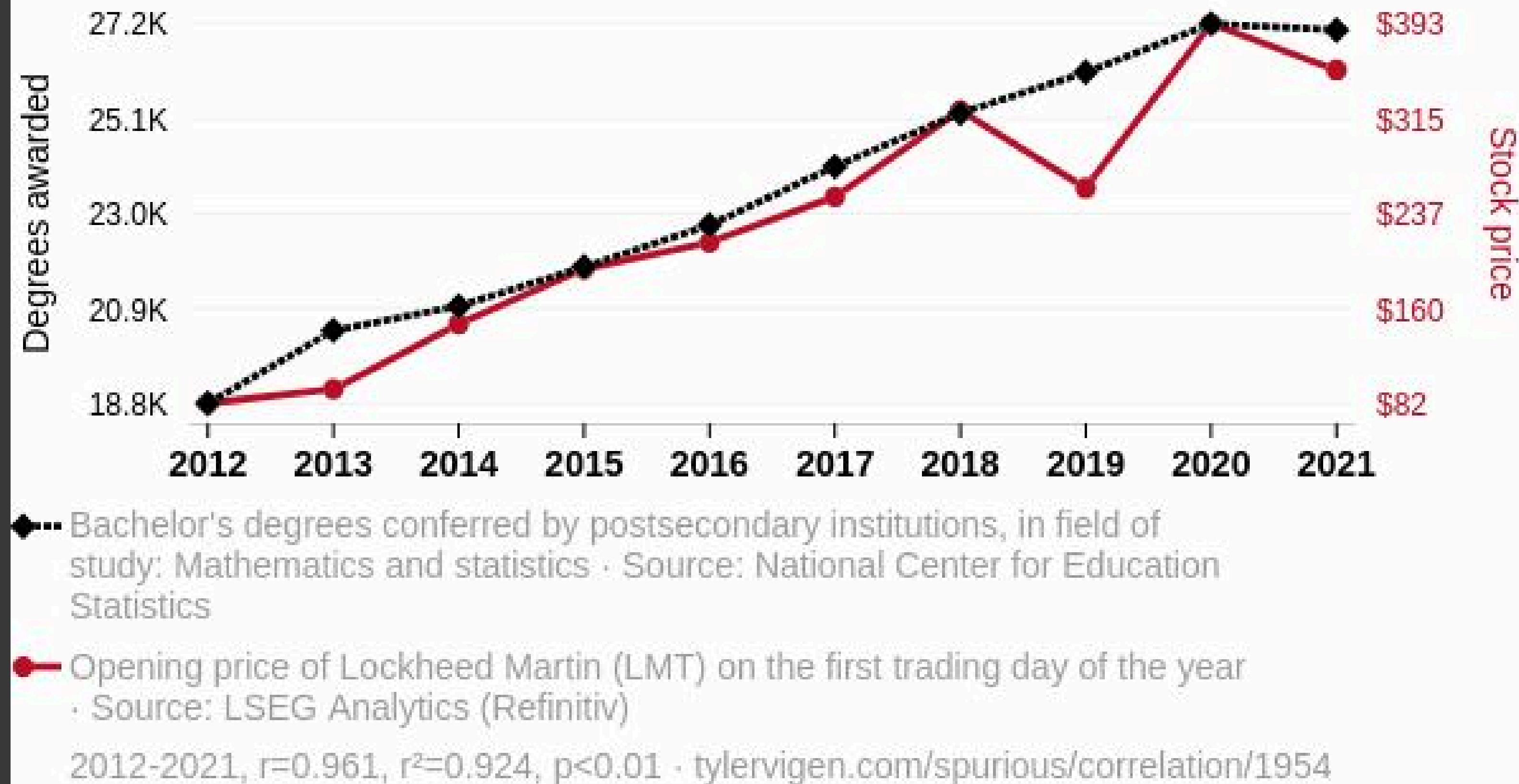
Spurious Correlations

Source: Spurious Correlations

Bachelor's degrees awarded in Mathematics and statistics

correlates with

Lockheed Martin's stock price (LMT)

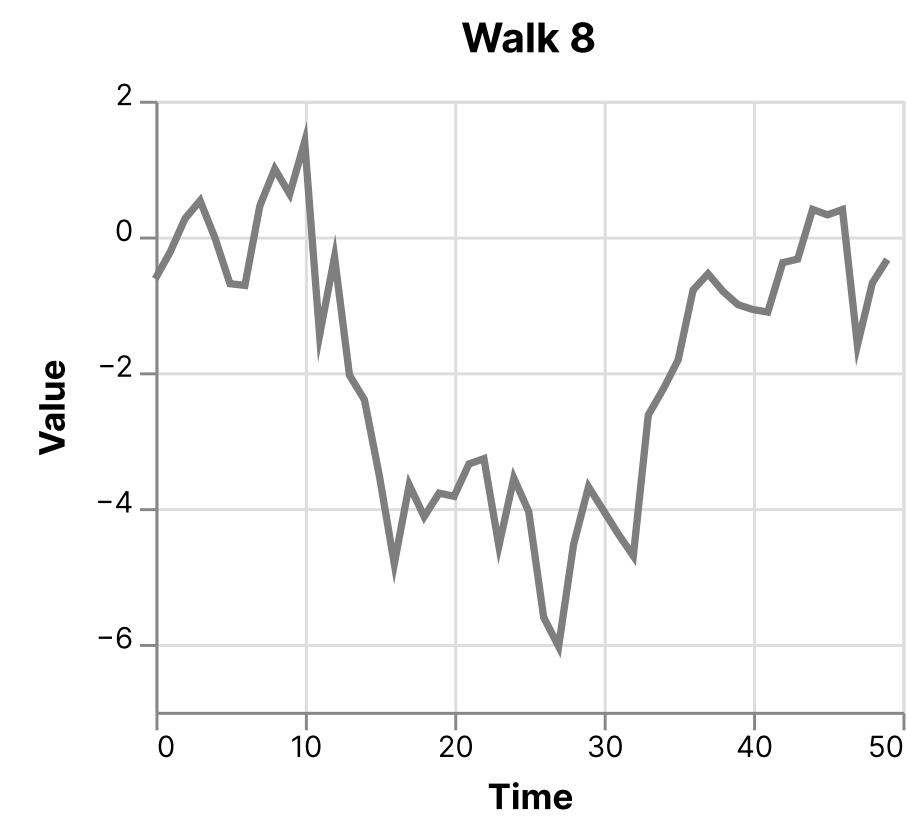
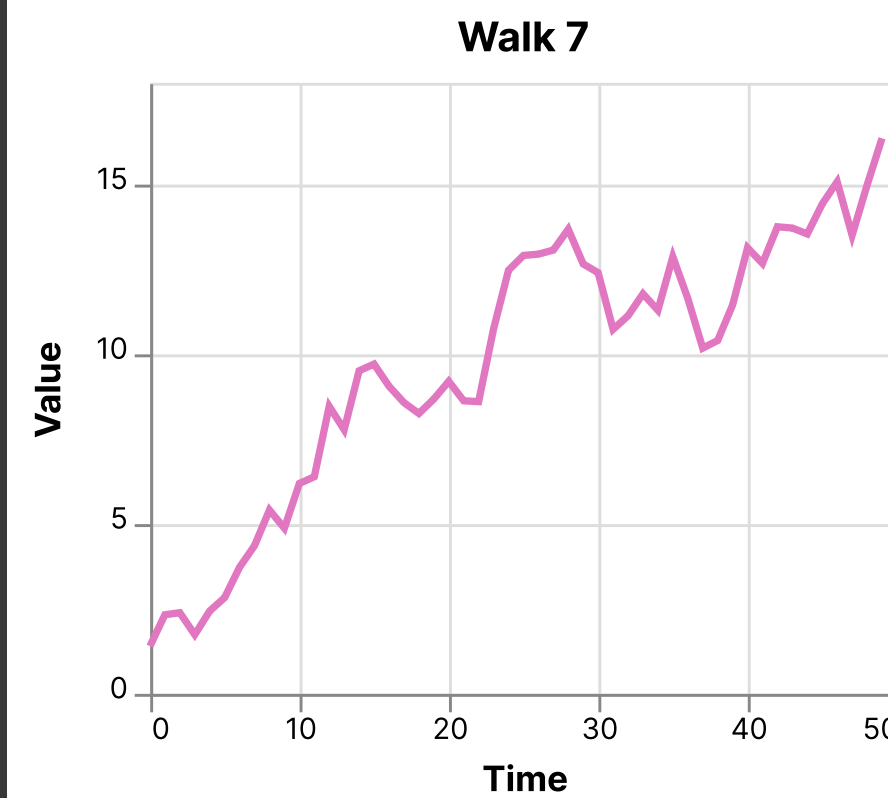
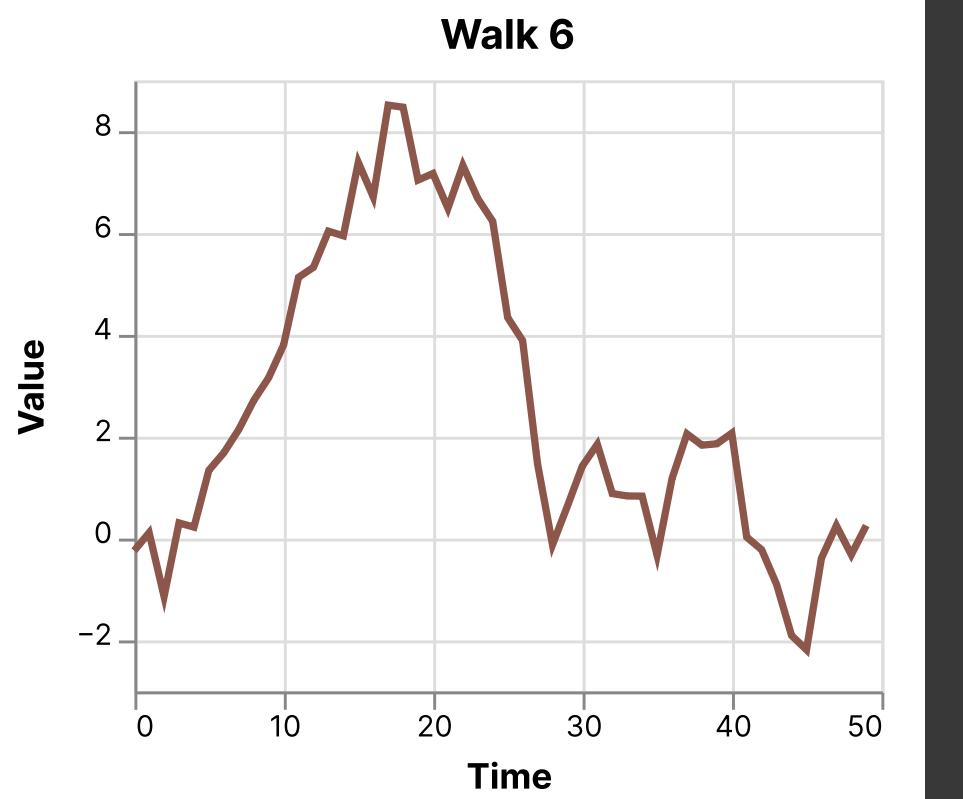
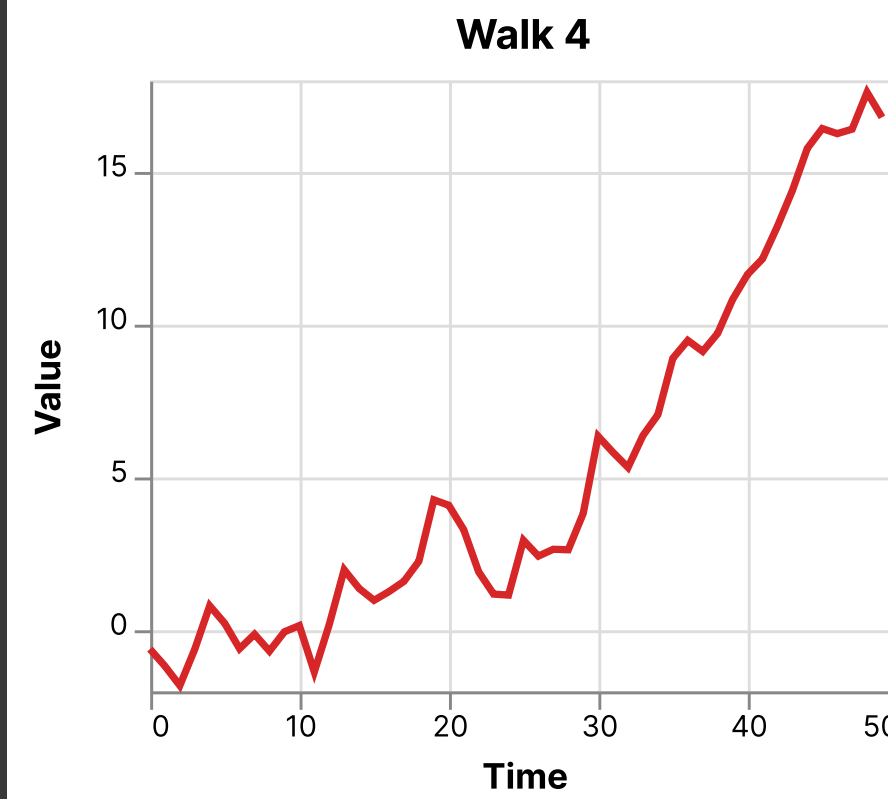
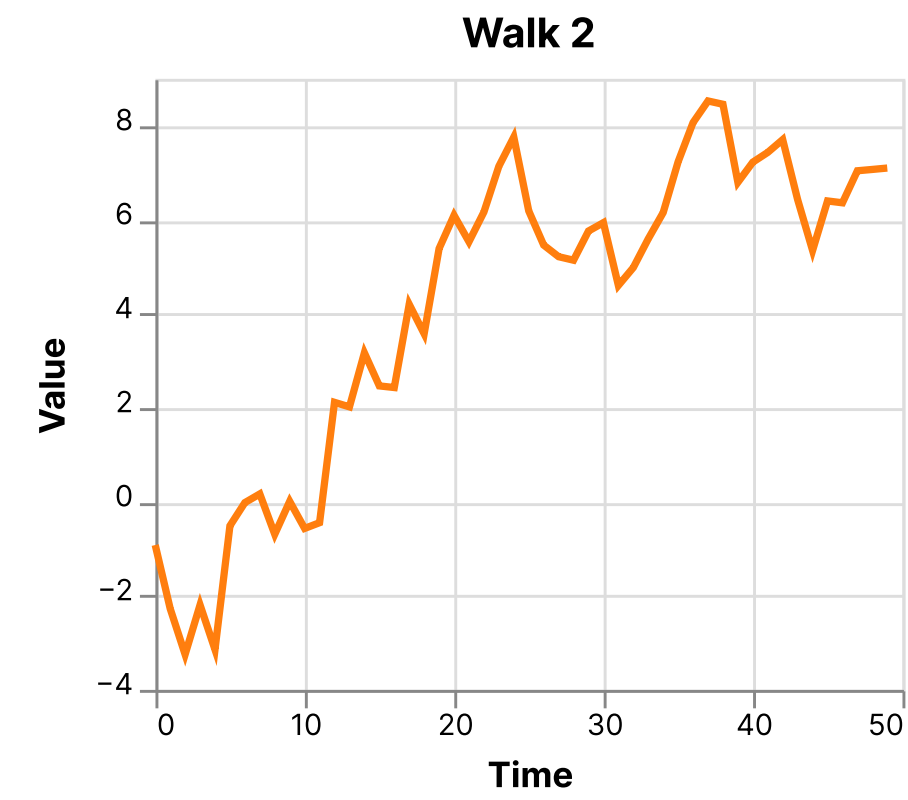
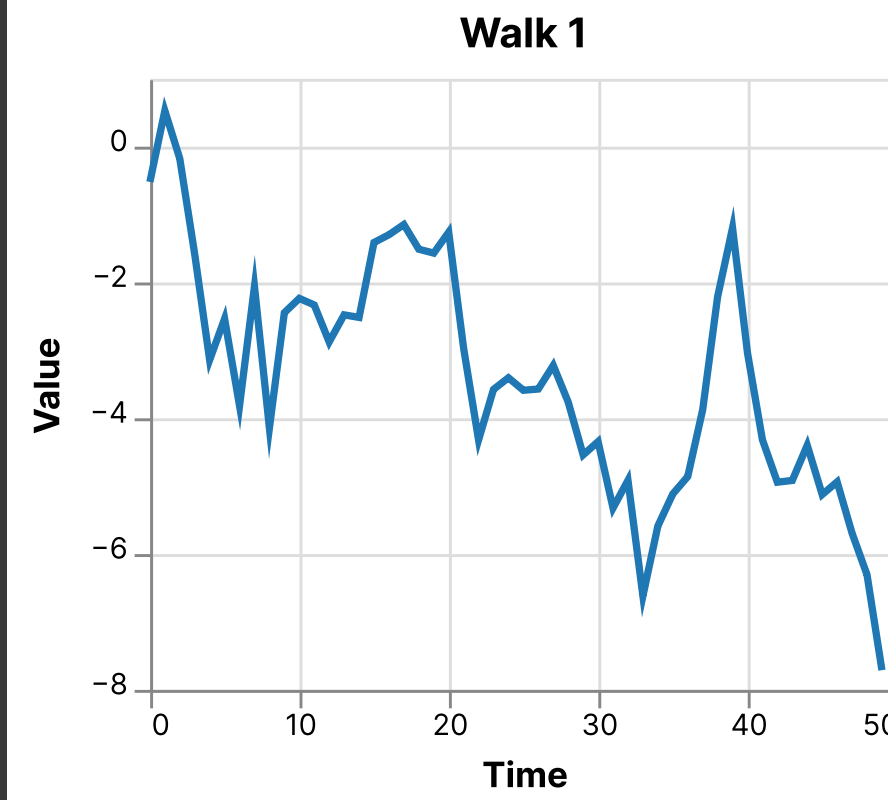


Spurious Correlations

Source: Spurious Correlations

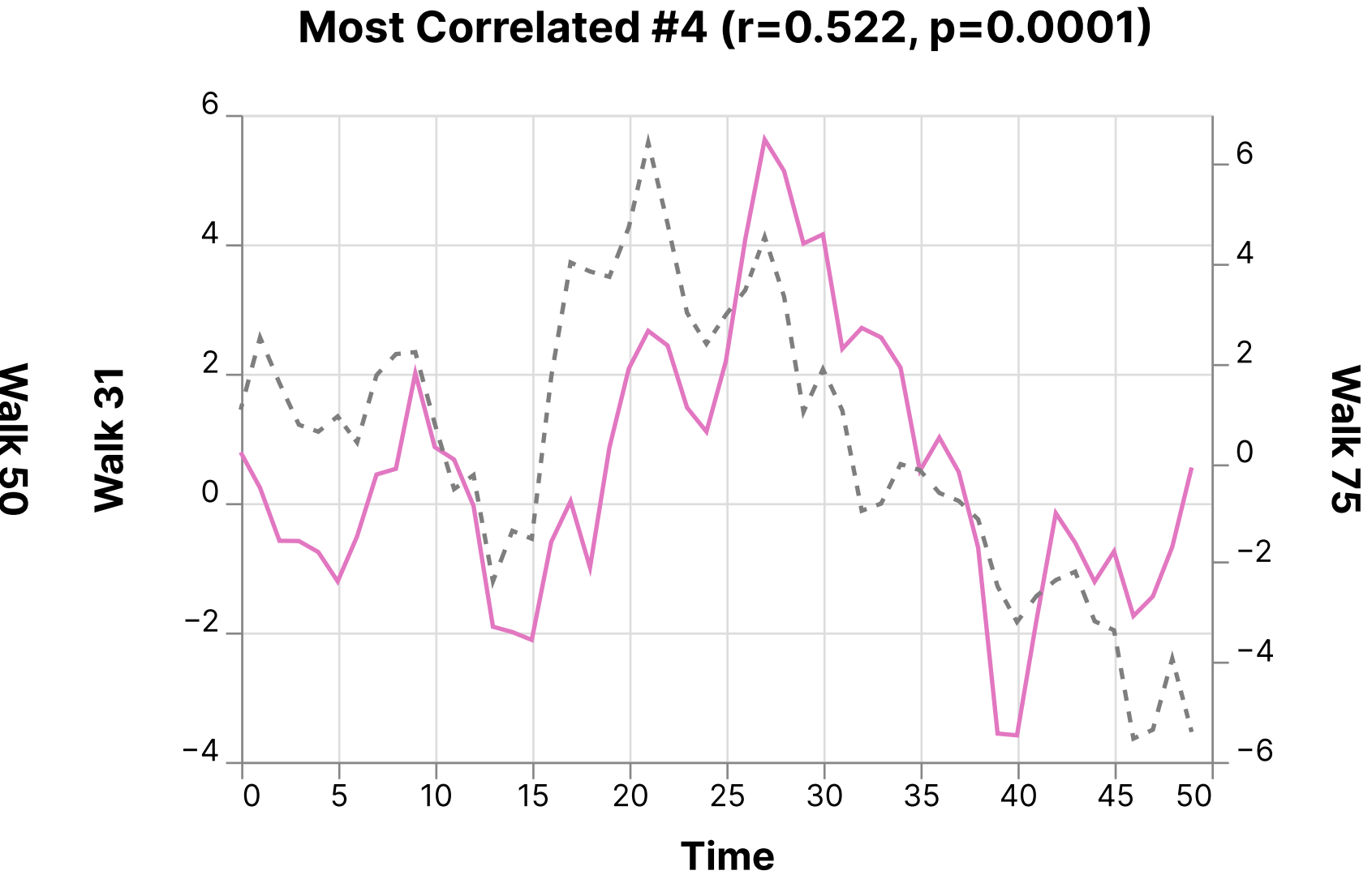
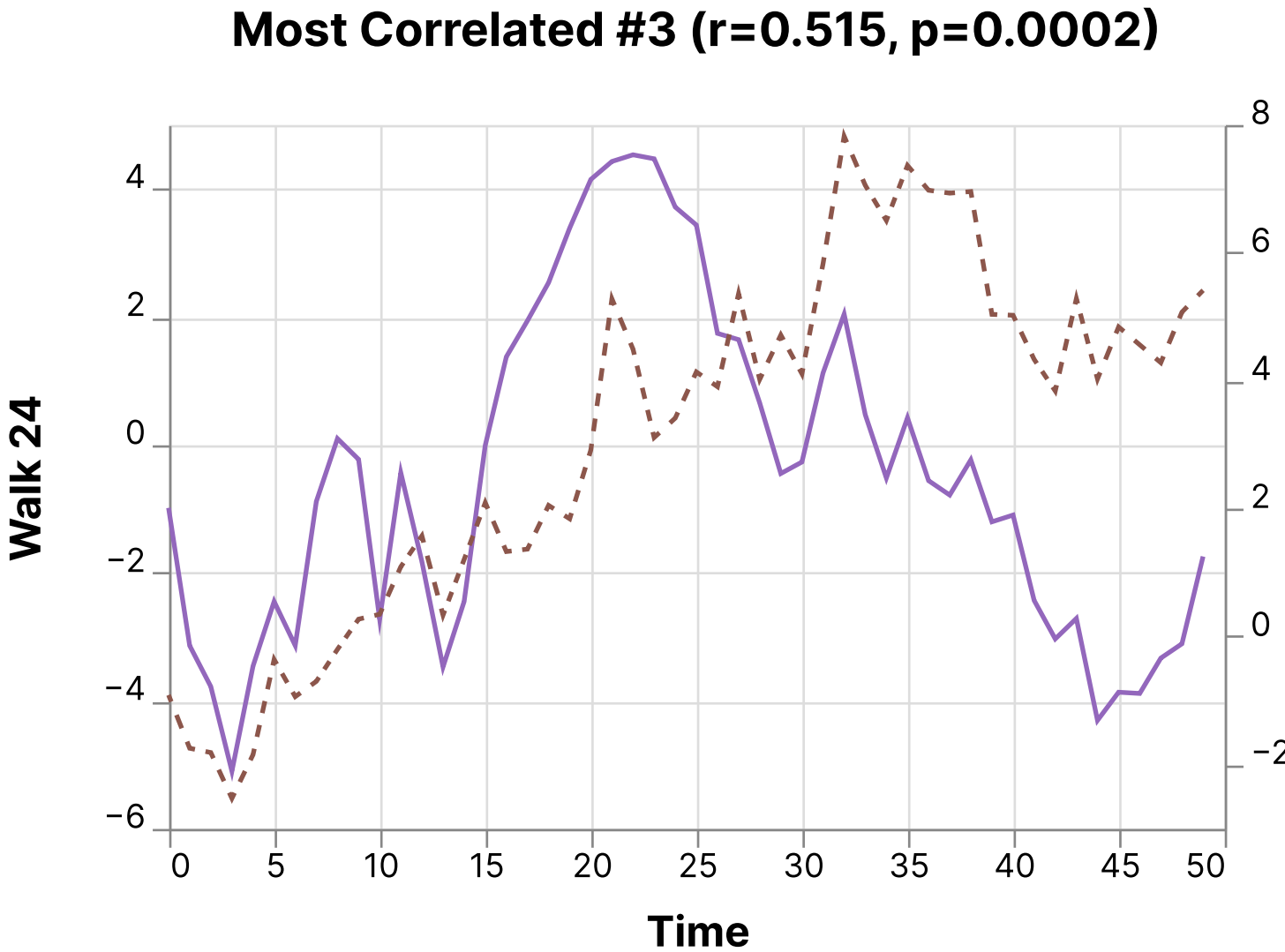
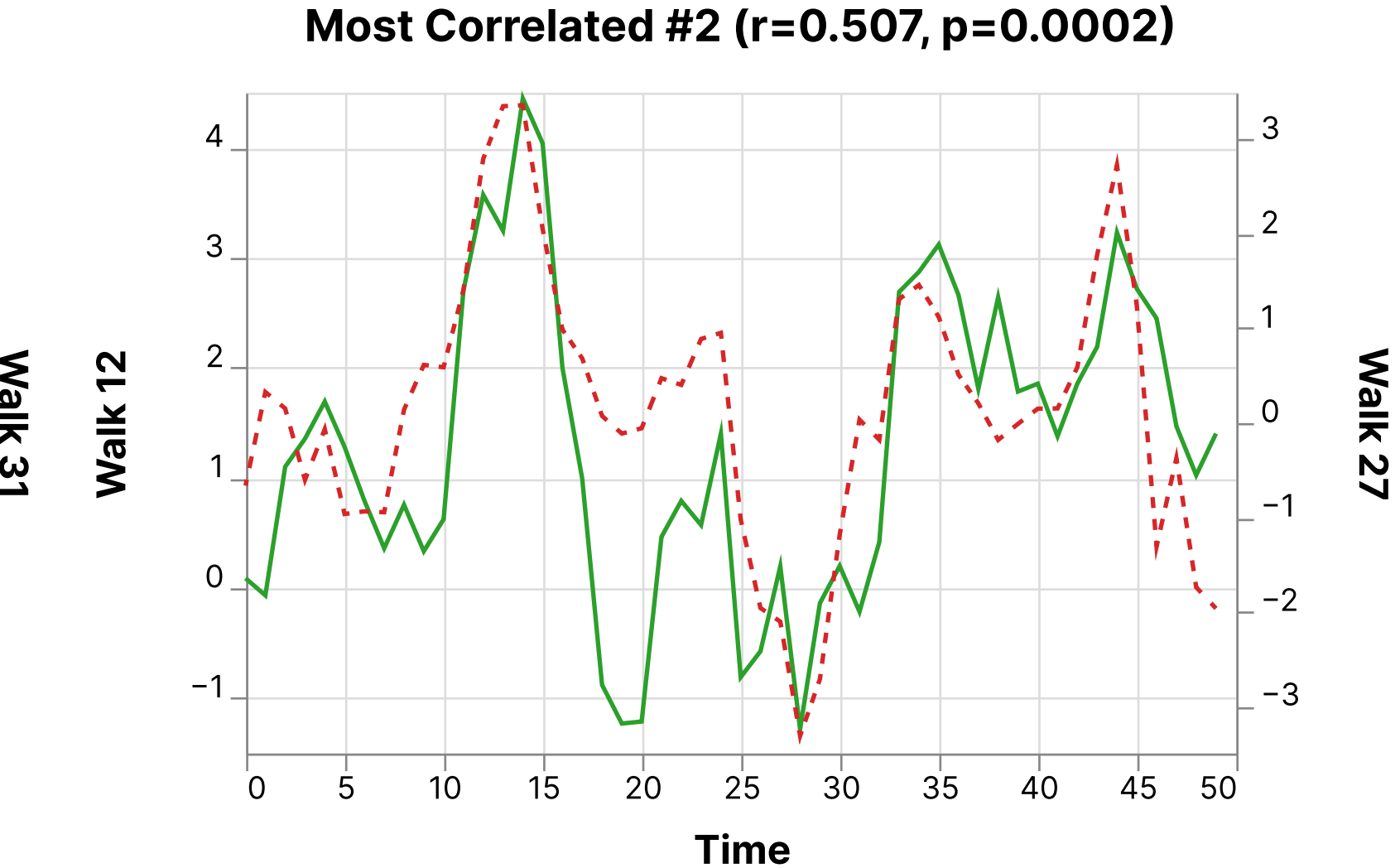
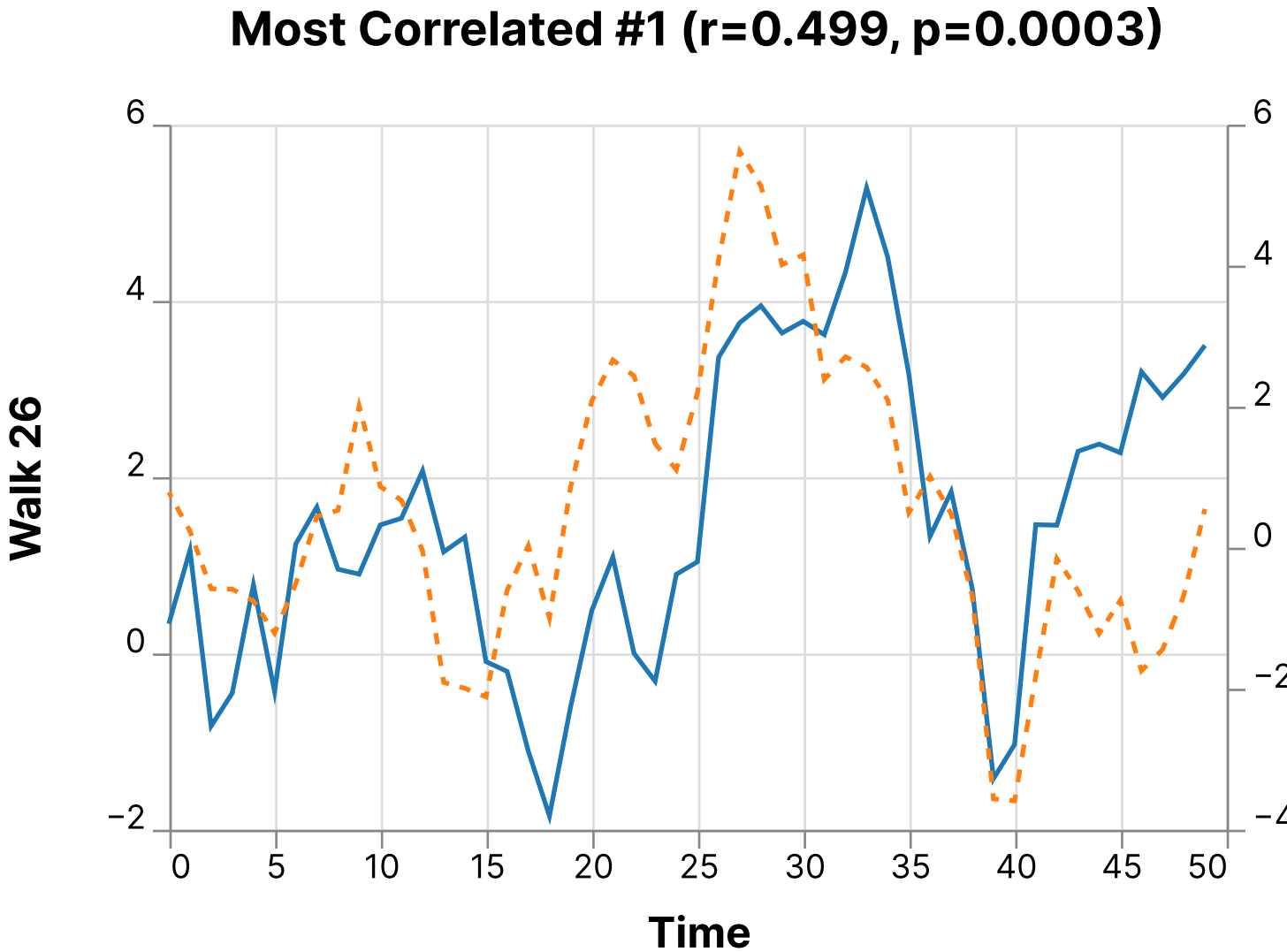
Multiple Comparisons

Random Trends

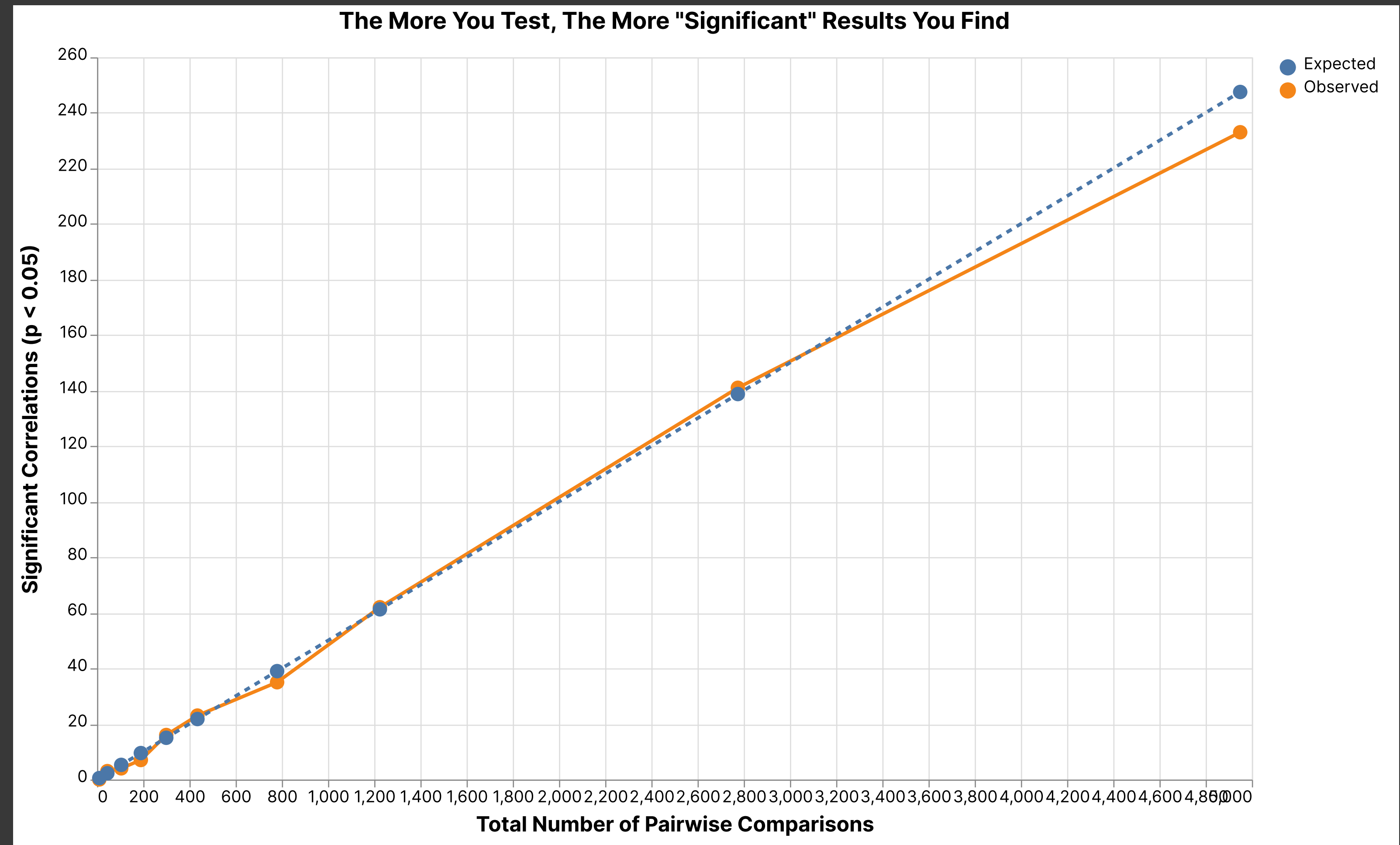


Multiple Comparisons

Most Correlated Pairs (positive correlations, based on de-trended data)

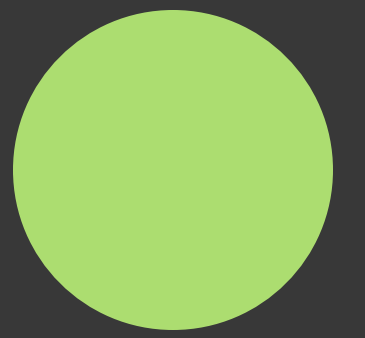


Multiple Comparisons



If $p = 0.05$, then 5% of all comparisons I do will be significant, whether or not they make sense!

Viz Research



Data Transformation + 
Encodings +
Perception +
Psychology

= Data Viz

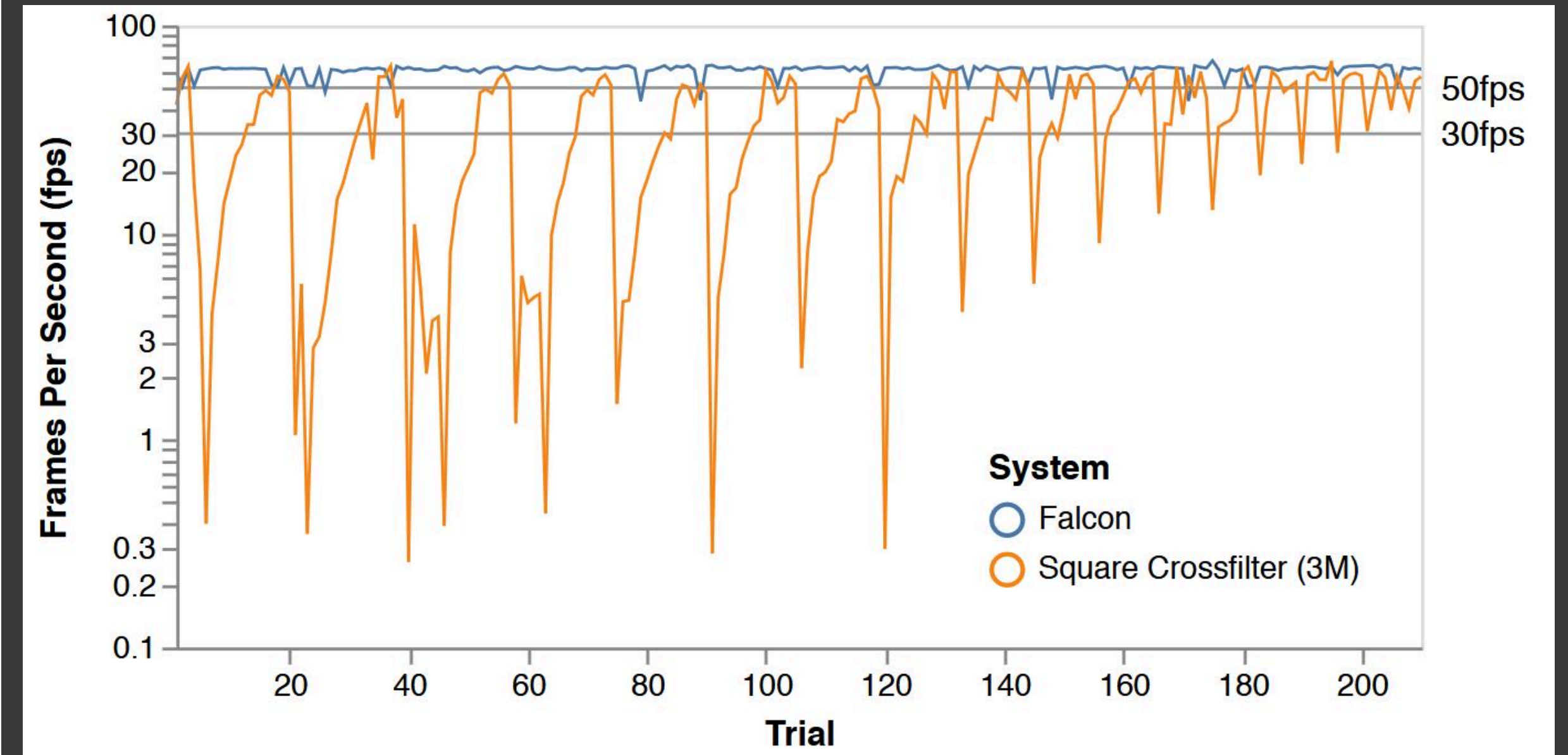
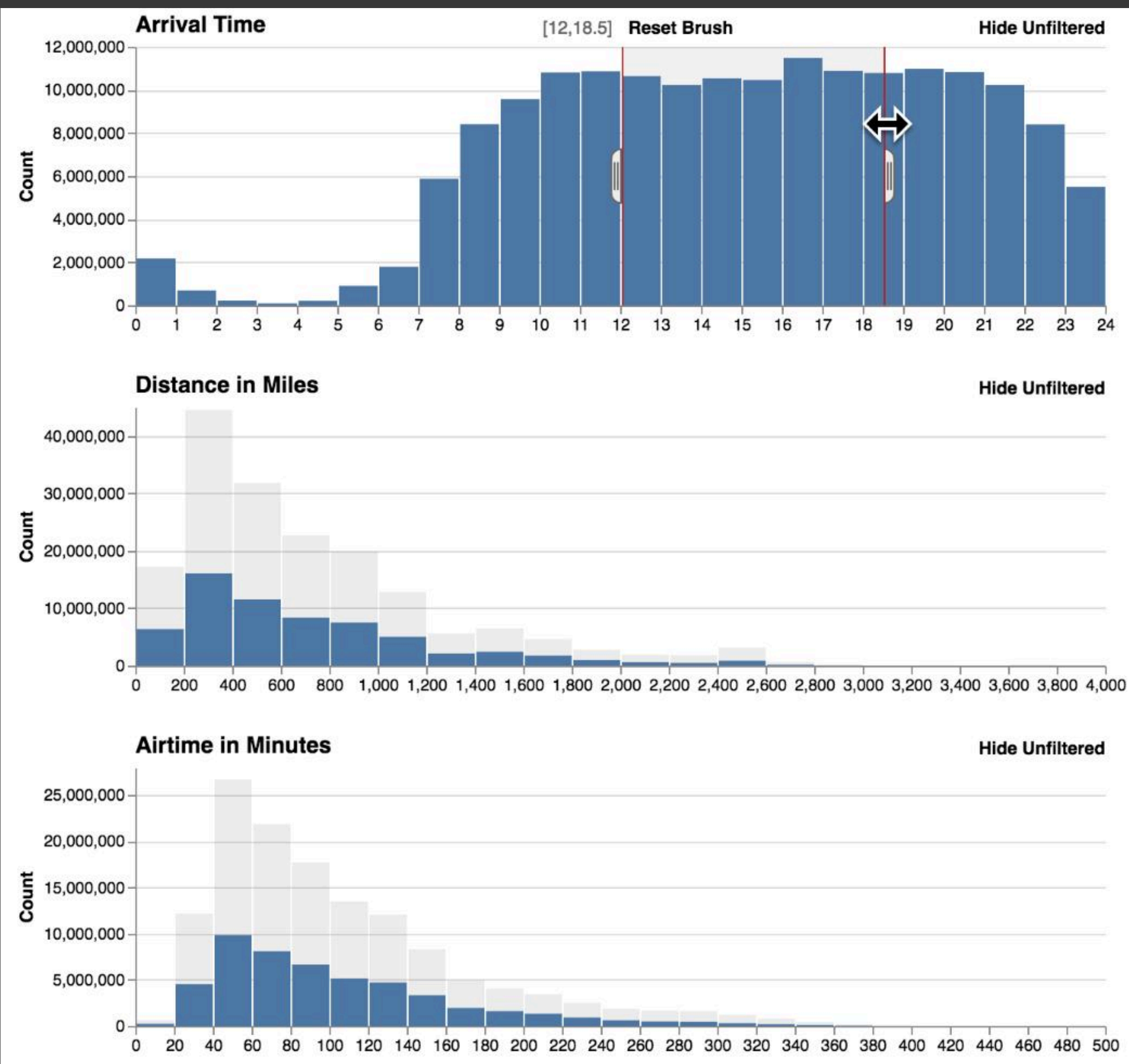
Data Transformation + 

Encodings +

Perception +

Psychology

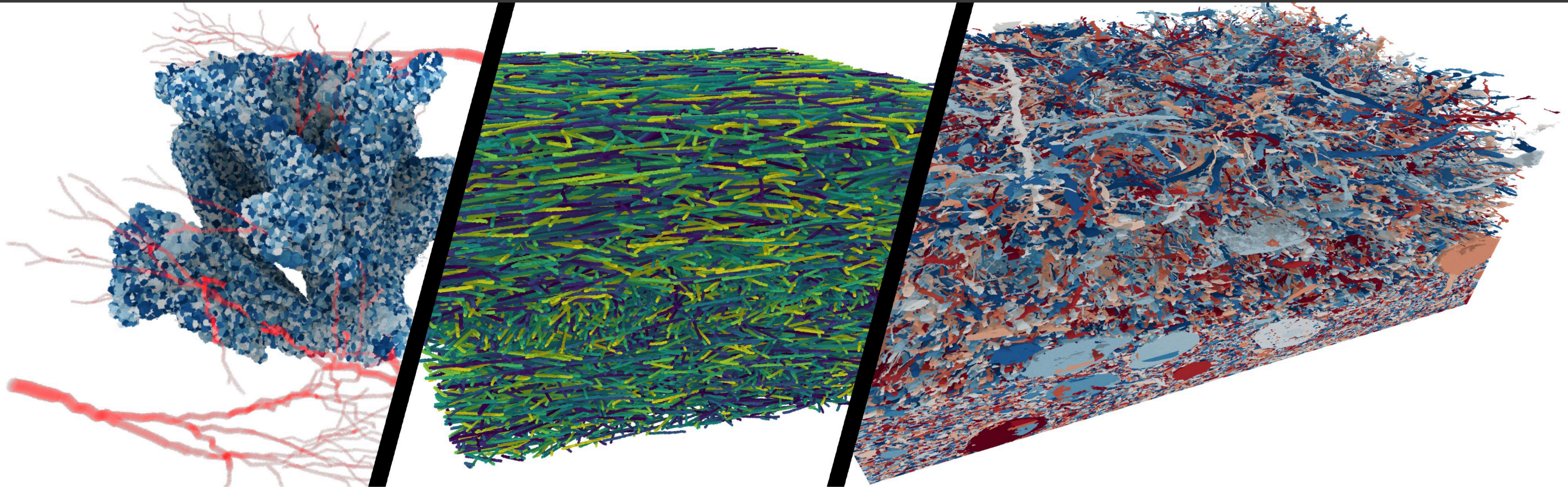
= Data Viz



Falcon: Balancing Interactive Latency and Resolution Sensitivity for Scalable Linked Visualizations (Moritz et al., CHI 2019)

Data Transformation + 
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= Data Viz



Original: 4000 MB
Ours: 112 MB (2.8%)

Original: 5694 MB
Ours: 51 MB (0.9%)

Original: 927.7 GB
Ours: 13.5 GB (1.5%)

Fast Compressed Segmentation Volumes for Scientific Visualization (Piochowiak et al, VIS 2023)

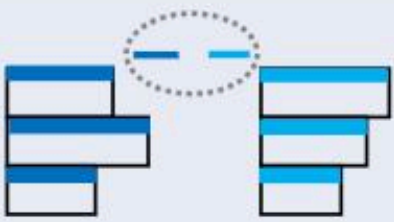
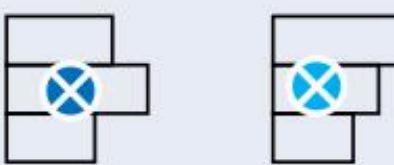
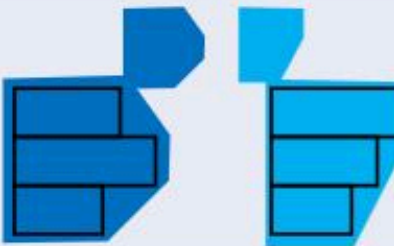
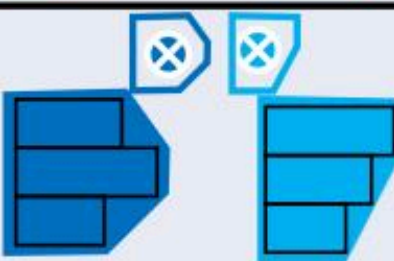
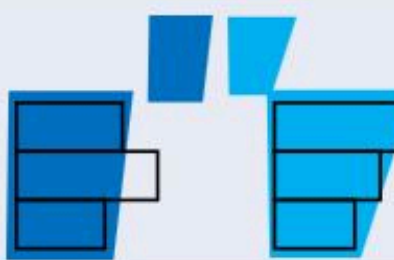
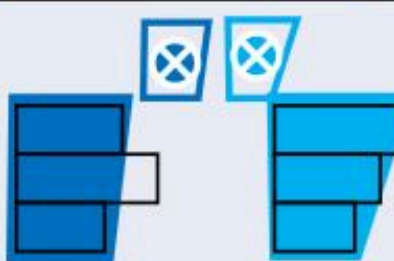
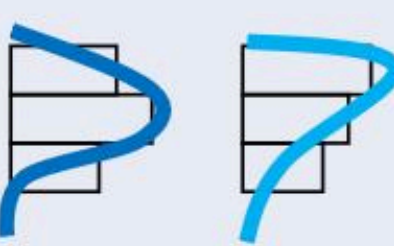
Data Transformation + 
Encodings +
Perception +
Psychology

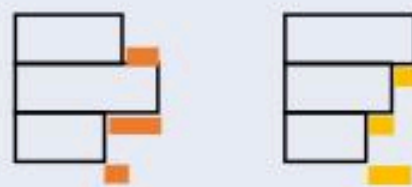
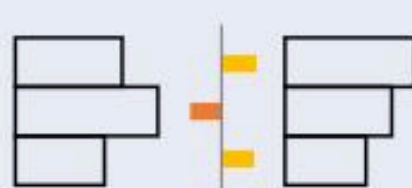
= Data Viz

Click on the set with the highest mean: ■ or ■



The Perceptual Proxies of Visual Comparison (Jardine et al., VIS 2019)

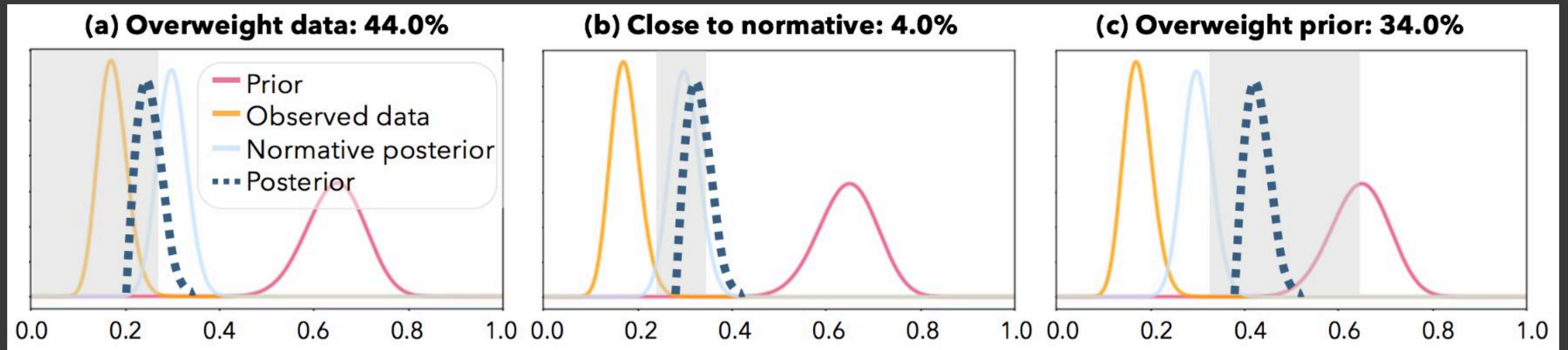
Global Proxies	Mean*		Extracts lengths of bars of each set, computes ensembles, chooses chart with longer ensemble.
	Centroid		Picks chart with largest centroid of the bar areas (along just relevant x axis).
	Hull Area		Calculates convex hull of chart, picks chart with bigger hull area.
	Hull Centroid		Calculates convex hulls, picks larger centroid (along relevant dimension only).
	Trap Area		Draws trapezoids between each chart's top and bottom bars, picks bigger area.
	Trap Centroid		Draws trapezoids between each chart's top and bottom bars, picks trapezoid with larger centroid.
	Symmetry Bias		Calculates skew (i.e. symmetry) of each set, chooses which chart is less skewed (i.e. more symmetric).

Focal Proxies	Range*		Extracts all pairwise deltas within a set, chooses set with the longest pairwise length difference.
	Biggest Mover Pair (Abs)		Between charts, finds largest delta between item pairs (a1-b1, a2-b2...), picks chart with largest positive delta.
	Biggest Mover Pair (Rel)		Same as Biggest Mover Pairwise, but scaled relative to the smaller item within the pair.
	Biggest First Item		Compare top items, picks chart with larger top item.
	Biggest Middle Item		Compare middle items, picks chart with larger middle item.
	Slope Min to Max		Finds each chart's min/max, computes slope between them. Picks the chart with the least-vertical slope.

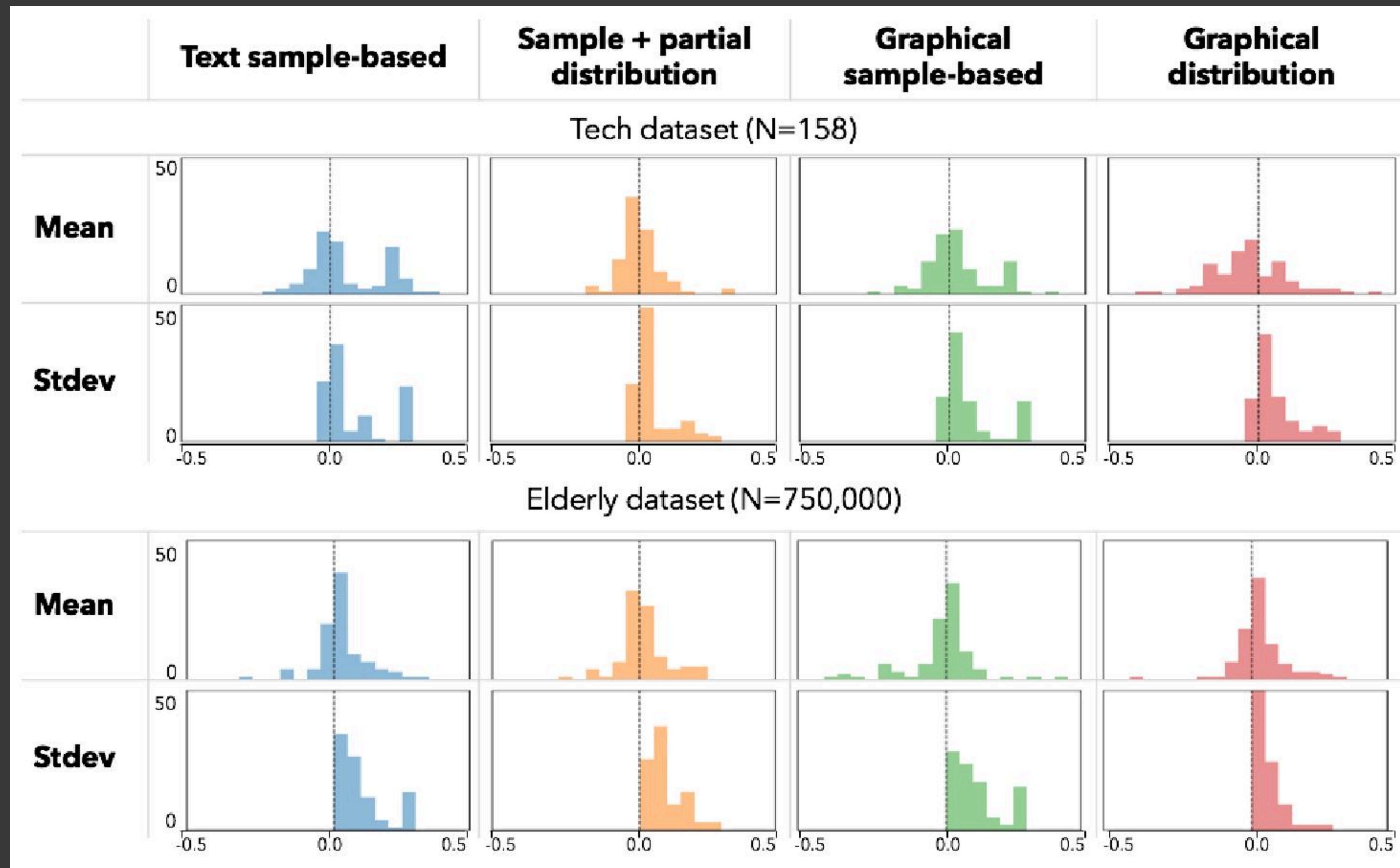
The Perceptual Proxies of Visual Comparison (Jardine et al., VIS 2019)

Data Transformation + 
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= Data Viz



A Bayesian Cognition Approach to Improve Data Visualization (Kim et al., 2019)

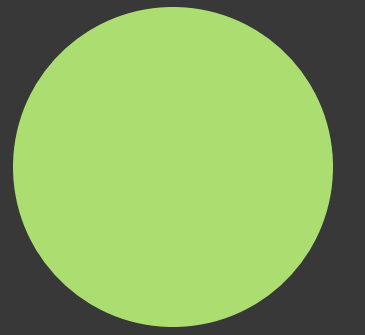


A Bayesian Cognition Approach to Improve Data Visualization (Kim et al., 2019)

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Additional Resources



Learning & Inspiration

- [Altair Cars Demo \(Code examples!\)](#)
- [NYT Graphics](#)
- [UW Visualization Curriculum](#)
- [Data Viz Project](#)
- [IEEE Vis](#)

Libraries

- [seaborn](#) (Python)
- [altair](#) (Python)
- [plotly](#) (Python)
- [vega-lite](#) (JSON)
- [D3](#) (JS)